

**AXIOMATIC
ARITHMETIC**
The Structure of Numbers
An Incomplete Manuscript

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Singapore

Axiomatic Arithmetic: The Structure of Numbers

An Incomplete Manuscript
Commit: c1065fc0

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For those who we've yet to meet

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Author's Note

The genesis for this book began back in late September 2024. The idea was to write a book that explains looks at how we can define all the numbers that are used today from the simple axioms of Zermelo-Fraenkel Set Theory.

However, for multiple reasons, I did not go ahead and finish the book. I finished the first two parts on the foundations of axiomatic arithmetic and the natural numbers, but I struggled to continue the momentum with the other parts of the ladder. I could not properly motivate the integers, the rationals, the real numbers, and the complex numbers. As such, this project was put on hiatus since December 2024.

Now, as the 2-year anniversary of beginning this book approaches, I felt that it would be a waste to let whatever I've written be hidden away much longer. Therefore, I am finally releasing this book to the world, as incomplete as it is. I hope whatever I've written is useful to someone. As the old proverb goes: *"One man's trash is another man's treasure"*. I hope you'll enjoy, despite the lack of completeness.

7 September, 2026

PART 0
Foundation



Wir müssen wissen. Wir werden wissen.
(We must know. We will know.)

— David Hilbert, 1930
([\[Smi14\]](#))

CHAPTER 1

Mathematical Logic

Mathematics is an inherently logical subject; the basis of all mathematics is in how we form logical statements and prove them. For one to begin with the exploration of axioms in mathematics, we have to start with the fundamentals – mathematical logic.

1.1. Mathematical Statements

The core of mathematics is in writing mathematical statements. Maybe your mathematics experience so far has mostly involved finding answers to problems. However, as we begin defining numbers from scratch, we must be adept at writing and communicating mathematical ideas. Precision is vital, so we must first consider the basic building blocks of all mathematical ideas: mathematical statements.

Definition 1.1.1: Statements

A *statement* (or *proposition*) is a sentence that is definitely true or definitely false, but not both.

In other words, a mathematical statement is a sentence we can prove or disprove. A mathematical statement is not an opinion, not an order, not an instruction, nor a feeling. A statement is either a fact or falsity. There is no alternate view that one can take to prove otherwise.

Example 1.1.2

The following are statements.

- A square is a quadrilateral.
- 16 is a square number.
- 9 is a prime.
- Every even number greater than 2 can be expressed as the sum of two primes.
- $3 + 8 = 12$.

The following are **not** statements.

- Would you like some water?
- The product of two numbers.
- Shoo!
- $1 + 2 + 3 + \cdots + n$.
- $x + 1 = 3$.

The reason why “ $x + 1 = 3$ ” is not a statement because its truth or falsity depends on the value of x . For example, if $x = 2$ then the statement is true, but if x is any other value the statement is false. We will examine these kinds of sentences later.

We may name statements using lowercase variables like p , q , and r .

Example 1.1.3

If we define the statements p , q , and r , where

p : every odd number is one more than an even number,

q : every triangle has sides of equal length, and

r : the only even prime number is 2,

then p is a true statement, q is a false statement, and r is a true statement.

Exercise 1.1: Statement Detective

Which of the following are statements?

- | | |
|--------------------------|---|
| (a) Hello! | (e) $8 = 2 \times 4$. |
| (b) 4 equals 5. | (f) There is a number n that is one less than a square. |
| (c) You passed the exam. | (g) $x^2 - 1 = (x - 1)(x + 1)$. |
| (d) $n \geq 4$. | |

1.2. Logical Connectives

Statements come in many different forms. We took a look at some examples and non-examples of statements above. Some statements can be

broken down into ‘simpler’ statements, while others cannot. We have names for these kinds of statements.

Definition 1.2.1: Atomic and Molecular Statements

An **atomic statement** is a statement that cannot be divided into other simpler statements. A statement that is not atomic is called a **molecular statement**.

Example 1.2.2

These are atomic statements.

- A square is a quadrilateral.
- 16 is a square number.
- 9 is prime.
- $3 + 8 = 12$.

These are molecular statements.

- $1 + 1 = 2$ and $2 + 2 = 4$.
- Either 8 is prime or not.
- 18 is a multiple of 3 and 6.
- 10 is even or odd.

To form new more complex (i.e., molecular) statements from simpler (i.e., atomic or molecular) statements, we can use **logical connectives**. These connectives, as implied by their name, connect statements together to form a more complex statement. To define these logical connectives, we elect to use truth tables.

Definition 1.2.3

A **truth table** is a table showing how the truth or falsity of a statement depends on the truth or falsity of the statements from which it is constructed from.

This definition will seem a little out of place without examples, so let’s define a few logical connectives.

1.2.1. Negation

Negation is probably the easiest logical connective to discuss. Negation simply turns any true statement into a false statement, and vice versa.

Example 1.2.4

The negation of “5 is an odd number”, which is true, is “5 is **not** an odd number”, i.e. “5 is an even number”, which is false.

We can capture the idea of negation in the following definition.

Definition 1.2.5: Negation

Let p be a statement. The *negation of p* , denoted $\neg p$ and read “not p ”, has the following truth table.

p	$\neg p$
T	F
F	T

We read the truth table as follows.

- The leftmost column(s) show the ‘source’ statements. In the above definition, we only have p as the singular ‘source’ statement.
- The rightmost column show the ‘output’ statement. In the above definition, we have $\neg p$ as the ‘output’ statement.
- We denote the possibility that a statement is true by “T” and denote the possibility that a statement is false by “F”.

Thus, the above table tells us that $\neg p$ is false when p is true and $\neg p$ is true when p is false.

Example 1.2.6

Let p be the statement “5 is an odd number”, which is a true statement. Then $\neg p$ is “5 is **not** an odd number” (i.e., “5 is an even number”), which is a false statement.

1.2.2. Conjunction

The conjunction connective is better remembered as “AND”.

Definition 1.2.7: Conjunction

Let p and q be statements. The *conjunction of p and q* , denoted $p \wedge q$ and read “ p and q ”, has the following truth table.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

In other words, the conjunction of two statements is true if and only if both the statements are true.

Example 1.2.8

We look at a series of statements and their conjunction.

- If p is the statement “3 is odd” (true) and q is the statement “4 is even” (true), then $p \wedge q$ is the statement “3 is odd **and** 4 is even” (true).
- If p is the statement “a square is a quadrilateral” (true) and q is the statement “a triangle is a quadrilateral” (false), then $p \wedge q$ is the statement “a square is a quadrilateral **and** a triangle is a quadrilateral” (false).
 - Alternatively, $p \wedge q$ can be written as “both squares and triangles are quadrilaterals” which is still false.
- If p is the statement “4 equals 5” (false) and q is the statement “5 is odd” (true), then $p \wedge q$ is the statement “4 equals 5 **and** 5 is odd” (false).
- If p is the statement “9 is a prime” (false) and q is the statement “ $3 + 8 = 12$ ” (false), then $p \wedge q$ is the statement “9 is prime **and** $3 + 8 = 12$ ” (false).

1.2.3. Disjunction

Like with the conjunction connective, the word “disjunction” is not particularly enlightening. The disjunction connective is thus better understood to be akin to “OR”.

Definition 1.2.9: Disjunction

Let p and q be statements. The *disjunction of p and q* , denoted $p \vee q$ and read “ p or q ”, has the following truth table.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Thus, the disjunction of two statements is true if and only if at least one of the statements is true.

Example 1.2.10

We look at a series of statements and their disjunction.

- If p is the statement “3 is odd” (true) and q is the statement “4 is even” (true), then $p \vee q$ is the statement “3 is odd **or** 4 is even” (true).
- If p is the statement “a square is a quadrilateral” (true) and q is the statement “a triangle is a quadrilateral” (false), then $p \vee q$ is the statement “a square is a quadrilateral **or** a triangle is a quadrilateral” (true).
- If p is the statement “4 equals 5” (false) and q is the statement “5 is odd” (true), then $p \vee q$ is the statement “4 equals 5 **or** 5 is odd” (true).
- If p is the statement “9 is a prime” (false) and q is the statement “ $3 + 8 = 12$ ” (false), then $p \vee q$ is the statement “9 is prime **or** $3 + 8 = 12$ ” (false).

Exercise 1.2: Combining Basic Connectives

Define the statements p , q , and r , where

p : 1 is a positive number,
 q : 1 is an even number, and
 r : 1 is an odd number.

Is the statement “ $\neg((p \vee q) \wedge r)$ is false” true?

1.2.4. Conditional/Implication

The conditional (or implication) connective is harder to motivate. Connecting the statements p and q with the conditional has essentially the same meaning as “if p then q ”. However, there is a notable quirk with the definition of the conditional connective which we will see below.

Definition 1.2.11: Conditional/Implication

Let p and q be statements. The statement connecting p and q with

the **conditional** or **implication** connective, denoted $p \implies q$ and typically read “ p implies q ”, has the following truth table.

p	q	$p \implies q$
T	T	T
T	F	F
F	T	T
F	F	T

A statement of the form “ p implies q ” is called a **conditional statement**, and p is called the **antecedent** while q is called the **consequent**.

Remark

The statement “ $p \implies q$ ” can be read many ways. For instance,

- p implies q ;
- if p then q ;
- q if p ;
- p only if q ;
- p is a sufficient condition for q ; and
- q is a necessary condition for p .

Let us motivate the reason why we define the truth table for the conditional as above. Suppose a mother tells her child that “if it is sunny, then we will go to the park”. In this case, the statement p is “it is sunny” and the statement q is “we will go to the park”, making the overall statement $p \implies q$. So what are the possibilities for the truth or falsity of this overall statement?

- If it is sunny and they went to the park, the promise of the overall statement was not broken. Thus, if p and q are both true then $p \implies q$ is also true.
- If it is sunny and they did not go to the park, clearly the mother broke her promise. So if p is true but q is false then we can say that $p \implies q$ is false.

- If instead it is not sunny (maybe it is cloudy) and they went to the park, at no point was the promise broken. The only thing that the mother said was that they would go to the park if it was sunny; it was not the **only** reason that they could go to the park. Perhaps the child was very well behaved and she wanted to reward her child. Perhaps they were on the way back from school and coincidentally pass by the park. Either way, if p is false but q is true, we say that $p \implies q$ is true.
- If it is not sunny and they did not go to the park, it is intuitive to say that the promise was not broken. It could be raining, and so going to the park would be a terrible idea! So if both p and q are false then $p \implies q$ is still true.

Thus, the only case when $p \implies q$ is false is if p is true but q is false.

Remark

A conditional statement that is true by virtue of the antecedent being false is said to be *vacuously true*.

Example 1.2.12

- The statement “if 3 is odd then 4 is even” is clearly true, with both antecedent and consequent being true.
- The statement “if 5 is prime then 6 is prime” is clearly false, with a true antecedent and a false consequent.
- The statement “if 0 equals 1 then 5 is odd” is true since the antecedent is false, meaning that the statement is vacuously true (with a true consequent).
- The statement “if 1 equals 2 then 1234 equals 5678” is true since the antecedent is false, meaning that this is a vacuously true statement (with a false consequent).

1.2.5. Biconditional

Sometimes we would want to form a statement of the form “if p then q , and if q then p ”. Using symbolic notation we could write it as “ $(p \implies q) \wedge (q \implies p)$ ”, but this is a little clunky to write every time we wish to write such a statement. This motivates the biconditional connective.

Definition 1.2.13: Biconditional

Let p and q be statements. The statement connecting p and q with the **biconditional** connective, denoted $p \iff q$ and typically read “ p if and only if q ”, has the following truth table.

p	q	$p \iff q$
T	T	T
T	F	F
F	T	F
F	F	T

A statement of the form “ p if and only if q ” is called a **biconditional statement**.

Remark

The statement “ $p \iff q$ ” can also be read many ways. For instance,

- p if and only if q ;
- p iff q (i.e., “iff” is a shorthand for “if and only if”);
- p is a necessary and sufficient condition for q ; and
- p is equivalent to q .

Exercise 1.3: (Bi)conditionals

Let t be the conditional statement “20 is a multiple of 5 if 20 ends in a 0 or 5”. Let r be the antecedent of t and s be the consequent of t .

- Write down the statement s in English.
- Define atomic statements p and q such that r is a molecular statement formed by p and q .
- Write the statement t using only atomic statements and logical connectives.
- Write the biconditional statement “ $s \iff r$ ” in English. Is “ $s \iff r$ ” true?

1.3. Logical Equivalence

Sometimes, the molecular statements that we are considering will have an equivalent form that is simpler to work with. This means that, regardless of the underlying truth values of the atomic statements making up the molecular statements, they hold the same truth value. This is the idea of logical equivalences.

Definition 1.3.1

Two statements p and q are *logically equivalent* if and only if they have identical truth values, regardless of the atomic statements that make up them.

We write $p \equiv q$ if and only if the statements p and q are logically equivalent.

Remark: For the Pedantic

Logical equivalences is really the same as biconditionals. That is, when we write " $p \equiv q$ " it is the same as " $p \iff q$ ". However, for this book, we will write " $p \equiv q$ " to mean " p and q have the same truth table", whereas we will write " $p \iff q$ " if we intend to mean that " $\text{if } p \text{ then } q \text{ and if } q \text{ then } p$ " (i.e., the 'traditional' biconditional meaning).

To show that two statements are logically equivalent, we typically consider their truth tables.

Example 1.3.2

Let p and q be statements. We show that $(p \implies q) \equiv (\neg p) \vee q$ by considering the truth tables of the two statements.

p	q	$\neg p$	$p \implies q$	$(\neg p) \vee q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

By inspection, we can see that $(\neg p) \vee q$ has the same truth values as $p \implies q$ regardless of the truth values of p and q . Thus, the $p \implies q$ is logically equivalent to $(\neg p) \vee q$.

Remark

We separate the intermediate value(s) from the rest by drawing a double line to the sides of the intermediate values.

Example 1.3.3

We show that

$$(p \iff q) \equiv (p \wedge q) \vee ((\neg p) \wedge (\neg q))$$

for any statements p and q . For brevity, let $r = p \wedge q$, $s = (\neg p) \wedge (\neg q)$, and $t = (p \wedge q) \vee ((\neg p) \wedge (\neg q))$. So we are to show that $(p \iff q) \equiv t$.

p	q	$\neg p$	$\neg q$	r	s	$p \iff q$	t
T	T	F	F	T	F	T	T
T	F	F	T	F	F	F	F
F	T	T	F	F	F	F	F
F	F	T	T	F	T	T	T

By inspection of the truth table we establish the required result.

Exercise 1.4: Contrapositive

Show that

$$(p \implies q) \equiv ((\neg q) \implies (\neg p))$$

for all statements p and q .

What **Exercise 1.4** reveals is an important relationship regarding conditional statements.

Definition 1.3.4: Contrapositive

Let p and q be statements. The *contrapositive* of the conditional statement $p \implies q$ is $(\neg q) \implies (\neg p)$.

Proposition 1.3.5

The contrapositive of a conditional statement is equivalent to the conditional statement.

Proof. See [Exercise 1.4](#). ■

There are other properties of logical connectives that we will note down in the following propositions.

Proposition 1.3.6: Commutativity of AND and OR

Let p and q be statements. Then $p \wedge q \equiv q \wedge p$ and $p \vee q \equiv q \vee p$.

Proof. We consider truth tables.

p	q	$p \wedge q$	$q \wedge p$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	F	F

p	q	$p \vee q$	$q \vee p$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	F

From the above truth tables, the result is established. ■

Proposition 1.3.7: Associativity of AND and OR

Let p , q , and r be statements. Then $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$ and $p \vee (q \vee r) \equiv (p \vee q) \vee r$.

Proof. We first show $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$.

p	q	r	$p \wedge q$	$q \wedge r$	$p \wedge (q \wedge r)$	$(p \wedge q) \wedge r$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	F	T	F	F
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

From the above truth table, the result is established. We now show $p \vee (q \vee r) \equiv (p \vee q) \vee r$.

p	q	r	$p \vee q$	$q \vee r$	$p \vee (q \vee r)$	$(p \vee q) \vee r$
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	T	T	T	T
T	F	F	T	F	T	T
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	F	T	T	T
F	F	F	F	F	F	F

This proves the second claim. ■

Proposition 1.3.8: Distributivity of AND and OR

Let p , q , and r be statements. Then $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ and $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

Proof. We first prove the first claim.

p	q	r	$p \wedge q$	$p \wedge r$	$q \vee r$	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	T	F	F
T	F	T	F	T	T	T	T
T	F	F	F	F	F	F	F
F	T	T	F	F	T	F	F
F	T	F	F	F	T	F	F
F	F	T	F	F	T	F	F
F	F	F	F	F	F	F	F

We now prove the second claim.

p	q	r	$p \vee q$	$p \vee r$	$q \wedge r$	$p \vee (q \wedge r)$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	F	T	T
T	F	T	T	T	F	T	T
T	F	F	T	T	F	T	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	F	F
F	F	T	F	T	F	F	F
F	F	F	F	F	F	F	F

This proves the second claim. ■

We now note an important set of properties regarding logical connectives.

Proposition 1.3.9: De Morgan's Laws

Let p and q be statements. Then $\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$ and $\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$.

Proof. See [Exercise 1.5](#) (later). ■

We end this section by looking at the order of precedence for logical connectives.

Connective	Precedence
$\neg$	1
$\wedge$ and $\vee$	2
$\implies$ and $\iff$	3

To disambiguate the order of operations, we use parentheses (i.e., brackets).

Exercise 1.5: De Morgan's Laws

Prove [Proposition 1.3.9](#).

Exercise 1.6: Simplifying Statements

Let p , q , and r be statements. Simplify the statement

$$((p \vee \neg q) \wedge \neg r) \vee p$$

into a statement that uses only three logical connectives in total.

1.4. Predicates and Quantifiers

It would certainly be useful to include variables in our statements. Suppose I wanted to claim that “if n is odd, then $n + 1$ is not odd”. This looks like a conditional statement, but because n is a variable, this is not a statement. In fact, such sentences are called predicates.

Definition 1.4.1: Predicates

A *predicate* is a sentence that contains a finite number of *free variables* and becomes a statement when specific values are substituted for the variables.

For notational clarity, and to avoid confusing predicates with statements, we denote predicates using uppercase letters.

Example 1.4.2

- Consider the sentence “ n is a multiple of 3”. This is not a statement since we have the free variable “ n ” within the sentence; this is a predicate. Denoting this predicate by $P(n)$, we see that $P(0)$ is a true statement, $P(1)$ is a false statement, $P(2)$ is a false statement, $P(3)$ is a true statement, and so on.
- Let the predicate $Q(n)$ be “ n is a factor of 9”. Then $Q(1)$, $Q(3)$, and $Q(9)$ are true statements.

Returning back to our opening example, we could let $P(n)$ denote the predicate “if n is odd, then $n + 1$ is not odd”. One observes that, no matter what integer we substitute for n , the statement $P(n)$ is always true. What we really want to do with “if n is odd, then $n + 1$ is not odd” is to quantify that this is true for all integers n .

There are two main quantifiers that we will look at here. The first is called the universal quantifier.

Definition 1.4.3: Universal Quantification

The *universal quantifier* is $\forall$ and is read “for all” or “every”.

A statement q of the form “ $\forall x, P(x)$ ” where $P(x)$ is a predicate is called a *statement*. The statement q is true if and only if $P(x)$ is true for all values of x .

Example 1.4.4

- We can express the (false) statement that “every number is greater than 0” as “ $\forall x, x > 0$ ”. Here the predicate $P(x)$ is “ $x > 0$ ”. This is clearly false since 0 is not greater than 0.
- We can express the (true) statement that “for every integer n , the integer $2n$ is even” as “ $\forall x, ((n \text{ is an integer}) \implies (2n \text{ is even}))$ ”.

Remark: For Those Who Know Sets

We can write “for every integer n , the integer $2n$ is even” as “ $\forall x \in \mathbb{Z}, (2n \text{ is even})$ ” instead.

The second quantifier is called the existential quantifier.

Definition 1.4.5: Existential Quantification

The *existential quantifier* is $\exists$ and is read “there exists” or “there is”.

A statement q of the form “ $\exists x, P(x)$ ” where $P(x)$ is a predicate is called a *existential statement*. The statement q is true if and only if $P(x)$ is true for at least one value of x .

Example 1.4.6

We can express the (true) statement that “there is a number greater than 0” as “ $\exists x, x > 0$ ”. Here the predicate $P(x)$ is “ $x > 0$ ”. This is clearly true since 1 is greater than 0.

Remark: In Aid of Clarity

We can write “ $\exists x, P(x)$ ” as “ $\exists x$ such that $P(x)$ ”. We can write “such that” as “s.t.” to make it more concise, i.e. “ $\exists x, P(x)$ ” is the same as “ $\exists x$ s.t. $P(x)$ ”.

It is critical to note that the truth or falsity of such quantified statements depends on the *domain of discourse*. Consider the quantified statement “ $\forall x, \exists y$ s.t. $(y < x)$ ”. Certainly if we are considering all the real numbers this is a true statement! We could easily set $y = x - 1$ and we would be done here. However, over the positive integers, such a trick no longer works. If $x = 1$, then $y = 1 - 1 = 0$ is not a positive integer. So it is essential to clarify the domain of discourse when writing down quantified statements.

Exercise 1.7: Fermat’s Last Theorem

Suppose our domain of discourse is the positive integers. Convert the statement “for all $n > 2$ the equation $x^n + y^n = z^n$ has no solution” into symbolic notation using quantifiers and connectives.

Let us now look back at the example “ $\forall x, \exists y \text{ s.t. } (y < x)$ ” using the positive integers as the domain of discourse. To prove that this statement is false, we gave the *counterexample* of $x = 1$. In fact, we proved the statement “ $\exists x \text{ s.t. } \forall y, (y \not< x)$ ”, i.e. “ $\exists x \text{ s.t. } \forall y, (y \geq x)$ ”. This is an example of negating a quantified statement.

Proposition 1.4.7: Negating Quantified Statements

Let $P(x)$ be a predicate. Then

- $\neg(\forall x, P(x)) \equiv \exists x \text{ s.t. } (\neg P(x))$; and
- $\neg(\exists x \text{ s.t. } P(x)) \equiv \forall x, (\neg P(x))$.

Proof. For the first equivalence, recall that the statement “ $\forall x, P(x)$ ” is true if and only if all x in the domain of discourse makes $P(x)$ true. Hence as long as one x makes $P(x)$ false, then the statement is false. Thus the negation would be “ $\exists x \text{ s.t. } (\neg P(x))$ ”.

Likewise, for the second equivalence, the statement “ $\exists x \text{ s.t. } P(x)$ ” is true if there is at least one x in the domain of discourse that makes $P(x)$ true. So if that statement is false, then there cannot be any x that makes $P(x)$ true. Hence the negation would be “ $\forall x, (\neg P(x))$ ”. ■

Example 1.4.8: Negating Quantified Statements

- Consider the statement “ $2n$ is even for all n ”, using the positive integers as the domain of discourse. One may write this using quantifiers as “ $\forall n, (2n \text{ is even})$ ”. The negation of this statement is thus “ $\exists n \text{ s.t. } (2n \text{ is not even})$ ”, i.e. “ $\exists n \text{ s.t. } (2n \text{ is odd})$ ”.
- Now consider the statement “there is a prime p such that $p = n^2 + 1$ for some n ”, where we again use the positive integers as the domain of discourse. We can write this symbolically as “ $\exists p \text{ s.t. } (\exists n \text{ s.t. } p = n^2 + 1)$ ”. Negating this statement would yield “ $\forall p, (\forall n, (p \neq n^2 + 1))$ ”.

We can also negate quantified statements that contain conditionals. We just have to appeal to the following fact.

Proposition 1.4.9: Negating Conditional Statements

Let p and q be statements. Then $\neg(p \implies q) \equiv p \wedge \neg q$.

Proof. See [Exercise 1.8](#) (later). ■

This fact is very useful when we are negating quantified conditional statements.

Example 1.4.10: Negating Quantified Conditionals

Consider the statement “ n^3 is odd if n is odd for all n ” where the domain of discourse is the integers. We can write this as

$$\forall n, ((n \text{ is odd}) \implies (n^3 \text{ is odd})).$$

The negation of this statement would hence be

$$\exists n \text{ s.t. } ((n \text{ is odd}) \wedge \neg(n^3 \text{ is odd})),$$

which in English would be “there exists n such that n is odd and n^3 is even”.

Exercise 1.8: Negating Conditional Statements

Prove **Proposition 1.4.9**.

Exercise 1.9: An Exercise in Symbolic Negation

Suppose our domain of discourse is the real numbers. Consider the statement “for all non-zero x , there exists a y such that $xy = 1$ ”.

- (i) Express the above statement symbolically.
- (ii) Negate the above statement, writing your answer in symbolic notation.

Lastly, let us take a look at a peculiar case of universal statements – ones where no element in the domain of discourse satisfies it.

Consider the universal statement “all my cats are dogs” where the domain of discourse are all the cats that I own. We can write this semi-symbolically as “ $\forall x, \text{Dog}(x)$ ” where the predicate $\text{Dog}(x)$ is true if and only if x is a dog. Clearly, this is absurd! This surely must be a false statement! But consider the case when I have no cats. Is this statement true?

Consider its negation, i.e. “there is a cat that is not a dog”. We can write this using the above symbols as “ $\exists x \text{ s.t. } \neg \text{Dog}(x)$ ”. Now recall that this existential statement is true if and only if there is an element that satisfies the predicate. In this case, one must give an example of x such that “ $\neg \text{Dog}(x)$ ” is true. However, I have no cats! It is impossible to give

an example that makes “ $\neg \text{Dog}(x)$ ” true. Thus, the statement “there is a cat that is not a dog” is false.

However, recall that “there is a cat that is not a dog” is the negation of “all my cats are dogs”. If the negation of that statement is false, that means that the original statement is true! Indeed, “all my cats are dogs” is true because I have no cats!

This is yet another example of a *vacuously true* statement. If the domain of discourse is empty, then all universal statements are automatically true.

Example 1.4.11: Vacuously True Universal Statements

- If the domain of discourse is the set of pennies that I have, and if I have no pennies, then “all my pennies are \$100 dollar bills” is vacuously true.
- Similarly, “none of my pennies are \$100 dollar bills” is also vacuously true.
- A child may say that “I ate every vegetable on my plate”. This is true when there were no vegetables on their plate in the first place.

1.5. Hierarchy of Mathematical Results

Mathematical statements often have a certain ‘tier’ attached to them. We look at the hierarchy of some of these ‘tiers’.

- **Definition:** A definition is a precise, formal, and abstract statement that introduce and clarify the meaning of mathematical terms, concepts, and objects.
- **Axiom:** A statement that is assumed to be true to act as a starting point for further reasoning.
- **Proposition:** A proposition can be thought of as a general proven result. “*Claims*” and “*Observations*” are equivalent to propositions.
- **Lemma:** A lemma can be thought of as a ‘small’ proven result that can help build other mathematical results. For example, Euclid’s lemma that “if a prime p divides the product ab of two integers a and b , then p must divide at least one of those integers a or b ” is used in the proof of the Fundamental Theorem of Arithmetic.

- **Theorem:** A theorem can be thought of as a ‘big’ proven result. For example, the Fundamental Theorem of Arithmetic is core to arithmetic as it describes how integers can be uniquely decomposed into its prime factors.
- **Corollary:** A corollary is said to be a ‘follow-up result’ from a theorem. These results usually follow ‘very quickly’ from a theorem or a proposition. For example, the AM-GM inequality is a corollary of the Pythagorean theorem.

CHAPTER 2

Introduction to Sets

The most fundamental object that we will be using in our exploration of the structure of numbers is the set. Sets are omnipresent throughout mathematics, so it is important that one is fluent in the language of sets. This chapter serves as an introduction to sets, as we will be formally defining what they are in a later chapter.

2.1. What Are Sets?

Definition 2.1.1: A Simple Definition of Sets

A *set* is a collection of things called *elements* of the set.

Of course, we will formally define what sets are in a future chapter, but this would suffice for now.

A set is often described by listing elements separated by commas, or by a characterizing property of its elements, within braces ($\{ \}$).

Example 2.1.2: Examples of Sets

- The collection $\{a, b, c\}$ is a set with 3 elements, namely a , b , and c . This is read “the set containing the elements a , b , and c ”.
- The collection $\{2, 3, 4, 5\}$ is a set with 4 elements, namely 2, 3, 4, and 5.
- The collection $\{\{1, 2, 3\}, \{4, 5\}, \{6, \{7\}\}\}$ is a set with 3 elements, which are also sets. Namely, it contains the set $\{1, 2, 3\}$, containing the elements 1, 2, and 3, the set $\{4, 5\}$, containing the elements 4 and 5, and the set $\{6, \{7\}\}$, containing the number 6 and the set $\{7\}$ which contains a single element 7.

What if a set has no elements?

Definition 2.1.3: Empty Set

The *empty set* is the set $\{ \}$ and is denoted by $\emptyset$. That is, $\emptyset = \{ \}$.

The empty set will come in useful when we are defining the natural numbers.

Let’s now look at set equality.

Definition 2.1.4: Set Equality

Two sets A and B are equal if and only if they contain the same elements.

To reflect that the sets A and B are equal to each other, we write $A = B$. If the sets A and B are not equal to each other, we instead write $A \neq B$.

Example 2.1.5

Consider the sets $\{1, 2, 3, 4\}$, $\{4, 3, 2, 1\}$, and $\{3, 4, 1, 2\}$. These sets are equal to each other even though their elements are listed in a different order. Thus we can write

$$\{1, 2, 3, 4\} = \{4, 3, 2, 1\} = \{3, 4, 1, 2\}.$$

However $\{1, 2, 3, 4\}$ and $\{1, 2, 3, 5\}$ are not equal since the elements in the two sets differ. Hence,

$$\{1, 2, 3, 4\} \neq \{1, 2, 3, 5\}.$$

It is important to emphasise that the order of elements in the sets do not matter.

We have specific notation to denote that an object is in a set.

Definition 2.1.6

Let A be a set. We write $x \in A$ if and only if x is an element of A . Otherwise we write $x \notin A$.

Example 2.1.7

Consider the set $A = \{1, 2, 3\}$. Then clearly 1, 2, and 3 are elements of the set A , and so we may write $1 \in A$, $2 \in A$, and $3 \in A$. We could also write this more compactly as $1, 2, 3 \in A$. On the other hand, one sees that 0, -1, and 4 are not elements of the set, so $0 \notin A$, $-1 \notin A$, and $4 \notin A$ (or $0, -1, 4 \notin A$).

To end this section, we introduce the idea of subsets.

Definition 2.1.8: Subsets

Let A and B be sets.

- A is a **subset** of B if and only if all elements of A are elements

of B . This is denoted by $A \subseteq B$. Symbolically,

$$A \subseteq B \iff \forall x, ((x \in A) \implies (x \in B)).$$

- A is a **proper subset** if and only if A is a subset of B and B contains at least one element not in A . In this case, we write $A \subset B$. Symbolically,

$$A \subset B \iff (A \subseteq B) \wedge (\exists x \text{ s.t. } ((x \notin A) \wedge (x \in B))).$$

Example 2.1.9

Consider the sets $A = \{1, 2\}$, $B = \{1, 2, 3\}$, $C = \{1, 4\}$, and $S = \{1, 2, 3\}$. Then $A \subseteq S$ since the elements of A , namely 1 and 2, also appear in S . Also $B \subseteq S$ since all elements of B appear in S . However $C \not\subseteq S$ since 4 is not an element of S .

We note that $B \not\subset S$ since S does not contain an element that is not in B . But $A \subset S$ since 3 is not in A .

Exercise 2.1: Simple Set Statements

Let S be a non-empty set where $S \neq \{\emptyset\}$. Determine whether the following are true or false.

- | | |
|---|--|
| (a) $\{1, 2\} \subset \{1, 2, 3, 4\}$ | (e) $S \in \{S, \emptyset\}$ |
| (b) $\{1, 2, 3\} \subseteq \{1, 2, 4\}$ | (f) $\{S\} \notin \{S, \emptyset\}$ |
| (c) $\emptyset \subseteq \emptyset$ | (g) $S \subseteq \{S, \emptyset\}$ |
| (d) $S \subset S$ | (h) $\{S\} \subseteq \{S, \emptyset\}$ |

2.2. Set-Builder Notation

For all the sets above, we were able to list down all the elements of the set. Sometimes, this is hard to do, especially when there are many elements in the set. In these cases, we use set-builder notation to describe the sets.

Definition 2.2.1: Set-Builder Notation

A set S defined using **set-builder notation** has the syntax

$$S = \{x \mid P(x)\},$$

where the elements of S are all x that satisfy the predicate $P(x)$. The vertical line can be read as “such that” or “satisfying”.

Remark

Some authors use the colon instead of a vertical line. For example, they may write $S = \{x : P(x)\}$.

Example 2.2.2: Set of Even Integers

Consider the set of even integers,

$$E = \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\},$$

but defining it in this way is quite imprecise. If we let $\mathbb{Z}$ be the set of integers, then we can say that

$$E = \{x \mid (x \in \mathbb{Z}) \wedge (\exists n \in \mathbb{Z} \text{ s.t. } x = 2n)\},$$

i.e. the set E is the set of all x such that x is an integer and that there is another integer n such that $x = 2n$.

It may be a little awkward to specify that “ $x \in \mathbb{Z}$ ” along with the actual condition “ $\exists n \in \mathbb{Z} \text{ s.t. } x = 2n$ ” in the predicate inside the set-builder notation. In such cases, we appeal to the following definition to have cleaner notation.

Definition 2.2.3: Set-Builder Notation with a Domain

Let D be a set. The set

$$\{x \in D \mid P(x)\}$$

is defined to be equal to the set

$$\{x \mid (x \in D) \wedge P(x)\}.$$

Example 2.2.4

Let $\mathbb{Z}$ be the set of integers and $\mathbb{Q}$ be the set of rational numbers.

- The set $\{x \in \mathbb{Z} \mid (x \geq -2) \wedge (x \leq 3)\} = \{-2, -1, 0, 1, 2, 3\}$.
- The set $\{x \in \mathbb{Q} \mid x^2 = 2\} = \emptyset$.

- The set $\{x \in \mathbb{Q} \mid \exists n \in \mathbb{Z} \text{ s.t. } 3x = n\}$ is the set of all rational numbers with a denominator of 3.

There is one more useful variant of set-builder notation – one where we can have a more complex expression on the left-hand side.

Definition 2.2.5: Set-Builder Notation with Expressions

Let $f(x)$ be an expression. The set

$$\{f(x) \mid P(x)\}$$

is defined to be equal to the set

$$\{y \mid \exists x \text{ s.t. } ((y = f(x)) \wedge P(x))\}$$

where $P(x)$ is a predicate.

Example 2.2.6

Recall that we can define the set of even integers as

$$E = \{x \mid (x \in \mathbb{Z}) \wedge (\exists n \in \mathbb{Z} \text{ s.t. } x = 2n)\}.$$

Using our new definitions, we can write E as

$$E = \{x \in \mathbb{Z} \mid \exists n \in \mathbb{Z} \text{ s.t. } x = 2n\}$$

or

$$E = \{2x \mid x \in \mathbb{Z}\}.$$

With set-builder notation we can define an important set – the power set of any set.

Definition 2.2.7: Power Set

Let A be a set. The *power set* of A is

$$\mathcal{P}(A) = \{S \mid S \subseteq A\},$$

i.e. $\mathcal{P}(A)$ is the set of all subsets of A .

Example 2.2.8: Examples of Power Sets

- $\mathcal{P}(\emptyset) = \{\emptyset\}$, since the empty set is always a subset of any set, including itself.

- $\mathcal{P}(\{1\}) = \{\emptyset, \{1\}\}.$
- $\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$
- $\mathcal{P}(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$

Exercise 2.2: Building Sets

Let $\mathbb{Z}$ be the set of integers and $\mathbb{Q}$ be the set of rational numbers.

- What is the set $\left\{ \frac{p}{q} \mid (p, q \in \mathbb{Z}) \wedge (q \neq 0) \right\}$?
- What is the set $\left\{ x \in \mathbb{Q} \mid \frac{1}{2}x \in \mathbb{Z} \right\}$? Describe it in English.
- What is the set $\left\{ x \mid (x \in \mathbb{Z}) \wedge \left(\frac{x+1}{2} \in \mathbb{Z} \right) \right\}$? Describe it in set-builder notation with an expression on the left-hand side.
- What is $\mathcal{P}(\{\emptyset\})$?

2.3. Set Operations

We have already seen a lot of notation regarding sets, and there are more to come. We look at some of the most important set operations here.

Definition 2.3.1: Set Union

Let A and B be sets. The *union* of A and B is

$$A \cup B = \{x \mid (x \in A) \vee (x \in B)\}.$$

Example 2.3.2

- $\{1, 2, 3\} \cup \{4, 5, 6\} = \{1, 2, 3, 4, 5, 6\}.$
- $\{1, 2, 3\} \cup \{2, 3, 4\} = \{1, 2, 3, 4\}.$ Recall that we do not keep duplicates in sets.
- $\{1, 2, 3\} \cup \emptyset = \{1, 2, 3\}.$

One may think of the set union as analogous to the disjunction operator. Likewise, the set intersection that we will define below is analogous to the conjunction operator.

Definition 2.3.3: Set Intersection

Let A and B be sets. The *intersection* of A and B is

$$A \cap B = \{x \mid (x \in A) \wedge (x \in B)\}.$$

Example 2.3.4

- $\{1, 2, 3\} \cap \{1, 2, 3\} = \{1, 2, 3\}.$
- $\{1, 2, 3\} \cap \{2, 3, 4\} = \{2, 3\}.$
- $\{1, 2, 3\} \cap \{4, 5, 6\} = \emptyset.$

The next set operation has no analogous operation in logic. The set difference can be interpreted as the set of items in the original set that is not in the set that we are subtracting off.

Definition 2.3.5: Set Difference

Let S and A be sets. The *set difference* of S and A is

$$S \setminus A = \{x \mid (x \in S) \wedge (x \notin A)\}.$$

Remark

Some authors will use the minus sign to denote the set difference, i.e. $S - A$ is the same as $S \setminus A$.

Example 2.3.6

- $\{1, 2, 3\} \setminus \{2, 3, 4\} = \{1\}.$
- $\{2, 3, 4\} \setminus \{1, 2, 3\} = \{4\}.$
- $\{1, 2, 3\} \setminus \{1, 2, 3\} = \emptyset.$

Another important operation that will come up in our investigation into numbers is the cartesian product. But first, we need to define n -tuples. One can interpret an n -tuple as an ordered sequence of n items, whereas a set containing n items is unordered. The notation of having n ordered items, especially for the case when $n = 2$, is very useful.

Definition 2.3.7: n -tuples

Let $A_1, A_2, \dots, A_n$ be non-empty sets, and let $a_1 \in A_1, a_2 \in A_2, \dots$, and $a_n \in A_n$, where $n \geq 2$. Then an **n -tuple**

$$(a_1, a_2, \dots, a_n) = \{\{a_1\}, \{a_1, a_2\}, \{a_1, a_2, a_3\}, \dots, \{a_1, a_2, \dots, a_n\}\}.$$

Remark: Special Names for Specific n -Tuples

A 2-tuple is called an (ordered) *pair*. A 3-tuple is called an (ordered) *triplet*. A 4-tuple is called an (ordered) *quadruplet*.

As mentioned, the important property of n -tuples is that they are ordered.

Proposition 2.3.8: n -Tuples are Ordered

Let $A_1, A_2, \dots, A_n$ be non-empty sets, and let $a_1, b_1 \in A_1, a_2, b_2 \in A_2, \dots$, and $a_n, b_n \in A_n$, where $n \geq 2$. Then $(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$ if and only if $a_1 = b_1, a_2 = b_2, \dots$, and $a_n = b_n$.

Proof. The reverse direction is trivial, so we show the forward direction only. Suppose $(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$. Thus

$$\{\{a_1\}, \{a_1, a_2\}, \dots, \{a_1, a_2, \dots, a_n\}\} = \{\{b_1\}, \{b_1, b_2\}, \dots, \{b_1, b_2, \dots, b_n\}\}$$

by definition of n -tuples. By equality of sets we thus obtain

$$\begin{aligned} \{a_1\} &= \{b_1\}, \\ \{a_1, a_2\} &= \{b_1, b_2\}, \\ \{a_1, a_2, a_3\} &= \{b_1, b_2, b_3\}, \\ &\vdots \\ \{a_1, a_2, \dots, a_n\} &= \{b_1, b_2, \dots, b_n\}. \end{aligned}$$

The first equality immediately gives $a_1 = b_1$. So if $\{a_1, a_2\} = \{b_1, b_2\} = \{a_1, b_2\}$ we must have $a_2 = b_2$. Therefore for $\{a_1, a_2, a_3\} = \{b_1, b_2, b_3\} = \{a_1, a_2, b_3\}$ we must have $a_3 = b_3$, and so on. Hence $a_1 = b_1, a_2 = b_2, \dots$, and $a_n = b_n$. ■

We usually abstract away the details of the n -tuple and just think of it as an ordered list of n items. That is, we do not really think about the underlying set of n -tuples when we use them; we just use their properties.

With n -tuples defined, we can define the cartesian product.

Definition 2.3.9: Cartesian Product of 2 Sets

Let A and B be sets. The *cartesian product* of A and B is

$$A \times B = \{(a, b) \mid (a \in A) \wedge (b \in B)\}.$$

Example 2.3.10

Let $A = \{1, 2, 3\}$ and $B = \{4, 5\}$. Then

$$A \times B = \{(1, 4), (2, 4), (3, 4), (1, 5), (2, 5), (3, 5)\}.$$

Also note

$$B \times A = \{(4, 1), (5, 1), (4, 2), (5, 2), (4, 3), (5, 3)\}.$$

Thus $A \times B \neq B \times A$ in general.

We can define the cartesian product of more than 2 sets.

Definition 2.3.11: General Cartesian Product

Let $A_1, A_2, \dots, A_n$ be sets where $n \geq 2$. The cartesian product of $A_1, A_2, \dots$, and A_n is

$$\begin{aligned} &A_1 \times A_2 \times \dots \times A_n \\ &= \{(a_1, a_2, \dots, a_n) \mid (a_1 \in A_1) \wedge (a_2 \in A_2) \wedge \dots \wedge (a_n \in A_n)\}. \end{aligned}$$

Remark

In particular, if A is a set, then define $A^2 = A \times A$, $A^3 = A \times A \times A$, $A^4 = A \times A \times A \times A$, and so on.

Example 2.3.12

Let $A = \{1, 2\}$, $B = \{3, 4\}$, and $C = \{5, 6\}$. Then

$$\begin{aligned} A \times B \times C = \{ &(1, 3, 5), (1, 3, 6), (1, 4, 5), (1, 4, 6), \\ &(2, 3, 5), (2, 3, 6), (2, 4, 5), (2, 4, 6)\}. \end{aligned}$$

Exercise 2.3: Set Operation Statements

Let $S = \{1, 2, 3, 4\}$, $T = \{2, 3, 5\}$, and $U = \{(2, 2), (3, 3), (5, 5)\}$. Determine whether the following statements are true or false.

- | | |
|--------------------------------------|--------------------------------|
| (a) $S \cup T = \{1, 2, 3, 4, 5\}$ | (e) $S \setminus T = \{1, 4\}$ |
| (b) $S \cup U = \{1, 2, 3, (5, 5)\}$ | (f) $S \setminus \{1, 4\} = T$ |
| (c) $S \cap T = \{2, 3\}$ | (g) $T^2 = U$ |
| (d) $T \cap U = \emptyset$ | (h) $U \subset (S \cup T)^2$ |

We end this section by looking at a way to quantify the number of items in a set.

Definition 2.3.13: Cardinality of a Set

Let A be a set. Then the *cardinality* of A , denoted $|A|$, is a measure of the size of A .

Example 2.3.14

- $|\{1, 2, 3\}| = 3$ since there are 3 elements in the set $\{1, 2, 3\}$.
- $|\{2, 3, 5, 7, 11\}| = 5$ since there are 5 elements in the set.
- $|\emptyset| = 0$ since the empty set contains no elements.

Note that this definition of cardinality of sets only works if there is a finite number of elements in the set. For infinite sets, there is a more technical definition, but we will address it when we come to it.

Exercise 2.4: Cardinality Challenges

Let $\mathbb{Z}$ be the set of integers.

- Let $S = \{n \in \mathbb{Z} \mid (-10 \leq n \leq 10) \wedge (\exists m \in \mathbb{Z} \text{ s.t. } n = 2m + 1)\}$. What is $|S|$?
- Find two sets A and B such that $|A| = 5$, $|B| = 6$, and $|A \cup B| = 7$. What is $|A \cap B|$?
- Let $|A| = n$. What is $|A^p|$, where p is a positive integer? What is $|\mathcal{P}(A)|$?

2.4. Quantified Statements with a Domain of Discourse

Recall that we introduced quantified statements in the previous chapter. For quantified statements, we need to specify a domain of discourse. With sets under our belt, we can define some notation to symbolically define a domain of discourse.

Definition 2.4.1: Quantified Statements with Domain

Let D be a set and $P(x)$ be a predicate. The universal statement

$$\forall x \in D, P(x)$$

is defined to be the equal to the statement

$$\forall x, ((x \in D) \implies P(x)).$$

Likewise, the existential statement

$$\exists x \in D \text{ s.t. } P(x)$$

is defined to be equal to the statement

$$\exists x \text{ s.t. } ((x \in D) \wedge P(x)).$$

In each case the domain of discourse is everything in the set D .

Example 2.4.2

Let $\mathbb{Z}$ denote the set of positive integers.

- $\forall x \in \mathbb{Z}, ((x \text{ is odd}) \implies (x + 1 \text{ is even}))$ means “for all integers x , if x is odd then $x + 1$ is even”, which is a true statement.
- $\exists x \in \mathbb{Z} \text{ s.t. } ((x \text{ is prime}) \wedge (\exists m \in \mathbb{Z} \text{ s.t. } x = m^2 + 1))$ means “there exists an integer x such that x is prime and $x = m^2 + 1$ for some integer m ”. This is true using the example $x = 5$ and $m = 2$.
- $\forall x \in \emptyset, (x > 0)$ is a vacuously true statement.
- $\exists x \in \emptyset \text{ s.t. } (x \leq 0)$ is a false statement, since we cannot produce an example of x from the empty set that is less than or equal to zero.

CHAPTER 3

Proofs

Proofs compel belief. Without mathematical proofs, the statements we make are meaningless words without justification. To prove a claim, we must provide a series of logical steps to arrive at a logical conclusion. In this chapter, we explore the basics of writing proofs.

For this chapter, we will assume that we have defined the natural numbers, the integers, the rationals, and the reals, as well as their respective operations. Otherwise, how can we give examples of proofs?

3.1. Direct Proof

In essence, a *direct proof* for the statement “if p then q ” would begin by assuming p is true and showing that this makes q true. We don’t need to worry about the case of p being false, since $p \implies q$ is vacuously true in the case when p is false.

Example 3.1.1: Odd Integers and Odd Squares

We prove that “if n is an odd integer then n^2 is odd” using direct proof. We first write out the proof step-by-step.

Proof. Suppose n is an odd integer.

Then n can be written in the form $n = 2k + 1$ where k is an integer.

Hence we see

$$\begin{aligned}n^2 &= (2k + 1)^2 \\&= 4k^2 + 4k + 1 \\&= 2(2k^2 + 2k) + 1.\end{aligned}$$

We see n^2 is one more than a multiple of 2.

Thus $n^2 = 2(2k^2 + 2k) + 1$ is odd. ■

Of course, proofs are often not written this way; we often make it more succinct and remove unnecessary paragraph breaks. We produce a more ‘natural’ proof below.

Proof. Suppose n is an odd integer. Then n can be written in the

form $n = 2k + 1$ where k is an integer. Hence

$$\begin{aligned} n^2 &= (2k + 1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1, \end{aligned}$$

so n^2 is one more than a multiple of 2, meaning $n^2 = 2(2k^2 + 2k) + 1$ is odd. ■

Example 3.1.2: Square Roots Maintain Inequality

Let x and y be positive real numbers. We prove that “if $x \leq y$ then $\sqrt{x} \leq \sqrt{y}$ ”.

Proof. Suppose $x \leq y$. This means $x - y \leq 0$. As x and y are positive real numbers, we may write that inequality as $(\sqrt{x})^2 - (\sqrt{y})^2 \leq 0$. We can factor the left hand side as $(\sqrt{x} + \sqrt{y})(\sqrt{x} - \sqrt{y}) \leq 0$. Now $\sqrt{x} + \sqrt{y}$ is always positive, so if the entire inequality is less than or equal to 0 we must have $\sqrt{x} - \sqrt{y} \leq 0$. Therefore $\sqrt{x} \leq \sqrt{y}$. ■

Exercise 3.1: Direct Proof

Let x be a positive real number. Prove that $x(1 - x) > 0$ if $0 < x < 1$.

3.2. Proof by Division Into Cases

Division into cases is a variant of direct proof. We split a difficult statement into different cases which are individually easier to prove, and then join them together in order to prove the full statement.

Example 3.2.1: An Expression That is a Multiple of 4

We look at a proof of the statement “ $1 + (-1)^n(2n - 1)$ is a multiple of 4 if n is an integer”.

Proof. Suppose n is an integer. Then, n is either odd or even. We look at two cases.

- If n is odd, then $(-1)^n = -1$ and $n = 2k + 1$ for some integer k . Thus

$$1 + (-1)^n(2n - 1) = 1 - (2(2k + 1) - 1) = 4(-k)$$

which is a multiple of 4.

- If n is even, then $(-1)^n = 1$ and $n = 2k$ for some integer k .
Thus

$$1 + (-1)^n(2n - 1) = 1 + (2(2k) - 1) = 4k$$

which is a multiple of 4.

Hence, in both cases, $1 + (-1)^n(2n - 1)$ is a multiple of 4. ■

Example 3.2.2: Sum of Integers with Opposite Parity

We look at a proof of the statement “if two integers have opposite parity, then their sum is odd”. Note that “opposite parity” means that one integer is odd and one integer is even.

Proof. Suppose m and n are two integers with opposite parity.

- If m is even and n is odd, then we may write $m = 2a$ and $n = 2b + 1$ where a and b are some integers. Thus

$$\begin{aligned} m + n &= (2a) + (2b + 1) \\ &= 2(a + b) + 1 \end{aligned}$$

which is one more than a multiple of 2, so $m + n$ is odd.

- On the other hand, if m is odd and n is even, then we may write $m = 2a + 1$ and $n = 2b$ where a and b are some integers. Thus

$$\begin{aligned} m + n &= (2a + 1) + (2b) \\ &= 2(a + b) + 1 \end{aligned}$$

which is one more than a multiple of 2, so $m + n$ is odd.

Hence, in either case, $m + n$ is odd. ■

Now, the two cases in the above proof are almost the same, except for whether m or n is the even integer. It is more productive to just do one case and indicate that the other case is nearly (or exactly) identical to the current case. The phrase “without loss of generality” is a way to signpost to the reader that the proof is treating only one of several nearly identical cases. We produce a more succinct version of the above proof.

Proof. Suppose m and n are two integers with opposite parity. Without

loss of generality, suppose m is even and n is odd. Therefore $m = 2a$ and $n = 2b + 1$ for some integers a and b . Hence

$$\begin{aligned} m + n &= (2a) + (2b + 1) \\ &= 2(a + b) + 1 \end{aligned}$$

which is one more than a multiple of 2, so $m + n$ is odd. ■

Exercise 3.2: Splitting into Cases

Prove that $m + n$ is even if the integers m and n have the same parity (i.e., both odd or both even).

3.3. Contrapositive Proof

We now look at a *contrapositive proof*. Recall that $p \implies q$ is logically equivalent to $\neg q \implies \neg p$. Thus, a contrapositive proof for “if p then q ” would begin by assuming $\neg q$ is true and deducing that this means that $\neg p$ is true.

Generally, we would want to prove in the direction from simple to complex. So if p is more complex than q , we may consider using a contrapositive proof.

Example 3.3.1: A Test for Odd Integers

Suppose n is an integer. We prove the statement “if $n^2 - 6n + 5$ is even then n is odd”. We note that a direct proof would be tedious and problematic. Using a contrapositive proof would be easier.

We first note that the contrapositive statement that we want to prove is “if n is **not** odd, then $n^2 - 6n + 5$ is **not** even”, that is, “if n is even, then $n^2 - 6n + 5$ is odd”.

Proof. We consider a proof by contrapositive.

Suppose n is even. Then $n = 2k$ where k is an integer. Note

$$\begin{aligned} n^2 - 6n + 5 &= (2k)^2 - 6(2k) + 5 \\ &= 4k^2 - 12k + 5 \\ &= (4k^2 - 12k + 4) + 1 \\ &= 2(2k^2 - 6k + 2) + 1 \end{aligned}$$

which means $n^2 - 6n + 5$ is odd. ■

Example 3.3.2: A Non-Obvious Inequality

Suppose x and y are real numbers. We prove the statement “ $x \leq y$ if $x^3 + xy^2 \leq x^2y + y^3$ ” using a contrapositive proof.

We first note that the contrapositive statement that we want to prove is “if $x > y$ then $x^3 + xy^2 > x^2y + y^3$ ”.

Proof. We consider a proof by contrapositive.

Assume $x > y$. Then $x - y > 0$. Also, since $x > y$, thus x and y are not both zero. Hence $x^2 + y^2 > 0$. Observe

$$(x - y)(x^2 + y^2) > 0 \times (x^2 + y^2) = 0$$

so

$$(x - y)(x^2 + y^2) = x^3 + xy^2 - x^2y - y^3 > 0$$

Therefore $x^3 + xy^2 > x^2y + y^3$. ■

Exercise 3.3: Contrapositive Proof

Suppose that a and b are integers. Prove that if $a(b^2 + 5)$ is even then either a is even or b is odd.

3.4. Proof by Contradiction

The third proof technique is called a *proof by contradiction*. This method can be used to prove any kind of statement. The basic idea is to assume that the statement we want to prove is false, and then show that this assumption leads to a contradiction. A proof by contradiction for the statement “ p ” would start by assuming that p is false, and then showing that this assumption would lead to a contradiction, which means that p is **not** false, i.e. p is true.

Remark: Contradiction as an Implication

In fact, what we are showing is that the statement “ $\neg p \implies (c \wedge \neg c)$ ” is true.

Usually, when writing a proof by contradiction, we would like to inform the reader that a proof by contradiction is being employed. Terms such as “by way of contradiction”, “towards a contradiction”, “suppose for the sake of contradiction” etc. may be used to signpost the use of a proof by contradiction.

Example 3.4.1: The Irrationality of $\sqrt{2}$

We present the classic proof by contradiction for the statement “ $\sqrt{2}$ is irrational”.

Proof. By way of contradiction, assume that $\sqrt{2} = \frac{a}{b}$ for some integers a and $b \neq 0$. Furthermore let this fraction be fully reduced; in particular, this means that a and b are not both even. Squaring both sides yields $2 = \frac{a^2}{b^2}$, meaning $a^2 = 2b^2$. Hence a^2 is even, so write $a = 2c$ where c is an integer. This leads to $2b^2 = (2c)^2 = 4c^2$ which implies $b^2 = 2c^2$. Hence b is even, which contradicts the fact that a and b are not both even.

Hence, $\sqrt{2}$ is irrational. ■

Note that it is not necessary to have the final statement that “ $\sqrt{2}$ is irrational” (or, more generally, “ p is true”) as it is implied from the proof by contradiction. However, stating it makes it clear what we have proven using the proof by contradiction.

Example 3.4.2: No Smallest Positive Rational Number

We prove the statement that “for every positive rational number x , there exists a positive rational number y such that $y < x$ ” by way of contradiction.

We note that the negation of the above statement is “there exists a rational number x such that for every positive rational number y we have $y \geq x$ ”.

Proof. Suppose for the sake of contradiction that there exists a rational number x such that for every positive rational number y we have $y \geq x$. Write $x = \frac{p}{q}$ where p and q are positive integers.

Now consider the rational number $\frac{p-1}{q}$. Clearly $\frac{p-1}{q} < \frac{p}{q} = x$. By assumption, every positive rational number y satisfies $y \geq x$. Hence, $\frac{p-1}{q}$ is non-positive, meaning $\frac{p-1}{q} \leq 0$. Since q is positive, hence $p-1 \leq 0$ which means $p \leq 1$. But as p is a positive integer, we conclude $p = 1$. Hence $x = \frac{1}{q}$.

We now consider the rational number $\frac{1}{q+1}$. Clearly $\frac{1}{q+1} < \frac{1}{q} = x$. By assumption we must conclude that $\frac{1}{q+1}$ is non-positive. However, $1 > 0$ and $q+1 > 0$, so $\frac{1}{q+1}$ is positive. Hence we have the fact that $\frac{1}{q+1}$ is positive and non-positive simultaneously, leading to a contradiction. ■

Remark: A Direct Proof

A direct proof would be easier. We provide a direct proof of the claim below.

Proof. Since x is a positive rational number, we can write $x = \frac{p}{q}$ where p and q are positive integers. Then set $y = \frac{p}{q+1}$. Clearly $\frac{p}{q+1} < \frac{p}{q} = x$ and $\frac{p}{q+1}$ is positive, hence we have found a y such that $y < x$. ■

Exercise 3.4: Proof by Contradiction

Prove that no integers a and b exist such that $2a + 4b = 1$.

We now look at a proof by contradiction for conditional statements. Recall that $\neg(p \implies q) \equiv p \wedge \neg q$. Hence, to prove the statement “if p then q ” via contradiction, we would start by assuming that $p \wedge \neg q$, and then showing that this assumption would lead to a contradiction, which means that $p \implies q$ is **not** false, i.e. $p \implies q$ is true.

Example 3.4.3: Irrational Product Must Have Irrational Factor

Suppose a and b are real numbers. We prove the statement “if a is rational and ab is irrational then b is irrational” using a proof by contradiction.

We note that the statement we want to contradict is “(a is rational and ab is irrational) **and** (b is **not** irrational)”, i.e. “ a is rational and b is rational and ab is irrational”.

Proof. By way of contradiction assume a is rational, b is rational, and ab is irrational. We may then write

$$a = \frac{m}{n} \text{ and } b = \frac{p}{q}$$

where $m, n, p, q \in \mathbb{Z}$ where $n, q \neq 0$. Hence

$$ab = \left(\frac{m}{n}\right) \left(\frac{p}{q}\right) = \frac{mp}{nq}$$

which is clearly rational. Therefore we have that ab is irrational (by assumption) and ab is rational (by our working above), a clear contradiction. ■

Example 3.4.4: A Property of Pythagorean Triples

Suppose a , b , and c are integers. We prove the statement that “if $a^2 + b^2 = c^2$ then at least one of a or b is even” using a proof by contradiction.

We note that the statement we want to contradict is “($a^2 + b^2 = c^2$) **and not** (at least one of a or b is even)”, i.e. “ $a^2 + b^2 = c^2$ **and** both a and b are odd”.

Proof. Seeking a contradiction, assume that $a^2 + b^2 = c^2$ and both a and b are odd. Thus we may write $a = 2m + 1$ and $b = 2n + 1$ where m and n are integers. Hence

$$\begin{aligned} a^2 + b^2 &= (2m + 1)^2 + (2n + 1)^2 \\ &= (4m^2 + 4m + 1) + (4n^2 + 4n + 1) \\ &= 4m^2 + 4n^2 + 4m + 4n + 2 \\ &= 2(2m^2 + 2n^2 + 2m + 2n + 1) \end{aligned}$$

which means that $c^2 = a^2 + b^2$ is even. Hence c is even, which means we may write $c = 2k$ where k is an integer. This leads to

$$c^2 = 4k^2 = 2(2m^2 + 2n^2 + 2m + 2n + 1) = a^2 + b^2.$$

Clearly $4k^2$ is a multiple of 4, while $2(2m^2 + 2n^2 + 2m + 2n + 1)$ is not. Yet, they are equal to each other, a contradiction. ■

Exercise 3.5: Proof by Contradiction for a Conditional Statement

Prove that $\frac{a+b}{2} \geq \sqrt{ab}$ if a and b are positive real numbers by way of contradiction.

Despite the power of proof by contradiction, it is best to use it only when the direct and contrapositive approaches do not seem to work.

Example 3.4.5: A Test for Odd Integers: Proof by Contradiction

Suppose n is an integer. We prove the statement “if $n^2 - 6n + 5$ is even then n is odd” using a proof by contradiction.

Proof. Working towards a contradiction, assume $n^2 - 6n + 5$ is even and n is **not** odd, i.e. n is even. Then $n = 2k$ for some integer k . Note

that

$$\begin{aligned}
 n^2 - 6n + 5 &= (2k)^2 - 6(2k) + 5 \\
 &= 4k^2 - 12k + 5 \\
 &= (4k^2 - 12k + 4) + 1 \\
 &= 2(2k^2 - 5k + 2) + 1
 \end{aligned}$$

which means that $n^2 - 6n + 5$ is odd. Hence, $n^2 - 6n + 5$ is even (by assumption) and $n^2 - 6n + 5$ is odd (as above), a contradiction. ■

While there is nothing wrong with this proof, notice that part of it assumes that n is even and concludes that $n^2 - 6n + 5$ is odd, which is the contrapositive approach done in [Example 3.3.1](#).

3.5. Basics of Mathematical Induction

Mathematical induction is a method for proving that a predicate $P(n)$ is true for all integers $n \geq m$, where m is an integer. That is, mathematical induction proves the infinitely many statements $P(m), P(m+1), P(m+2), \dots$, where m is an integer.

A proof by induction consists of two steps.

- The first, the **base case**, proves the statement for m without assuming any knowledge of other cases, where m is an integer. In other words, the base case proves the statement $P(m)$.
- The second, the **induction step**, proves that if the statement holds for any given case $n = k$, then it must also hold for the next case $n = k + 1$. In other words, $\forall k \in \mathbb{N}, (P(k) \implies P(k+1))$. The assumption that $P(k)$ being true is called the **induction hypothesis**.

These two steps establish that the predicate $P(n)$ holds for all integers $n \geq m$.

Let's look at some examples of mathematical induction.

Example 3.5.1: Sum of the First n Positive Integers

We prove the famous identity

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

for all positive integers n using mathematical induction.

Proof. When $n = 1$, the left hand side is 1 and the right hand side is also $\frac{1(1+1)}{2} = 1$. Thus the base case is true.

Now assume that the statement holds for some positive integer k , meaning

$$1 + \cdots + k = \frac{k(k+1)}{2}.$$

We need to show that the statement holds for $k + 1$, meaning

$$1 + \cdots + k + (k + 1) = \frac{(k + 1)(k + 2)}{2}.$$

We see that

$$\begin{aligned} & 1 + \cdots + k + (k + 1) \\ &= (1 + \cdots + k) + (k + 1) \\ &= \frac{k(k + 1)}{2} + (k + 1) && \text{(induction hypothesis)} \\ &= \frac{k(k + 1)}{2} + \frac{2(k + 1)}{2} \\ &= \frac{k(k + 1) + 2(k + 1)}{2} \\ &= \frac{(k + 1)(k + 2)}{2} \end{aligned}$$

which proves the case for $k + 1$. Hence $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$. ■

Example 3.5.2: n th-Power of $1 + x$ Is At Least $1 + nx$

Suppose $x > -1$. We will prove that $(1 + x)^n \geq 1 + nx$ if n is a non-negative integer.

Proof. When $n = 0$, the left hand side is $(1 + x)^0 = 1 = 1 + 0x$ which is exactly the right hand side. Thus the base case is true.

Assume that the statement holds for some non-negative integer k , i.e. $(1 + x)^k \geq 1 + kx$. We show that the statement holds for $k + 1$, i.e. $(1 + x)^{k+1} \geq 1 + (k + 1)x$.

We first note that since $x > -1$, thus $1 + x > 0$. We start with the

induction hypothesis.

$$\begin{aligned}
 (1+x)^k &\geq 1+kx \\
 (1+x)^k(1+x) &\geq (1+kx)(1+x) && (\text{since } 1+x > 0) \\
 (1+x)^{k+1} &\geq 1+x+kx+kx^2 \\
 &= 1+(k+1)x+kx^2 \\
 &> 1+(k+1)x. && (\text{since } kx^2 > 0)
 \end{aligned}$$

Hence we see $(1+x)^{k+1} \geq 1+(k+1)x$, meaning that the statement is true for $k+1$.

Therefore $(1+x)^n \geq 1+nx$ if n is a non-negative integer. ■

Observe that in the above example that the base case uses $n = 0$ instead of $n = 1$. This demonstrates that we don't necessarily have to start the proof at $n = 1$.

Exercise 3.6: Mathematical Induction

Prove using mathematical induction that $a^2 - 1$ is a multiple of 8 for all positive odd integers a .

There are, of course, more advanced methods of mathematical induction, such as strong induction and structural induction. However, we will cover them when the time comes. For now, the simplest form of mathematical induction would suffice.

3.6. Proving Non-Conditional Statements

3.6.1. Biconditional Statements

Recall that a biconditional statement is a statement like " $p \iff q$ ", i.e., " p if and only if q ". We prove such a statement by proving that $p \implies q$ (which we will henceforth call the *forward direction*) and $q \implies p$ (which we henceforth call the *reverse direction*). Each of these statements may be proved using any of the proof techniques that we covered.

Example 3.6.1: Parity of n Matches n^2

We will prove the biconditional statement "the integer n is even if and only if n^2 is even".

Proof. We prove the forward direction (n is even implies n^2 is even)

first by using direct proof. Assume that n is even. Then we may write $n = 2k$ where k is an integer. Hence $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$ which is even.

We now prove the reverse direction (n^2 is even implies n is even) via a contrapositive proof. Suppose n is not even, meaning n is odd. Hence $n = 2k + 1$ where k is an integer. Observe $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ which is odd. ■

Example 3.6.2: Multiple of 6 is Multiple of 2 and 3

Suppose n is an integer. We will prove “ n is a multiple of 6 if and only if n is a multiple of 2 and 3”.

Proof. We prove the forward direction by using direct proof. Assume n is a multiple of 6, meaning $n = 6k$ for some integer k . Clearly $6k = 2(3k)$ and $6k = 3(2k)$, so n is a multiple of 2 and a multiple of 3.

We now prove the reverse direction, again using direct proof. Assume n is a multiple of 2 and 3, so we may write $n = 2a$ and $n = 3b$ for some integers a and b . Then $2a = 3b$. Hence $a = \frac{3}{2}b$ and $b = \frac{2}{3}a$. Since a and b are integers, hence we conclude b is a multiple of 2 and a is a multiple of 3. Write $a = 3p$ and $b = 2q$ where p and q are integers. Hence $n = 2(3p) = 6p$ and $n = 3(2q) = 6q$. In both cases we see n is a multiple of 6. ■

Exercise 3.7: Biconditional Proof

Let n be an integer. Prove that n is one more than a multiple of 5 if and only if n is of the form $5k - 4$ where k is an integer.

3.6.2. Existence Statements

Some statements only assert the existence of something. These are called *existence statements* and one only has to provide a particular example that shows it is true.

Example 3.6.3: Existence of an Even Prime Number

The statement “there exists an even prime number” is proven readily by noticing that 2 is an even prime number.

Example 3.6.4: Two Different Sums of the Same Perfect Cube

The statement “an integer that can be expressed as the sum of two perfect cubes in two different ways exists” is proven by giving the example 1729, since $1729 = 1^3 + 12^3 = 9^3 + 10^3$.

Note that while an example suffices to prove an existence statement, a single example does not prove a (universal) conditional statement.

Exercise 3.8: Existence Proof

Prove that there is a positive integer that is one less than a perfect cube and two less than a perfect square.

Existence proofs fall into two categories.

- **Constructive proofs** provide an explicit example that proves the statement.
- **Non-constructive proofs** prove that an example exists without providing it.

So far, we have only seen constructive proofs. Let us now look at an example of a non-constructive proof.

Example 3.6.5: Irrational to Power of Irrational Can be Rational

We prove the classic statement that “there exist irrational numbers x and y such that x^y is rational” using a non-constructive proof.

Proof. We know that $\sqrt{2}$ is irrational from [Example 3.4.1](#). Consider the number $\sqrt{2}^{\sqrt{2}}$. There are two possibilities for the rationality of this number.

- If $\sqrt{2}^{\sqrt{2}}$ is rational, then set $x = y = \sqrt{2}$. We have found two irrational numbers such that their exponentiation (i.e., $x = \sqrt{2}^{\sqrt{2}}$) is rational.
- If instead $\sqrt{2}^{\sqrt{2}}$ is irrational, set $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$. Observe that

$$x^y = \left(\sqrt{2}^{\sqrt{2}} \right)^{\sqrt{2}} = (\sqrt{2})^{\sqrt{2} \times \sqrt{2}} = (\sqrt{2})^2 = 2$$

is a rational number. So we again found two irrational numbers such that their exponentiation is rational.

Hence, in both cases, we have an irrational number to an irrational power that results in a rational number. ■

Notice that in the above example we did not explicitly prove whether the number $\sqrt{2}^{\sqrt{2}}$ is rational or irrational; we just showed that either case leads to a case where two irrational numbers, when exponentiated, results in a rational number.

Exercise 3.9: A Constructive Proof

Let $x = \sqrt{2}$ and $y = 2 \log_2 3$. It may be assumed that $\sqrt{2}$ is irrational.

- (i) Prove that y is irrational.
- (ii) Produce a constructive proof that there exist irrational x and y such that x^y is rational.

CHAPTER 4

The Language of Axiomatic Set Theory

Writing clear, concise, and logical sentences is essential in mathematics. We need a consistent way to write statements to arrive at the axioms required for defining the foundations of mathematics. That is a job reserved for formal language, a structure we will explore in this chapter.

4.1. An Introduction to Formal Languages

In English, one may consider a sentence a sequence of words and think of an English word as a sequence of letters. However, not every sequence of letters is a valid word, and not every sequence of words is a grammatically sound sentence. The idea of formal language is to generalise this idea of valid words and valid sentences.

Definition 4.1.1: Formal Language

A formal language $\mathcal{L}$ is a structure comprising two things.

- A set Σ of symbols, called the *alphabet* of $\mathcal{L}$.
- A *formal grammar* comprising *formation rules* that specify which *strings*, which are sequences of symbols from Σ , belong to $\mathcal{L}$.

We will explore more about alphabets, strings, formation rules, formal grammar, and their related constructs in the following pages.

4.1.1. Alphabets and Strings

In English, our (primary) alphabet is A to Z, along with the two cases each letter can come in (i.e., uppercase or lowercase). These 54 characters define all the valid letters we can use in English. We use these valid letters to form words via concatenating letters together. Likewise, we can define symbols that make up our alphabet in a formal language and define strings that use the symbols present in our alphabet.

Definition 4.1.2: Alphabet

Let $\mathcal{L}$ be a formal language. The **alphabet** of $\mathcal{L}$, usually denoted Σ , is the set of symbols used to construct strings in $\mathcal{L}$.

Definition 4.1.3: String

Let $\mathcal{L}$ be a formal language. A **string** is either a finite concatenation of symbols from Σ in $\mathcal{L}$ or the empty string ε .

The **length** of a string is the number of symbols making up the string, with the empty string having length 0.

Remark: Words

Some authors will call strings **words** formed from Σ . However, we will stick with the term “string” here as the term “empty word” is awkward to use. Also, this may cause confusion with our definition of words later on.

Example 4.1.4: A Simple Alphabet

Let $\Sigma = \{a, b\}$. Then ε (the empty string), “a”, “b”, “aa”, “ab”, “ba”, “bb”, “aaa”, “aab”, “aba”, “baa”, etc. are all valid strings. Notice that we have not defined any rules on what strings belong in our formal language – these are just strings formed from Σ .

4.1.2. Grammar and Words

It is undoubtedly unhelpful to say that every string, essentially a combination of letters in our alphabet, is meaningful within our language. In English, for example, the ‘words’ “hjbhix”, “jsadbhf”, and “jkdswjaaa” should not count as words! We want to have some restrictions on what strings belong in our formal language and which ones do not. That is the job of formation rules that make up our formal grammar.

Definition 4.1.5: Formation Rule

Let $\mathcal{L}$ be a formal language with alphabet Σ . A **formation rule** of $\mathcal{L}$ is a rule describing which strings formed from Σ belong to $\mathcal{L}$ and which do not. Such strings are called the **words** of $\mathcal{L}$.

Remark: Well-Formed Formulae

It is common to see words called *well-formed formulae* (or *well-formed formulas*) in other contexts. However, we will stick to using the term “words” as this will hopefully reduce the confusion when we actually introduce the well-formed formulae that are in the formal language of axiomatic set theory.

Definition 4.1.6: Formal Grammar

Let $\mathcal{L}$ be a formal language. The *formal grammar* of $\mathcal{L}$ is the set of formation rules of $\mathcal{L}$.

With formation rules and formal grammar defined, we can finally look at some examples of formal languages.

Example 4.1.7: Simple Formal Languages

Let $\Sigma = \{x, y\}$.

- Let L_1 be the formal language using the alphabet Σ such that L_1 contains strings of length at most 3 and ends with the symbol y . Thus one sees that

$$L_1 = \{y, xy, yy, xxy, xyy, yxy, yyy\}.$$

Note that the empty string ε is not a word in L_1 since it does not end with the symbol y .

- Let L_2 be the formal language using the alphabet Σ such that L_2 contains strings that read the same forwards as they do backwards (i.e., palindromes). So L_2 contains at least words

$$\varepsilon, "x", "y", "xx", "yy", "xxx", "xyx", "yxy", "yyy", "xxxx",$$

among infinitely many others. Note that the empty string will vacuously belong in L_2 since it has no characters and thus read the same forwards as it does backwards.

Before we look at the next example, let us look at the idea of arity.

Definition 4.1.8: Arity

The *arity* is the number of arguments or operands taken in by a function or operation. An *n-ary* function or operation is a function

or operation of arity n .

Example 4.1.9

- The arity of the function $f(x) = x^2$ is 1 since it only takes in one argument. Thus $f(x)$ is a 1-ary (i.e., *uniary*) function.
- The function $f(x, y, z) = x + yz$ is 3-ary (i.e., *ternary*) as it takes in three arguments.
- The arity of the plus operator is 2 as it takes in two operands. Thus the plus operator is a 2-ary (i.e., *binary*) operator.
- The arity of a constant (e.g., 3, 7, 11) is 0 since it does not accept any operands/arguments. They can be considered outputs of a 0-ary (i.e., *nullary*) operation.

Let us now look at a way of writing down expressions without using parentheses. In fact, some programming languages (e.g. Lisp) use a similar format of writing down statements.

Example 4.1.10: Łukasiewicz Notation

Our standard way of writing expressions, such as $a + b$, is said to use infix notation (since the operator sits **in between** the two operands a and b). In 1920, the Polish mathematician Jan Łukasiewicz introduced prefix and postfix notation, which we now call *Łukasiewicz notation* (or *Polish notation*) and *reversed Łukasiewicz notation* (or *reversed Polish notation*) respectively. Using our earlier example of $a + b$, it would be written $+ab$ in Łukasiewicz notation and $ab+$ in reversed Łukasiewicz notation. We may add spaces to aid clarity when writing in (reversed) Łukasiewicz notation, i.e. $+ a b$ and $a b +$.

We can use formal language to describe Łukasiewicz notation. Suppose we have an alphabet Σ consisting of symbols. Some of these symbols could be operators (e.g., $+$, $-$, $\times$, $\div$). Define non-operator symbols have arity 0 (i.e., nullary). The formal grammar of the formal language L describing Łukasiewicz notation contains only one formation rule as follows.

If $s \in \Sigma$ is n -ary and $\phi_1, \phi_2, \phi_3, \dots, \phi_n$ are words in L , then

$$s\phi_1\phi_2\phi_3 \dots \phi_n$$

is also a word in L .

Notably, any non-operator symbol is a word in L .

- “+ 2 3” in L is equivalent to “2 + 3” when written in infix notation.
- “+ 2 3 × 4” in L is equivalent to $(2 + 3) \times 4$ when written in infix notation.
- “+ − ÷ 2 3 ÷ 4 5 6” in L is the same as “ $((2 \div 3) - (4 \div 5)) + 6$ ” when written in infix notation.

Exercise 4.1: An Exercise in Languages

- (a) Suppose L is a formal language that contains the strings “ a ”, “ $abba$ ”, and “ $abracadabra$ ”. What symbols must be in the alphabet of L ? Assume each symbol is a string of length 1.
- (b) Write the expression $4 + ((5 - 6) \times 7)$, which is in infix notation, using Łukasiewicz notation and reversed Łukasiewicz notation, without using brackets (parentheses).
- (c) Let L be the formal language for reversed Łukasiewicz notation. What is the formal grammar of L ?

Exercise 4.2: Language Construction Puzzle

Let $\Sigma = \{a, b\}$. Suppose L is a formal language with a formal grammar comprising the following rules.

0. ε and “ ba ” are words.
1. If ϕ is a word, then so is $a\phi$.
2. If ϕ is a word, then so is ϕb .

Which of the following strings are words in L ?

- (a) “ $abab$ ” (b) “ $aaabbb$ ” (c) “ $aababa$ ” (d) “ $ababbb$ ”

4.2. The Language of Axiomatic Set Theory

Using formal language when talking about sets helps us alleviate any ambiguity of statements. For the following, let $\mathcal{L}$ denote the formal lan-

guage of axiomatic set theory.

To start, we need to define the alphabet.

Definition 4.2.1: The Alphabet of Axiomatic Set Theory

The alphabet of $\mathcal{L}$ consist of two main types of symbols.

- **Logical Symbols:** These symbols have some intrinsic meaning to them. These include
 - an infinite set of *variables*, often denoted by lowercase letters at the end of the English alphabet and can be distinguished using subscripts (e.g., x, y, z, x_0, y_1, z_0), although we may use any English letter if it serves to aid comprehension;
 - the equality symbol (i.e., $=$), denoting equality;
 - the logical connectives $\wedge, \vee, \neg, \implies$, and $\iff$;
 - the universal quantifier $\forall$ and the existential quantifier $\exists$; and
 - parentheses (i.e., $($ and $)$) and other punctuation symbols.
- **Non-Logical Symbols:** These represent predicates, functions and constants. In particular,
 - for every integer $n \geq 0$ we have an infinite collection of n -ary *predicate symbols*, which we can write, for example, as

$$P_0^n, P_1^n, P_2^n, P_3^n, P_4^n, \dots;$$

and

- for every integer $n \geq 0$ we have an infinite collection of n -ary *function symbols*, which we can write, for example, as

$$f_0^n, f_1^n, f_2^n, f_3^n, f_4^n, \dots,$$

where the nullary function symbols are called *constants*.

If the arity of a non-logical symbol is clear from context, we may elect to omit the superscript.

Remark: The Alphabet of First-Order Logic

The alphabet defined here is almost the same as the alphabet used in first order logic, except that we include “ $\in$ ” in the language.

Don't worry too much about what functions (and functions symbols) are. For now, one can interpret functions as a black box that takes in a number of inputs (i.e., the arity of the function) and returns a single output. We will cover more about functions when we get to it.

Exercise 4.3: Strings in $\mathcal{L}$

Which of the following are strings in $\mathcal{L}$?

- | | | |
|--------------|---------------------------|---|
| (a) x | (e) $p \vee q$ | (i) $\forall$ |
| (b) xy | (f) $p \wedge q$ | (j) $\forall x(\exists y(x \in y))$ |
| (c) $xy = x$ | (g) $\implies (p \neg q)$ | (k) $\forall \iff \exists$ |
| (d) $=$ | (h) $\in \in \in$ | (l) $\exists \wedge \forall \in \neg()$ |

We can now move on to defining the words in $\mathcal{L}$. There are two main types of words in $\mathcal{L}$, namely *terms* and *well-formed formulae*.

Definition 4.2.2: Term

A *term* in $\mathcal{L}$ is formed by applying the following two rules a finite number of times.

T0. Any variable is a term.

T1. If f is an n -ary function symbol and $t_1, t_2, \dots, t_n$ are terms, then the string $f(t_1, t_2, \dots, t_n)$ is also a term. In particular, constants are nullary function symbols and thus are terms.

Example 4.2.3: Terms in $\mathcal{L}$

- The symbols x, y, a, b , etc. can all be considered variables and thus are terms by **T0**.
- Suppose f is a binary function symbol and x and y are variables. Thus $f(x, y)$ is a term by **T0** and **T1**.
- The constants 3, 7, 11, etc. are nullary function symbols and thus are terms by **T1**.

Definition 4.2.4: Well-Formed Formula (WFF)

A *well-formed formula* (often abbreviated as WFF) in $\mathcal{L}$ is formed by applying the following five rules a finite number of times.

- F1.** If t_1 and t_2 are terms, then the strings $t_1 = t_2$ and $t_1 \in t_2$ are WFF.
- F2.** If P is an n -ary predicate symbol and $t_1, t_2, \dots, t_n$ are terms, then the string $P(t_1, t_2, \dots, t_n)$ is a WFF.
- F3.** If ϕ is a WFF, then the string $\neg\phi$ is also a WFF.
- F4.** If both ϕ and ψ are WFFs, then the strings $\phi \wedge \psi$, $\phi \vee \psi$, $\phi \implies \psi$, and $\phi \iff \psi$ are all WFFs.
- F5.** If x is a variable and ϕ is a WFF, then the strings $\forall x\phi$ (meaning “for all x , ϕ holds”) and $\exists x\phi$ (meaning “there exists x such that ϕ holds”) are both WFFs.

WFFs that only use the rules **F1** and **F2** have a special name.

Definition 4.2.5: Atomic Formulae

A WFF that uses only the rules **F1** and **F2** as described in [Definition 4.2.4](#) is called an *atomic formula*.

In other words, atomic formulae do not use any binary operators such as $\wedge$, $\vee$, $\implies$, or $\iff$. They also do not include any quantifiers such as $\forall$ or $\exists$.

In addition, it should be noted that parentheses can be included as appropriate to properly parse WFFs.

Example 4.2.6: WFFs in $\mathcal{L}$

- We know that both x and y , as variables, are terms by **T0**. Thus the strings $x = y$ and $x \in y$ are WFFs by **F1**. In fact, both the strings mentioned are atomic formulae.
- Suppose P is a predicate symbol and x and y are variables. Thus the string $P(x, y)$ is a WFF by **F2**. Notice that the string $P(x, y)$ is also an atomic formula.
- Since $x = y$ is a WFF, the string $\neg(x = y)$, which can also be written as $x \neq y$ in a more familiar notation, is also a WFF by **F3**. However the WFF $\neg(x = y)$ is not an atomic formula.
- As both $x = y$ and $P(x, y)$ are WFFs by above observations, thus $(x = y) \wedge P(x, y)$ is also a WFF by **F4**. Again, this WFF is not an atomic formula.

- We know that $x = x$ is a WFF. Thus $\forall x(x = x)$ and $\exists x(x = x)$ are both WFFs (but not atomic formulae) by F5.

Exercise 4.4: What's in $\mathcal{L}$?

Characterise the following strings as terms, WFFs, or neither. If the string is a WFF, determine if it is an atomic formula.

- (a) $f(x, 7)$ where x is a variable and the function f is a binary function symbol.
- (b) $\forall x(P(x) \iff Q(x, x))$ where x is a variable, P is a unary predicate symbol, and Q is a binary predicate symbol.
- (c) $3(f(x)) = 7$ where x is a variable and the f is a unary function symbol.
- (d) $\neg(\exists x(\forall y(f(x) = f(y))))$ where x and y are variables and f is a unary function symbol.
- (e) $\forall x(\exists y(x \in y))$ where x and y are variables.
- (f) $P(x = y, x, f(x))$ where x and y are variables, f is a unary function symbol, and P is a ternary predicate symbol.

Notice that variables may appear in WFFs. Some of the variables appear to be ‘controlled’ by, say, a quantifier, while others are not. This is an important observation.

Definition 4.2.7: Free and Bound Variables

Let x be a variable and χ be a WFF. The variable x is said to be either *bound* or *free* based on the following rules.

- V1. If χ is an atomic formula, then x is free in χ if and only if x appears in χ . There are no bound variables in atomic formulae.
- V2. If χ takes the form $\neg\phi$ where ϕ is a WFF, then
 - x is free in χ if and only if x is free in ϕ ; and
 - x is bound in χ if and only if x is bound in ϕ .
- V3. If χ takes the form $\phi \wedge \psi$, $\phi \vee \psi$, $\phi \implies \psi$, or $\phi \iff \psi$, where ϕ and ψ are WFFs, then
 - x is free in χ if and only if x is free in either ϕ or ψ ; and

- x is bound in χ if and only if x is bound in either ϕ or ψ .

V4. If χ takes the form $\forall y\phi$ or $\exists y\phi$, where y is a variable and ϕ is a WFF, then

- x is free in χ if and only if x is free in ϕ and x is a different variable from y ; and
- x is bound in χ if and only if x is bound in ϕ or x is the same variable as y .

The variable y is bound in χ .

Example 4.2.8

- Consider the WFF " $x = y$ ". This is an atomic formula, and thus both x and y are free in it by **V1**.
- In the WFF " $\neg(x = y)$ ", since x and y are both free in the WFF " $x = y$ ", thus both x and y are free in " $\neg(x = y)$ " by **V2**.
- The WFF " $\forall x(x = y)$ " has a bound variable x and a free variable y by **V4**.
- The WFF " $\exists a(P(x) \implies Q(x, a))$ " has a bound variable a and a free variable x by both **V3** (applied to the WFF " $P(x) \implies Q(x, a)$ ") and **V4**.
- In the WFF " $\forall x\forall y\exists z(P(x, y) \implies (Q(x, z) \vee Q(y, z)))$ ", the variables x , y , and z are all bound. Note that the variable w is neither bound nor free since it does not appear in the WFF.

Note that a variable could be both bound and free in the same WFF. This is rare, but possible.

Example 4.2.9

Consider the WFF $P(x) \implies \exists xQ(x)$. On the left side of the WFF, x is free. However, on the right side, x is bound. However, to avoid confusion, we would usually use a different variable to express the existential statement on the right. For example, $P(x) \implies \exists yQ(y)$. Now x is free on both sides of the WFF (and thus free in the WFF).

Exercise 4.5: Bound and Free Variables

In the following WFFs, determine if the variable x is bound, free, both, or neither.

- (a) $\forall x(\neg P(x))$ where P is a unary predicate symbol.
- (b) $(x = f(y)) \implies \forall x \exists y (y = x)$ where y is a variable and f is a unary function symbol.
- (c) $P(x) \implies (\exists y (Q(x, y)))$ where y is a variable, P is a unary predicate symbol, and Q is a binary predicate variable.
- (d) $P(a) \implies (P(b) \implies (P(c) \implies P(d)))$ where a, b, c , and d are all variables and P is a unary predicate symbol.

Explicitly stating which variables are free in a WFF is tedious when there are many variables to list. We have a shorthand for this.

Definition 4.2.10: WFFs with Free Variables

Let ϕ be a WFF and let the variables $x_1, x_2, \dots, x_n$ be free in ϕ . This fact can be written as $\phi(x_1, x_2, \dots, x_n)$.

Note that the string " $\phi(x_1, x_2, \dots, x_n)$ " is **not** technically a WFF in $\mathcal{L}$. However, if this string does appear, it means to substitute ϕ with its actual statement, replacing the $x_1, x_2, \dots, x_n$ with the corresponding values as listed in the parentheses. The result of this process will create a WFF.

Example 4.2.11

Let ϕ be the WFF " $x = y$ ". Note

$$\forall x \phi(x, x)$$

is **not** technically a WFF. However, using our interpretation above, we will substitute the appropriate variables to get

$$\forall x (x = x)$$

which is a WFF.

CHAPTER 5

Introduction to Axiomatic Set Theory

Paradoxes are the bane of creating a consistent system in mathematics. However, set theory, as we have introduced it, is full of paradoxes. For example, as stated so far, a set could contain itself. However, this leads to Russell's paradox of the set of all sets that are not members of themselves. To that end, we look at a more axiomatic definition of what sets are, inspired by the work of Ernst Zermelo and Abraham Fraenkel, to create a system that is (hopefully) devoid of paradoxes.

5.1. Zermelo-Fraenkel Set Theory

The modern study of set theory was initiated by Richard Dedekind and Georg Cantor in the 19th century. Unfortunately, these early systems are rife with paradoxes, prompting the creation of more robust axiomatic systems. Zermelo-Fraenkel set theory with the axiom of choice (often abbreviated as ZFC) is the most is still the best-known and most studied, and it will be what we use to define numbers from the ground up.

ZFC has 10 main axioms.

1. **Axiom of Extensionality:** Two sets are equal if they have the same elements.
2. **Axiom of the Empty Set:** The empty set exists.
3. **Axiom of Pairing:** Any two objects have a pair.
4. **Axiom of Union:** The union of the elements in a set exists.
5. **Axiom Schema of Specification:** An axiom describing how a set can be created using set-builder notation.
6. **Axiom of Regularity:** Every non-empty set A contains an element B which is disjoint from A . This is used to prevent paradoxes about sets containing themselves.
7. **Axiom of Infinity:** An infinite set exists.
8. **Axiom Schema of Replacement:** The image of any function exists.

9. **Axiom of Choice:** It is always possible to create a new set by choosing one element from each set in any collection of non-empty sets.
10. **Axiom of Power Set:** The power set of any set exists.

Remark: Axiom of Choice

The Axiom of Choice is a controversial axiom, and some authors may choose to omit it. ZFC without the axiom of choice included is usually denoted ZF. However, for our exploration of numbers here, we require the axiom of choice.

Remark: Choice of Axioms

The names and statements of the axioms in ZFC are, frustratingly, inconsistent.

- Some authors call the axiom schema of specification the axiom of subsets.
- Some call the axiom of foundation the axiom of regularity.
- Many authors will group the axiom of the empty set along with the axiom of infinity.

The 10 axioms that we look here are what should be the most illuminating and essential ones.

An important thing to note when we are working in ZFC is that **everything is a set**. The foundational object that we are discussing is the set, and we are defining everything in terms of sets. We are simply specifying that the properties of these objects (sets) are. Thus awkward notation like $a \in b$ or $1 \cup 2$ is readily explained away when we remember that all objects are sets.

To avoid overloading content at this juncture, we will only look at the first six of the ten axioms of ZFC. The rest will be introduced in later chapters as required.

5.2. Axiom of Extensionality

The first axiom that we will look at is the axiom of extensionality.

Axiom 5.2.1: Extensionality

$$\forall A \forall B (\forall x ((x \in A \iff x \in B) \implies A = B))$$

Let us understand what this means. For all A and B , remembering the golden rule that everything is a set, if for all x , x is a member of A if and only if x is a member of B , then A and B are equal. In other words, if any element in A is in B , and vice versa, then A and B are equal, and this holds for all sets.

Some may interpret this axiom as the foundational property of sets. Sets are equal if they have the same members. That is what all they are. All the other axioms simply tell us what sets exist.

5.3. Axiom of the Empty Set

The next axiom starts to give us an idea of what sets could exist.

Axiom 5.3.1: Empty Set

$$\exists A (\neg \exists x (x \in A))$$

Simply put, there exists a set A such that there does not exist an x that is in A . This applies to all x , meaning that there must be nothing in A . Hence we see that A is the empty set $\emptyset$.

5.4. Axiom of Pairing

The axiom of pairing gives us another example of what sets exist.

Axiom 5.4.1: Pairing

$$\forall x \forall y \exists S (\forall z (z \in S \iff ((z = x) \vee (z = y))))$$

In English, this means that for all x and y , there exists a set S such that any element z in S must be either x or y . Clearly, this must mean that the set S in question is the set containing only x and y , i.e. the set $\{x, y\}$. This is why this is called the axiom of pairing – it guarantees that the unordered pair exists.

We now consider sets with one element.

Definition 5.4.2: Singleton

A *singleton* is a set containing one element. We denote a singleton containing the element x by $\{x\}$.

Axiom 5.4.1 allows us to definitively prove within ZFC that singletons exist.

Proposition 5.4.3: Singletons Exist

Singletons exist. Formally,

$$\forall x \exists S (\forall z (z \in S \iff z = x)).$$

Proof. **Axiom 5.4.1** says that

$$\forall x \forall y \exists S (\forall z (z \in S \iff ((z = x) \vee (z = y)))).$$

In particular we may choose $y = x$. Thus the statement reduces to

$$\forall x \exists S (\forall z (z \in S \iff z = x))$$

as required. ■

Axiom 5.4.1 also allows us show that sets have ‘no repeated elements’. For example, we can show that the sets $\{a, a\}$ and $\{a\}$ are the same set.

Theorem 5.4.4: The ‘No Repeated Elements’ Property

$\{a, a\} = \{a\}$ for all a . Formally,

$$\forall a (\{a, a\} = \{a\}).$$

Proof. First, note for any a that the sets $\{a, a\}$ and $\{a\}$ do indeed exist by **Axiom 5.4.1** and **Proposition 5.4.3** respectively. Next, recall that **Axiom 5.2.1** tells us that

$$\forall A \forall B (\forall x ((x \in A \iff x \in B) \implies A = B)).$$

In particular, let $A = \{a, a\}$ and $B = \{a\}$.

- If $x \in A$, then x is forced to be a . Clearly this means that $x \in B$.
- If $x \in B$, then x is either a (the first one) or a (the second one). In either case we have $x = a$. Thus $x \in A$ also.

Hence the condition “ $x \in A \iff x \in B$ ” is satisfied. So that axiom tells us that $A = B$, i.e. $\{a, a\} = \{a\}$. ■

5.5. Axiom of Union

Remember how we can take the union of sets? This next axiom explicitly gives us that operation within ZFC.

Axiom 5.5.1: Union

$$\forall A \exists U \forall x (x \in U \iff \exists S ((x \in S) \wedge (S \in A)))$$

We can interpret this in English as saying that given any set A , there exists a set U (the union) such that any element x in U must be in some set S in A (recalling that everything is thought of as a set in ZFC).

The combination of [Axiom 5.4.1](#) and [Axiom 5.5.1](#) allows us to create a set from an arbitrary list of items.

Theorem 5.5.2: Set of Any List of Items

For any $a_1, a_2, \dots, a_n$, the set $\{a_1, a_2, \dots, a_n\}$ exists.

Proof. See [Exercise 5.1](#) (later). ■

Remark: Infinite Sets

Note that the theorem only works for a finite number of items. The argument by induction does not work for an infinite number of items, which is partly the reason why we need to define the axiom of infinity (to be introduced later).

A related result is the union of an arbitrary (but finite) number of sets.

Theorem 5.5.3: Union of Finite Number of Sets Exists

For any finite number of sets $A_1, \dots, A_n$, the set

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \dots \cup A_n$$

exists.

Proof. We consider induction on n .

The base case when $n = 1$ is trivial since $\bigcup_{i=1}^1 A_i = A_1$ exists.

Suppose the induction hypothesis holds for some number of sets $k \geq 1$, i.e. the set $\bigcup_{i=1}^k A_i$ exists for any sets $A_1, \dots, A_k$. We show that the set $\bigcup_{i=1}^{k+1} A_i$ exists for any sets $A_1, \dots, A_k, A_{k+1}$.

By induction hypothesis the set $\bigcup_{i=1}^k A_i$ exists. By [Axiom 5.4.1](#) we can form a set

$$\left\{ \bigcup_{i=1}^k A_i, A_{k+1} \right\}.$$

Thus by [Axiom 5.5.1](#), the set

$$\left(\bigcup_{i=1}^k A_i \right) \cup A_{k+1} = \bigcup_{i=1}^{k+1} A_i$$

exists, proving the case for $k + 1$.

Hence, by mathematical induction, the theorem is proven. ■

Exercise 5.1: Set of Any List of Items

Prove [Theorem 5.5.2](#).

5.6. Axiom Schema of Specification

This axiom schema, also known as the axiom of subsets, allows us to restrict the elements of a set to those that satisfy a specific condition.

Axiom 5.6.1: Schema of Specification

Let ϕ be a WFF. Then

$$\forall A \forall w_1 \forall w_2 \cdots \forall w_n \exists S (\\ \forall x (x \in S \iff (x \in A \wedge \phi(x, w_1, w_2, \dots, w_n, A))) \\)$$

where S is not free in ϕ .

In other words, for every set A and for every w_1 to w_n there exists a set S such that x is in S if and only if x is in A and that ϕ is satisfied. In essence, this axiom schema defines set builder notation, but in a more rigorous manner.

Note that we require one instance of the above for every WFF ϕ . Thus, this is an axiom schema and not just an axiom.

This axiom schema can be used to prove several things.

5.6.1. Intersection

First we will show that the intersection of two sets exists.

Theorem 5.6.2: Intersection of Sets is a Set

Let A and B be sets. Then their intersection $A \cap B$ is also a set.

Proof. We first use [Axiom 5.6.1](#) to create an axiom for the WFF ϕ , where ϕ is “ $x \in B$ ”. The axiom generated is

$$\forall A \exists S (\forall x (x \in S \iff (x \in A \wedge x \in B))).$$

Since this applies to all A , this must apply to the A mentioned in the statement of the theorem. Thus there must exist a set S such that $x \in S$ if and only if $x \in A$ and $x \in B$. This is exactly the definition of the set intersection. ■

Henceforth the string “ $A \cap B$ ” that appear in axioms and other results will refer to the intersection of the sets A and B given by [Theorem 5.6.2](#).

5.6.2. Cartesian Product

This axiom, along with the previous axioms, can be used to prove that the Cartesian product exists. Recall that the Cartesian product of two sets A and B is the collection of 2-tuples where the first element comes from A and the second comes from B .

Lemma 5.6.3: Cartesian Product of Two Sets is a Set

Let A and B be non-empty sets. Then their Cartesian product,

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\},$$

is indeed a set. In particular, it is a subset of $\mathcal{P}(\mathcal{P}(A \cup B))$.

Proof. We just need to show that the collection of all ordered pairs is indeed a set. Let $a \in A$ and $b \in B$. Note [Axiom 5.5.1](#) says that the union of A and B (i.e., the set $A \cup B$) exists. Thus one sees clearly that $a, b \in A \cup B$.

Now [Proposition 5.4.3](#) tells us that the set containing only a (i.e., $\{a\}$) exists. Furthermore [Axiom 5.4.1](#) tells us that the set containing only a and b (i.e., $\{a, b\}$) exists. Clearly both $\{a\}$ and $\{a, b\}$ are subsets of $A \cup B$. Since the power set of $A \cup B$ (i.e., $\mathcal{P}(A \cup B)$) exists by [Axiom 9.5.1](#), thus we can write $\{a\}, \{a, b\} \in \mathcal{P}(A \cup B)$.

Recall that the ordered pair (a, b) is defined to be the set $\{\{a\}, \{a, b\}\}$. One sees that

$$\{\{a\}, \{a, b\}\} \in \mathcal{P}(\mathcal{P}(A \cup B)).$$

and so $(a, b) = \{\{a\}, \{a, b\}\}$ exists and is an element of $\mathcal{P}(\mathcal{P}(A \cup B))$. Thus, the Cartesian product of A and B can be formed using [Axiom 5.6.1](#). In particular, one sees

$$A \times B = \{(a, b) \in \mathcal{P}(\mathcal{P}(A \cup B)) \mid a \in A \wedge b \in B\}$$

Thus the collection of all such (a, b) is indeed a set by [Theorem 5.5.2](#). ■

With this result we can prove that the more general Cartesian product exists.

Theorem 5.6.4: Cartesian Product is a Set

Let $A_1, A_2, \dots, A_n$ be non-empty sets where $n \geq 2$. Then the Cartesian product $A_1 \times A_2 \times \dots \times A_n$ is indeed a set.

Proof. We make the following claim.

The Cartesian product of the sets $A_1, A_2, \dots, A_n$ is a subset of $\mathcal{P}(\mathcal{P}(A_1 \cup A_2 \cup \dots \cup A_n))$.

We will prove this using induction on n , which will immediately prove the theorem.

The base case of $n = 2$ is given by [Lemma 5.6.3](#).

Now assume that the induction hypothesis holds for some $k \geq 2$, i.e. the Cartesian product $A_1 \times \dots \times A_k$ is a subset of $\mathcal{P}(\mathcal{P}(A_1 \cup \dots \cup A_k))$. We will show that the Cartesian product $A_1 \times \dots \times A_k \times A_{k+1}$ is a subset of $\mathcal{P}(\mathcal{P}(A_1 \cup \dots \cup A_k \cup A_{k+1}))$.

We know that $\{a_1\}, \{a_1, a_2\}, \{a_1, a_2, a_3\}, \dots, \{a_1, a_2, \dots, a_k\} \in \mathcal{P}(A_1 \cup A_2 \cup \dots \cup A_k)$ from the induction hypothesis. Note that $A_1 \cup \dots \cup A_k \cup A_{k+1}$ exists by [Axiom 5.5.1](#), and that $a_1, \dots, a_k, a_{k+1} \in A_1 \cup \dots \cup A_k \cup A_{k+1}$.

Note that $\{a_{k+1}\}$ and $\{a_1, a_2, \dots, a_k\}$ are both subsets of $\mathcal{P}(A_1 \cup \dots \cup A_k \cup A_{k+1})$, where the former exists due to [Proposition 5.4.3](#) and the latter exists due to the induction hypothesis. Hence their union, namely $\{a_1, a_2, \dots, a_k, a_{k+1}\}$, exists, and is a member of $\mathcal{P}(A_1 \cup \dots \cup A_k \cup A_{k+1})$.

Therefore, the object

$$\{\{a_1\}, \{a_1, a_2\}, \{a_1, a_2, a_3\}, \dots, \{a_1, a_2, \dots, a_k\}, \{a_1, a_2, \dots, a_k, a_{k+1}\}\},$$

which is $(a_1, \dots, a_k, a_{k+1})$, is a set. In particular, it is in $\mathcal{P}(\mathcal{P}(A_1 \cup \dots \cup A_k \cup A_{k+1}))$. Therefore the collection of all such objects, i.e. the Cartesian product

$A_1 \times \cdots \times A_k \times A_{k+1}$, which can be defined using set-builder notation as

$$\{(a_1, \dots, a_{k+1}) \in \mathcal{P}(\mathcal{P}(A_1 \cup \cdots \cup A_{k+1})) \mid a_1 \in A_1 \wedge \cdots \wedge a_{k+1} \in A_{k+1}\}$$

via [Axiom 5.6.1](#), exists because it is a subset of $\mathcal{P}(\mathcal{P}(A_1 \cup \cdots \cup A_{k+1}))$. Hence the case for $k + 1$ is also true.

The theorem is proven by induction. ■

5.7. Axiom of Regularity

The last axiom that we will look at in this chapter is one that helps alleviate some issues with the ‘intuitive’ definition of sets.

5.7.1. Why Regularity?

Before we introduce the axiom, let us explore why we need this axiom.

Consider our original definition of a set, which is just a collection of items. For example we can think of the set $\{1, 2, 3\}$ as the collection containing the items 1, 2, and 3. The issue comes to when we have sets in sets; in particular, sets that reference themselves.

Our axioms so far do not restrict us putting sets in themselves. In fact, without regularity, the set A given by

$$A = \{x \mid x \in x\},$$

i.e. the set containing sets that contain themselves, would certainly not raise any eyebrows. An infinitely nested set (which we can uglily write down as $\{\{\{\dots\}\}\}$) would still exist in A . However, things get weird if we continue tugging at this string.

Consider the set R (sometimes called *the Russell set*) given by

$$R = \{x \mid x \notin x\}.$$

We can note down some examples of sets in R . For one, the empty set is in R since the empty set does not contain itself. The set $\{1\}$ is in R since $\{1\}$ does not contain itself. However, the infinitely nested set $\{\{\{\dots\}\}\}$ would not be in R since it would contain itself. The trouble is this: does R contain R ?

- Suppose R contains R . This means $R \in R$. However by definition of R , this must mean that R is not in R (since R contains all sets that are not members of themselves, including R).

- Suppose now R does not contain R . This means $R \notin R$. But the definition of R means that R necessarily contains R (since R contains all sets that are not members of themselves, including R).

This is a paradox! If R contains R then R does not contain R ; if R does not contain R then R contains R . This is known as **Russell's Paradox**.

5.7.2. The Axiom

To avoid issues of sets containing themselves (and paradoxes like those mentioned earlier), we introduce the axiom of regularity in an attempt to regulate what sets can contain.

Essentially, the axiom states that any non-empty set must contain at least one element (which is a set in ZFC) that has no elements in common with the original set. More formally, we can write this axiom as

$$\forall A (A \neq \emptyset \implies \exists x (x \in A \wedge x \cap A = \emptyset)).$$

Note, however, that this is not a valid string in $\mathcal{L}$ – the set intersection symbol is a valid symbol in $\mathcal{L}$. Thus, we need to be more creative on how we state this axiom.

One way is to notice that “ $x \cap A = \emptyset$ ” precisely means that the intersection $x \cap A$ has no elements. That means that there does not exist any element that is both in x and in A , i.e. “ $\neg \exists y (y \in x \wedge y \in A)$ ”. On a related note, the condition “ $A \neq \emptyset$ ” means that the set A contains something, i.e. “ $\exists x (x \in A)$ ”.

With this we can state the axiom of regularity in a more formal language.

Axiom 5.7.1: Regularity

$$\forall A (\exists x (x \in A) \implies \exists x ((x \in A) \wedge \neg (\exists y (y \in x \wedge y \in A))))$$

The axiom of regularity leads to this important observation.

Proposition 5.7.2: No Recursive Inclusions

Let A and B be sets. Then either $A \notin B$ or $B \notin A$.

Proof. Seeking a contradiction, suppose that $A \in B$ and $B \in A$. So $A \in B \cap \{A, B\}$ and $B \in A \cap \{A, B\}$. Thus one sees that both $A \cap \{A, B\}$ and $B \cap \{A, B\}$ are non-empty.

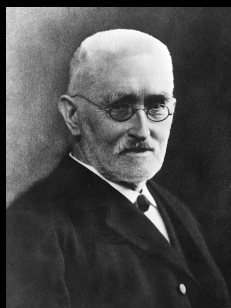
Now, by [Axiom 5.7.1](#), we must have a $X \in \{A, B\}$ such that $X \cap \{A, B\} = \emptyset$.

- If $X = A$, this means $A \cap \{A, B\} = \emptyset$, contradicting that $A \cap \{A, B\}$ is non-empty.
- If instead $X = B$, this means $B \cap \{A, B\} = \emptyset$, contradicting that $B \cap \{A, B\}$ is non-empty.

In either case, a contradiction arises. Thus, we cannot have $A \in B$ and $B \in A$, meaning that either $A \notin B$ or $B \notin A$ as required. ■

With the axiom of regularity, our foundations for ZFC are set. We are now ready to define numbers, starting with the naturals.

PART I
Natural



I became convinced that studying the algebraic relationship of numbers is most conveniently based on a concept that is directly connected with the simplest arithmetic properties.

— Richard Dedekind, 1871

([\[Kle07\]](#), p. 66)

CHAPTER 6

Defining Natural Numbers

To define all the numbers, we have to start with the most basic of numbers – the natural numbers. The natural numbers are the most basic of all numbers. They are considered our counting numbers, and understanding how they can be defined using the axioms of ZFC is essential for continuing to create the rest of the numbers.

6.1. The Need for Abstraction

So why go through all this trouble to define the natural numbers? Surely everyone can agree what a natural number is?

Let us ask a simple question: what is 3? Seriously. Think about what 3 really means. In elementary school, we were taught that 3 is just a number quantifying a certain number of things. There are many 3s – 3 circles, 3 squares, 3 cars, 3 stars, 3 apples, 3 lines, 3 words, 3 points, 3 boxes, etc.

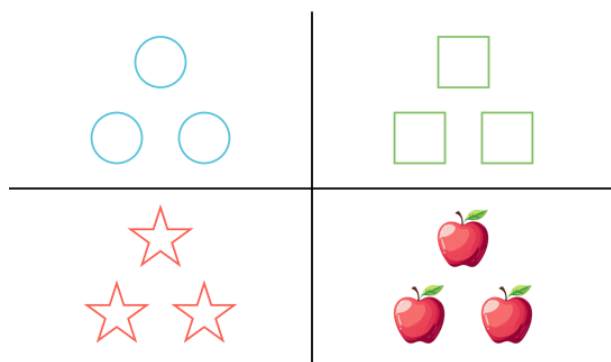


Figure 6.1. Various Examples of 3

What they all encapsulate is that there is a certain number of things. And **that** number is 3. It is an abstraction above a specific ‘type’ of 3. Not 3 circles, or 3 squares, or any of the other types that 3 could be; 3 is a pure symbol symbolising a pure count.

Of course, when it comes to asking what 3 is, at a fundamental level, we need to have a representative for the concept of 3. This is just like how Physics moved away from defining the SI base units using physical

objects and their properties, and instead defining them using fundamental physical constants of the Universe. Although they have shifted the burden away from a specific instance of the physical artefact, they still need one for practical use. Likewise, although we can abstract away the idea of 3, we still need a representative for 3.

In this chapter, we will be focusing on defining the natural numbers using a set-theoretic approach. In particular, we will use ZFC to construct representatives for **all** the natural numbers. That way, we will have no ambiguity when we discuss what 3, or 12, or 83, or 1289, or 89338, etc., really means at a fundamental level.

6.2. Defining Zero

The act of defining numbers has to start somewhere. The natural numbers all represent a certain quantity of things – 3 things, 12 things, 83 things, 1289 things, 89338 things, etc. The problem with defining these other numbers without a good starting point is that there can be an argument of what these numbers really mean at a fundamental level. What is a “thing”? How do we count these “things”? How can we ensure that others count these “things” the same way? That is why we will start with zero – the absence of anything.

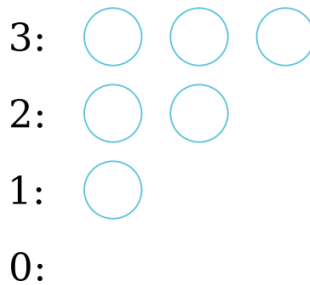


Figure 6.2. Counting Down To Zero

While other numbers (say 1) can have ambiguity of what they mean (e.g., 1 car, 1 square, 1 point, 1 circle, etc.), 0 necessarily means that there is nothing. It represents the absence of anything, and it does not matter what the type of “thing” is. We can agree that, whatever the “thing” is, they are not there, and the quantity of that number of items is 0.

Let us appeal to the notion of sets to further elaborate on this. There are many sets that contain 3 items. For example, the sets $\{a, b, c\}$, $\{1, 2, 3\}$, $\{x, y, z\}$, $\{0, 1, 2\}$, $\{+, -, /\}$, and $\{\text{hi, hello, bye}\}$ all contain 3 items. But

there is one set that contains zero items – the empty set. Any other set containing zero items must necessarily be equal to the empty set, since the items in a set uniquely define a set by the ZFC axiom of extensionality ([Axiom 5.2.1](#)).

As such, we may define the number zero to be the empty set.

Definition 6.2.1: Zero

The number zero is defined to be the empty set. That is,

$$0 = \emptyset.$$

This is definitely a little odd. Why are we defining a number to be a set? Recall that **everything is a set** under the axioms of ZFC. Thus every number must necessarily be a set. We might as well define 0, which represents the absence of anything, to be the empty set, the set devoid of everything. Surely any other choice of set that defines 0 would be absurd; why would we pick a non-empty set containing something to represent the lack of anything? Under this light, setting $0 = \emptyset$ is completely natural.

6.3. Successor

Okay, we have defined zero. But what about the rest of the natural numbers?

For the rest of the numbers, our key idea is to use the natural numbers defined before then to define that number itself. That is, to define a natural number n , we want to only use the sets $0, 1, 2, \dots, n - 1$ to define n . Appealing to the idea that each natural number n should be a set containing n items, it seems natural that we should define

$$n = \{0, 1, 2, \dots, n - 1\}.$$

This set contains n items, and so it can be used to represent the natural number n .

However, instead of directly defining what n is, we will do it in a more roundabout way – using something known as the successor of a set.

Definition 6.3.1: Successor

Let A be a set. The *successor* of A is

$$\mathcal{S}(A) = A \cup \{A\}.$$

In other words, the successor of a set A is the set containing the elements of A and A itself.

Example 6.3.2: Successor

- $\mathcal{S}(\emptyset) = \emptyset \cup \{\emptyset\} = \{\emptyset\}$ since the empty set is, of course, empty.
- $\mathcal{S}(\{a\}) = \{a\} \cup \{\{a\}\} = \{a, \{a\}\}.$
- $\mathcal{S}(\{a, b, c\}) = \{a, b, c\} \cup \{\{a, b, c\}\} = \{a, b, c, \{a, b, c\}\}.$
- $\mathcal{S}(\mathcal{S}(\{a\})) = \mathcal{S}(\{a, \{a\}\}) = \{a, \{a\}, \{\{a, \{a\}\}\}.$

Exercise 6.1: Successor Practice

What is $\mathcal{S}(\mathcal{S}(\{x, y\}))$?

Usually the successor of a set is only used for defining the natural numbers. Observe that a very nice pattern emerges when we repeatedly find the successor of 0.

- $\mathcal{S}(0) = \mathcal{S}(\emptyset) = \{\emptyset\}$. This set has 1 element.
- $\mathcal{S}(\mathcal{S}(0)) = \mathcal{S}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$. This set has 2 elements.
- $\mathcal{S}(\mathcal{S}(\mathcal{S}(0))) = \mathcal{S}(\{\emptyset, \{\emptyset\}\}) = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$. This set has 3 elements.

We see that applying finding the n th successor of 0 exactly gives a set with n elements. This repeated application of the successor defines the natural numbers.

Definition 6.3.3: Defining A Natural Number

The natural number n is defined to be the n th successor of 0, and The set that underlies n is called the *counter set of n* . In other words,

$$n = \overbrace{\mathcal{S}(\mathcal{S}(\mathcal{S}(\cdots \mathcal{S}(0) \cdots)))}^{n \text{ applications}}.$$

Alternatively,

$$n = \begin{cases} \emptyset & \text{if } n = 0 \\ \mathcal{S}(n-1) & \text{if } n \geq 1 \end{cases}$$

Remark: Undefined Operations

The alternative definition of n suffers from the fact that we have not yet defined the subtraction (i.e., $-$) and greater-than-or-equals-to (i.e., $\geq$) operations.

The counter sets of natural numbers can be drawn diagrammatically.

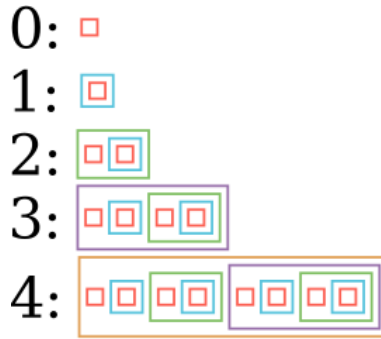


Figure 6.3. The First 5 Counter Sets

What we observe is interesting. The counter set of 1 contains the empty set (the counter set of 0), so we may write $1 = \{0\}$. The counter set of 2 contains both the empty set and the counter set of 1, so we may write $2 = \{0, 1\}$. We observe too that $3 = \{0, 1, 2\}$ and $4 = \{0, 1, 2, 3\}$. This observation allows us to make the definition of the natural numbers more concise.

Definition 6.3.4: Defining A Natural Number, More Concisely

Define the natural number

- 0 to be the empty set; and
- n to be the set $\{0, 1, 2, \dots, n - 1\}$ for any other natural number n .

Exercise 6.2: Analysing Counter Sets

- Write down the counter set of 5 in terms of 0, 1, 2, 3, and 4.
- Write down the counter set of 5, using only the curly braces $\{$ and $\}$ and commas.

- (iii) How many **open** curly braces (i.e., $\{$) would be used to define the counter set of 65536? Assume we only use the curly braces $\{$ and $\}$ as well as commas.

6.4. The Set of Natural Numbers

With the natural numbers defined, we are almost ready to define the set of natural numbers. However, we need to return back to ZFC to include an important axiom.

6.4.1. Axiom of Infinity

As defined so far, ZFC does not explicitly state that a set of an infinite number of elements exists. Sure, we can create a set based of a list of items ([Theorem 5.5.2](#)), but this only works for a **finite** number of items. We can take the union of a list of sets ([Theorem 5.5.3](#)), but this works only for a **finite** number of sets. Regardless of what operations we can do so far, we are limited to a finite number of items and finite sets. Repeatedly applying operations on finite sets will still make the resultant set finite.

Thus, we need to make a leap of faith and bring into existence **one** infinite set. That is the job of the axiom of infinity – an infinite set exists.

Loosely, we can state the axiom as

$$\exists I((\emptyset \in I) \wedge \forall x(x \in I \implies S(x) \in I)),$$

that is, there exists a set I (an infinite set) such that the empty set is an element of I , and, for any element (which is a set, according to ZFC) in I , its successor is also in I . One sees that this would mean that there is an infinite number of elements in I – we can just repeatedly take the successor of elements in I to create something that was not present in I . However, this is not stated in our formal language for axiomatic set theory. Can we make it more formal?

The axiom of infinity asserts that an infinite set I exists satisfying the following conditions:

- I1. the empty set is an element of I ;
- I2. for any element in I , its successor is also in I .

Condition I1 is easy to state as

$$\exists O(O \in I \wedge \neg(\exists x(x \in O))).$$

This states that there is a set O in I such that nothing exists in O , which is precisely the definition of the empty set in the axiom of the empty set ([Axiom 5.3.1](#)). Note that we use the letter O instead of the standard symbol for the empty set (i.e., $\emptyset$) to avoid confusion.

Condition **I2** is slightly trickier. Recall that the successor of a set A is the set $A \cup \{A\}$. So an element in the successor must be an element of A , or A itself. So if S is the successor of A , we can write this down as

$$x \in S \iff (x \in A \vee x = A).$$

Therefore **I2** can be written as

$$\forall x(x \in I \implies \exists S(S \in I \wedge \underbrace{\forall y(y \in S \iff (y \in x \vee y = x))}_{\text{Definition of Successor}})).$$

This means that any element in I must have its successor also in I .

We can now put **I1** and **I2** together to form the axiom of infinity.

Axiom 6.4.1: Infinity

The infinite set I exists. In particular,

$$\begin{aligned} \exists I(& \\ & \exists O(O \in I \wedge \neg(\exists x(x \in O))) \\ & \wedge \forall x(x \in I \implies \exists S(S \in I \wedge \forall y(y \in S \iff (y \in x \vee y = x)))) \\ &). \end{aligned}$$

6.4.2. A Consequence of the Axiom of Infinity

It should come as no surprise that the set I that is asserted by [Axiom 6.4.1](#) is exactly the set of natural numbers. Indeed, [Axiom 6.4.1](#) states that the empty set (which is the counter set for 0) is in I and that any successor of any set in I also belongs in I , which defines the rest of the natural numbers.

Definition 6.4.2: The Set of Natural Numbers

The *set of natural numbers*, denoted $\mathbb{N}$, is the set I given by the axiom of infinity ([Axiom 6.4.1](#)). Equivalently, it is the set containing all the natural numbers, i.e.

$$\mathbb{N} = \{0, 1, 2, 3, 4, 5, \dots\}.$$

One can imagine the size of $\mathbb{N}$ as the set with the ‘smallest’ infinity. Indeed, we will explore other sets with a larger ‘size’ than $\mathbb{N}$ in future parts. But $\mathbb{N}$ is special as it is the only infinite set asserted by an axiom; the rest are all consequences of the ZFC axioms, and not an axiom itself.

The rest of this part will be spent exploring the properties of $\mathbb{N}$. We will explore a rigorous definition of counting using natural numbers, comparing the sizes of sets, and defining the operations on $\mathbb{N}$.

CHAPTER 7

Relations I

One of mathematicians' most essential abilities is noticing and describing relationships. We come across many patterns and links that could be characterised by relations in our everyday lives, like a relation between a parent and child or between an employer and an employee. The idea of relationships between sets is essential, and we will look at their properties here.

7.1. What is a Relation?

A high-level explanation of a relation is simply a way to 'create a connection' between items in two sets. However, the abstract definition of a relation is much less enlightening.

Definition 7.1.1: Relation

Let X and Y be sets. A **relation from X to Y** is a subset of $X \times Y$.

Let R be a relation, $x \in X$, and $y \in Y$. We say that **x is related to y by R** (or that **x is R -related to y**), denoted $x R y$, if and only if $(x, y) \in R$. We write $x \not R y$ if and only if $(x, y) \notin R$.

If there is no confusion of what relation we are talking about, we can just say that **x is related to y** .

Remark: n -ary Relations

Strictly speaking, what we have defined above is a **binary relation**, since it only involves the Cartesian product of two sets. A more general type of relation that could be studied is a **n -ary relation**, which is a subset of a Cartesian product of n sets. In particular, for sets $A_1, A_2, \dots, A_n$, an n -ary relation on those sets is a subset of $A_1 \times A_2 \times \dots \times A_n$. However, we will solely be working with binary relations in this book, and thus the term "relation" shall only be used to describe binary relations.

This is clearly way too abstract of a definition to see how this captures the idea of 'creating connections' between set elements, so let's take a look at some concrete examples.

Example 7.1.2: The “lived in” Relation

Let M , a subset of the set of all mathematicians, be

$$\{\text{Peano, Cantor, Dedekind, Russell, Euler, Hilbert}\}$$

and let C , a subset of the set of all countries, be

$$\{\text{Italy, Switzerland, Russia, Germany, France}\}.$$

Then we can define R to be the “lived in” relation. In particular,

$$R = \{(\text{Peano, Italy}), (\text{Cantor, Russia}), (\text{Cantor, Germany}), (\text{Dedekind, Germany}), (\text{Euler, Switzerland}), (\text{Euler, Germany}), (\text{Euler, Russia}), (\text{Hilbert, Germany})\}.$$

We can show this relation diagrammatically.

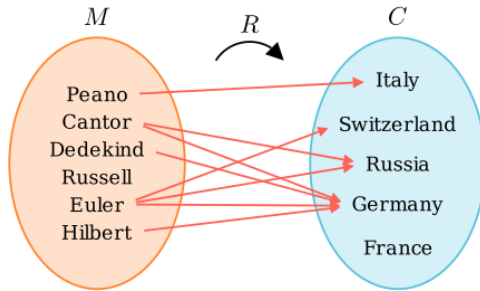


Figure 7.1. Arrow Diagram of R

On the left is the set M , and on the right is the set C . The arrows shows how they are related. For example, the arrow from “Peano” points to “Italy” since Giuseppe Peano lived in Italy. Notice that Bertrand Russell did not live in any of the countries in C , and that no mathematician in M lived in France.

Example 7.1.3: The “is less than” Relation

Let $A = \{0, 1, 2\}$ and $B = \{1, 2, 3\}$. Define the relation R such that

$$x R y \iff x < y$$

for all $x \in A$ and $y \in B$. Then $0 R 1$, $0 R 2$, $0 R 3$, $1 R 2$, $1 R 3$, and

2 R 3. However we see $1 \not R 1$, $2 \not R 1$, and $2 \not R 2$. Thus

$$R = \{(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3)\}.$$

Again, we may show this relation diagrammatically.

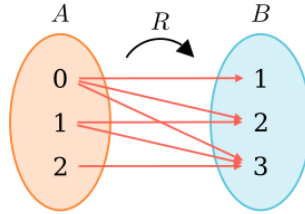


Figure 7.2. Arrow Diagram of R

Relations may also be performed on the same set.

Definition 7.1.4: Relation on a Set

A *relation R on a set A* is a relation from A to A . That is, R is a subset of A^2 .

Remark: n -ary Relation on a Set

Likewise, a n -ary relation on a set A is a subset of A^n .

Example 7.1.5: Relation on a Set

Let $A = \{3, 4, 5, 6, 7\}$ and define a relation R on A such that

$$x R y \iff x - y \text{ is an even number}$$

for all $x, y \in A$. Then one sees that

$$R = \{(3, 3), (3, 5), (3, 7), (4, 4), (4, 6), (5, 3), (5, 5), (5, 7), (6, 4), (6, 6), (7, 3), (7, 5), (7, 7)\}.$$

We can, of course, show this relationship using the standard arrow diagram, but we can also use a directed graph (a graph with directed edges) to represent it.

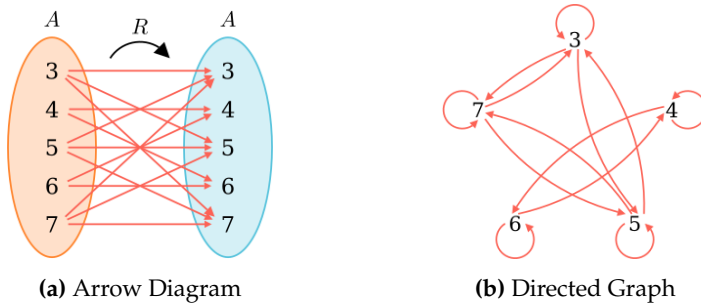


Figure 7.3. Diagrammatic Representations of R

Exercise 7.1: Relation Discriminator

Let $A = \{2, 3, 4, 5\}$ and $B = \{2, 4, 6, 8\}$. Which of the following sets are relations from A to B ? Which of the following sets are relations on A ?

- | | |
|--|--|
| (a) $R_1 = \emptyset$ | (d) $R_4 = \{2, 3, 4\}$ |
| (b) $R_2 = \{(2, 2), (3, 3), (4, 4)\}$ | (e) $R_5 = \{(2, 4), (4, 6), (6, 8)\}$ |
| (c) $R_3 = \{(2, 2), (3, 4), (4, 8), (5, 6), (5, 8)\}$ | (f) $R_6 = A^2$ |
| | (g) $R_7 = B^2$ |

We end this section by looking at the equality of relations.

Definition 7.1.6: Equality of Relations

Let A, B, C , and D be sets. Let R be a relation from A to B and S a relation from C to D . The two relations R and S are equal if and only if $A = C$, $B = D$, and the set R equals the set S .

Remark: A Relation as an Ordered Triple

In light of the above definition of relation equality, we could define a relation from the set A to the set B to be an ordered triple (A, B, R) , where $R \subseteq A \times B$. We stick to our definition of a relation as a subset of $A \times B$ instead.

7.2. Properties of Relations

Let us now look at some properties of relations.

Definition 7.2.1: Domain, Codomain, and Range (Relations)

Let X and Y be sets and R be a relation from X to Y .

- The *domain* of R is the set $\{x \in X \mid \exists y \in Y \text{ s.t. } x R y\}$.
- The *codomain* of R is the set Y .
- The *range* of R is the set $\{y \in Y \mid \exists x \in X \text{ s.t. } x R y\}$.

Example 7.2.2

Let $A = \{1, 2, 3\}$ and $B = \{2, 4, 9\}$. Let the relation R be a relation from A to B such that

$$x R y \iff x^2 = y.$$

In particular, we see that

$$R = \{(2, 4), (3, 9)\}.$$

The domain of R is

$$\begin{aligned} & \{a \in A \mid \exists b \in B \text{ s.t. } a R b\} \\ &= \{a \in A \mid \exists b \in B \text{ s.t. } a^2 = b\} \\ &= \{2, 3\}. \end{aligned}$$

The codomain of R is the set B , which is $\{2, 4, 9\}$.

The range of R is

$$\begin{aligned} & \{b \in B \mid \exists a \in A \text{ s.t. } a R b\} \\ &= \{b \in B \mid \exists a \in A \text{ s.t. } a^2 = b\} \\ &= \{4, 9\}. \end{aligned}$$

Let us now look at the inverse of a relation.

Definition 7.2.3: Inverse of a Relation

Let X and Y be sets and R be a relation from X to Y . The inverse of R , denoted R^{-1} , is the relation from Y to X where $y R^{-1} x$ if and only if $x R y$ for any $x \in X$ and $y \in Y$. In other words,

$$R^{-1} = \{(y, x) \in Y \times X \mid (x, y) \in R\}.$$

Example 7.2.4: The “is less than” Relation and its Inverse

In [Example 7.1.3](#), we looked at the relation R from the set $A = \{0, 1, 2\}$ to $B = \{1, 2, 3\}$ such that $x R y \iff x < y$ for all $x \in A$ and $y \in B$. In particular, we found that

$$R = \{(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3)\}.$$

Therefore

$$\begin{aligned} R^{-1} &= \{(1, 0), (2, 0), (3, 0), (2, 1), (3, 1), (3, 2)\} \quad (\text{Swap } x \text{ and } y) \\ &= \{(1, 0), (2, 0), (2, 1), (3, 0), (3, 1), (3, 2)\}. \quad (\text{Reordering}) \end{aligned}$$

As usual, we may show these relations diagrammatically.



Figure 7.4. Arrow Diagrams of R and R^{-1}

One sees that the arrow diagram for R^{-1} is formed by reversing the arrows on the arrow diagram of R .

Exercise 7.2: Properties of the “Divides” Relation

(Assume multiplication of integers is defined.)

Let $S = \{2, 3, 4, 5, 6, 7, 8\}$. Define a relation D on S such that

$$x D y \iff \exists m \in S \text{ s.t. } y = mx$$

for all $x, y \in S$.

- (i) List the elements in D . Hence list the elements in D^{-1} .
- (ii) List the elements in the
 - (a) domain of D ;
 - (b) codomain of D ; and
 - (c) range of D .

Exercise 7.3: Inverse Relation of Inverse Relation

Let A and B be sets. Let R be a relation from A to B . Prove that

$$(R^{-1})^{-1} = R.$$

7.3. Composition of Relations

We may sometimes want to find the overall relationship between two sets after a sequence of relations. This motivates the concept of composing relations.

Definition 7.3.1: Composition of Relations

Let A, B, C , and D be sets. Let R and S be relations from A to B and C to D respectively. The **composition of R with S** , written $S \circ R$, is a relation from A to D where

$$a(S \circ R)d \iff \exists x \in B \cap C \text{ s.t. } ((a R x) \wedge (x S d))$$

for all $a \in A$ and $d \in D$. In other words,

$$S \circ R = \{(a, d) \in A \times D \mid \exists x \in B \cap C \text{ s.t. } ((a, x) \in R \wedge (x, d) \in S)\}.$$

Example 7.3.2: The “lived in continent” Relation

In [Example 7.1.2](#), we defined the sets

$$\begin{aligned} M &= \{\text{Peano, Cantor, Dedekind, Russell, Euler, Hilbert}\}, \\ C &= \{\text{Italy, Switzerland, Russia, Germany, France}\}, \text{ and} \\ R &= \{(\text{Peano, Italy}), (\text{Cantor, Russia}), (\text{Cantor, Germany}), \\ &\quad (\text{Dedekind, Germany}), (\text{Euler, Switzerland}), \\ &\quad (\text{Euler, Germany}), (\text{Euler, Russia}), (\text{Hilbert, Germany})\}, \end{aligned}$$

where R is, in particular, a relation from M to C called the “lived in” relation. Now let the set A , a subset of all continents (“areas”), be

$$\{\text{Europe, North America, Asia}\}.$$

Define the relation S from C to A to be the “country of” relation

where

$$S = \{(\text{Italy, Europe}), (\text{Switzerland, Europe}), (\text{Russia, Asia}), (\text{Germany, Europe}), (\text{France, Europe})\}.$$

Thus the composition of R with S (i.e., $S \circ R$) is the “lived in continent” relation. In particular,

$$S \circ R = \{(\text{Peano, Europe}), (\text{Cantor, Europe}), (\text{Cantor, Asia}), (\text{Dedekind, Europe}), (\text{Euler, Europe}), (\text{Euler, Asia}), (\text{Hilbert, Europe})\}.$$

Like in [Example 7.1.2](#), we can show this relation diagrammatically.

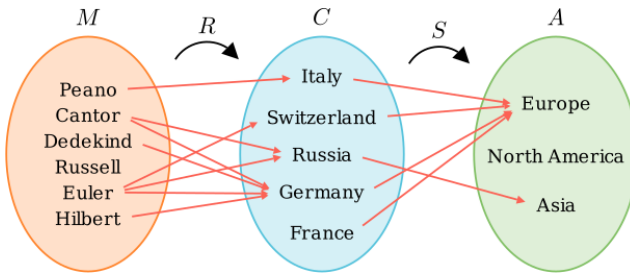


Figure 7.5. Arrow Diagram of R and S

Removing the intermediate set C yields the following arrow diagram.

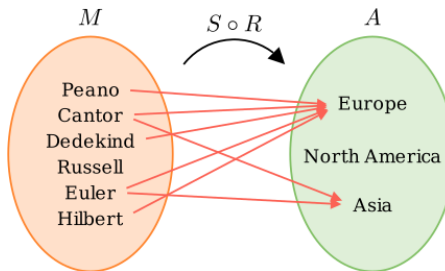


Figure 7.6. Arrow Diagram of $S \circ R$

Example 7.3.3: Composition of a Relation on a Set

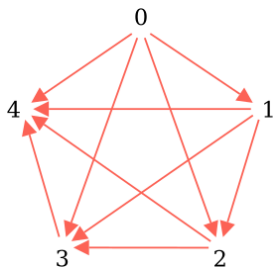
Let $A = \{0, 1, 2, 3, 4\}$. Define the relation R on A such that $x R y \iff x < y$ for all $x, y \in A$. Thus

$$R = \{(0, 1), (0, 2), (0, 3), (0, 4), (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}.$$

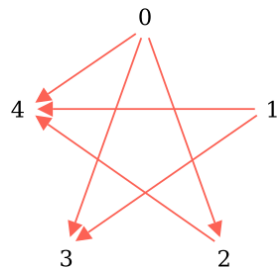
Subsequently, one sees that

$$R \circ R = \{(0, 2), (0, 3), (0, 4), (1, 3), (1, 4), (2, 4)\}.$$

We can draw the directed graphs of both relations.



(a) Directed Graph of R



(b) Directed Graph of $R \circ R$

Figure 7.7. Directed Graphs of R and $R \circ R$

Exercise 7.4: Properties of Compositions

Let A , B , and C be 3 distinct non-empty sets. Let R and S be relations from A to B and B to C respectively such that R and S are both non-empty. Which of the following statements is/are always true?

- (a) The domain of $S \circ R$ can be empty.
- (b) The domain of $S \circ R$ is A .
- (c) The domain of $S \circ R$ can be A .
- (d) The codomain of $S \circ R$ can be empty.
- (e) The codomain of $S \circ R$ is C .
- (f) The codomain of $S \circ R$ can be B .

- (g) The range of $S \circ R$ can be empty.
- (h) The range of $S \circ R$ is C .
- (i) The range of $S \circ R$ can be C .

Let us now consider the inverse of a composed relation.

Example 7.3.4: The “lived in continent” Relation’s Inverse

Recall in [Example 7.3.2](#) we have the sets

$$\begin{aligned} M &= \{\text{Peano, Cantor, Dedekind, Russell, Euler, Hilbert}\}, \\ C &= \{\text{Italy, Switzerland, Russia, Germany, France}\}, \text{ and} \\ A &= \{\text{Europe, North America, Asia}\} \end{aligned}$$

and the relations

$$\begin{aligned} R &= \{(\text{Peano, Italy}), (\text{Cantor, Russia}), (\text{Cantor, Germany}), \\ &\quad (\text{Dedekind, Germany}), (\text{Euler, Switzerland}), \\ &\quad (\text{Euler, Germany}), (\text{Euler, Russia}), (\text{Hilbert, Germany})\} \end{aligned}$$

and

$$\begin{aligned} S &= \{(\text{Italy, Europe}), (\text{Switzerland, Europe}), (\text{Russia, Asia}), \\ &\quad (\text{Germany, Europe}), (\text{France, Europe})\} \end{aligned}$$

from M to C and C to A respectively. We found that their composition was

$$\begin{aligned} S \circ R &= \{(\text{Peano, Europe}), (\text{Cantor, Europe}), (\text{Cantor, Asia}) \\ &\quad (\text{Dedekind, Europe}), (\text{Euler, Europe}), (\text{Euler, Asia}) \\ &\quad (\text{Hilbert, Europe})\}. \end{aligned}$$

The inverse is thus

$$\begin{aligned} (S \circ R)^{-1} &= \{(\text{Europe, Peano}), (\text{Europe, Cantor}), (\text{Asia, Cantor}) \\ &\quad (\text{Europe, Dedekind}), (\text{Europe, Euler}), (\text{Asia, Euler}) \\ &\quad (\text{Europe, Hilbert})\}. \end{aligned}$$

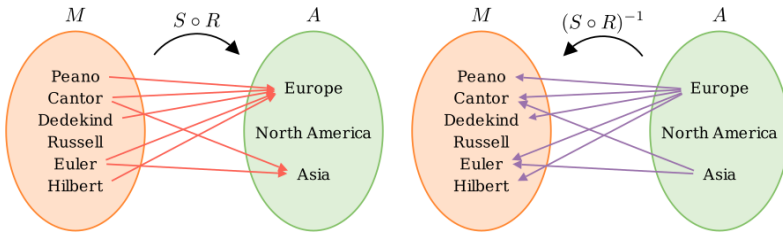


Figure 7.8. Arrow Diagrams of $S \circ R$ and $(S \circ R)^{-1}$

Now consider the relations R^{-1} and S^{-1} . Their arrow diagrams can be combined as one.

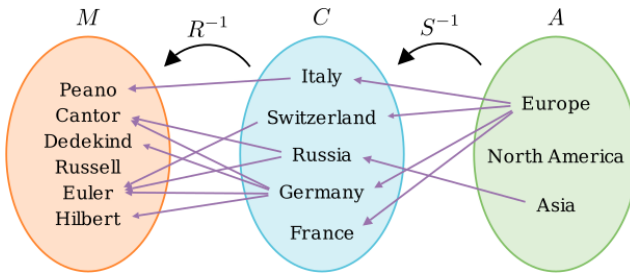


Figure 7.9. Arrow Diagram of R^{-1} and S^{-1}

From this diagram, we can see another method for finding $(S \circ R)^{-1}$ – by first using the inverse of S and then the inverse of R . In other words, we see that $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$.

This observation that $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$ is a key property of the inverse of a composition of two relations, informally called the *shoes and socks theorem*. The intuition behind this property is similar to that of putting on socks and then shoes. To remove them both, you first have to take off your shoes (i.e., the second thing that you put on) and then take off your socks (i.e., the first thing that you put on). Likewise, to undo the effect of two relations R and then S , we first have to undo S and then undo R .



Theorem 7.3.5: Shoes and Socks (Relations)

Let A, B, C , and D be sets, and let R and S be relations from A to B and C to D respectively. Then

$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}.$$

Proof. See [Exercise 7.6](#) (later). ■

Another property that the composition enjoys is associativity.

Theorem 7.3.6: Associativity of Composition (Relations)

Let A, B, C, D, E , and F be sets. Let R, S , and T be relations from A to B, C to D , and E to F respectively. Then

$$(T \circ S) \circ R = T \circ (S \circ R).$$

Proof. First, note that if either $B \cap C$ or $D \cap E$ is empty, then $(T \circ S) \circ R = T \circ (S \circ R) = \emptyset$. So assume neither $B \cap C$ nor $D \cap E$ are empty.

We note that $(T \circ S) \circ R \subseteq A \times F$ and $T \circ (S \circ R) \subseteq A \times F$. Now suppose $a \in A$ and $f \in F$. We see

$$\begin{aligned} (a, f) &\in (T \circ S) \circ R \\ \iff a ((T \circ S) \circ R) f \\ \iff \exists x \in B \cap C \text{ s.t. } (a R x \wedge x (T \circ S) f) \\ \iff \exists x \in B \cap C \text{ s.t. } (a R x \wedge (\exists y \in D \cap E \text{ s.t. } (x S y \wedge y T f))) \\ \iff \exists x \in B \cap C, y \in D \cap E \text{ s.t. } (a R x \wedge x S y \wedge y T f) \\ \iff \exists y \in D \cap E \text{ s.t. } ((\exists x \in B \cap C \text{ s.t. } (a R x \wedge x S y)) \wedge y T f) \\ \iff \exists y \in D \cap E \text{ s.t. } (a (S \circ R) y \wedge y T f) \\ \iff a (T \circ (S \circ R)) f \\ \iff (a, f) \in T \circ (S \circ R). \end{aligned}$$

So any element in $(T \circ S) \circ R$ is also in $T \circ (S \circ R)$ and vice versa, thereby meaning that $(T \circ S) \circ R = T \circ (S \circ R)$. ■

Since there is no distinction between $(T \circ S) \circ R$ and $T \circ (S \circ R)$, we can write “ $T \circ S \circ R$ ” without any ambiguity.

Exercise 7.5: Composition Commutativity?

Let A and B be sets. Let R and S be relations from A to B and B to A respectively. Which of the following is/are true?

- (a) It is always the case that $S \circ R = R \circ S$.
- (b) It is never the case that $S \circ R = R \circ R$.
- (c) It is possible for $S \circ R = R \circ S$.

Exercise 7.6: Shoes and Socks

Prove **Theorem 7.3.5**.

Exercise 7.7: Composition of Relation and its Inverse

Let A and B be sets and let R be a relation from A to B . Determine whether the following statement is true or false.

$$R^{-1} \circ R \subseteq \{(x, y) \in A^2 \mid x = y\}.$$

If it is true, prove it. Otherwise, give a counterexample.

CHAPTER 8

Functions

Picture a machine that takes in inputs and produces outputs. It could, for instance, double every number put into it. It may return the sum of all of its inputs, the product of all its inputs, or the largest number among all the inputs. That is what a function is – a relation that maps inputs to outputs. It is this kind of particular relation that we look at in this chapter.

8.1. What is a Function?

Definition 8.1.1: Function

A *function* f from a set X to a set Y is a relation from X to Y satisfying two properties.

F1. $\forall x \in X, \exists y \in Y \text{ s.t. } (x f y)$

F2. $\forall x \in X, \forall y_1, y_2 \in Y, ((x f y_1) \wedge (x f y_2)) \implies (y_1 = y_2)$

We denote such a function by $f : X \rightarrow Y$.

The condition **F2** essentially says that if a single $x \in X$ is related to two (or more) elements in Y , then they must be equal. This can be thought of as the “uniqueness” property of a function.

We can succinctly write both **F1** and **F2** in one statement using the *uniqueness quantifier* $\exists!$. A statement of the form $\exists! x \text{ s.t. } P(x)$, where $P(x)$ is a predicate, is logically equivalent to the statement

$$\exists x \text{ s.t. } (P(x) \wedge (\forall y (P(y) \implies x = y))),$$

i.e. there exists an x that satisfies the predicate and any y satisfying the predicate must be equal to x . Thus, we can combine **F1** and **F2** into the single condition

$$\forall x \in X, \exists! y \in Y \text{ s.t. } x f y.$$

This formulation of the function condition is what we will be using most of the time.

As mentioned, we may think of a function as a machine or process that takes in inputs and produce outputs. In particular, the ‘machine’ for the function $f : X \rightarrow Y$ is a machine that **must** produce an output for any

valid input (i.e., an input that comes from X) that is provided to it, and it must produce the same output for the same input that is provided to it.

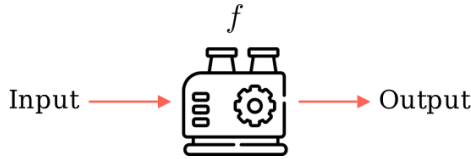


Figure 8.1. Function as a Machine

Since there is only one output for any given input, we can denote it unambiguously.

Definition 8.1.2: Image of an Element Under a Function

Let X and Y be sets and $f : X \rightarrow Y$ be a function. Suppose $x \in X$. The unique element of Y that is related to x under f is called the *image of x under f* and is denoted $f(x)$.

Let us look at some examples.

Example 8.1.3: The Successor Function

Let's look at a function that we already looked before, the successor function. Before, we defined it for sets. We can restrict it to the natural numbers. That is, the function $S : \mathbb{N} \rightarrow \mathbb{N}$ is defined by $S(n) = n \cup \{n\}$ (remember, we use the counter set for any natural number n).

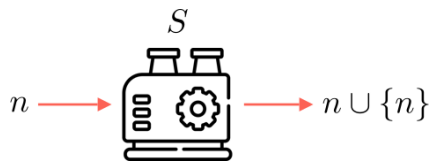


Figure 8.2. Successor Function as a Machine

Example 8.1.4: The Identity Function

Let A be a set. The identity function $\text{id}_A : A \rightarrow A$ is the function such that $\text{id}_A(x) = x$ for all $x \in A$. That is, this function does not transform the input into it in any way.

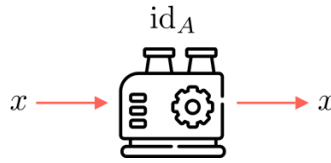


Figure 8.3. Identity Function as a Machine

Let us look at some examples of images under the identity function. Suppose $A = \{1, 2, 3\}$. Then

- $\text{id}_A(1) = 1$;
- $\text{id}_A(2) = 2$; and
- $\text{id}_A(3) = 3$.

Exercise 8.1: Empty Function?

Suppose A is a set. Consider the relation f from $\emptyset$ to A where $f = \emptyset$. Is f a function?

Let's now turn our attention to determining the equality of two functions. Recall that two relations R and S from the set A to the set B and the set C to the set D respectively (i.e., $R \subseteq A \times B$ and $S \subseteq C \times D$) are equal if and only if $A = C$, $B = D$, and $R = S$ as sets (**Definition 7.1.6**). Since functions are also relations, this definition applies to functions.

Definition 8.1.5: Equality of Functions

Let A , B , C , and D be sets. The functions $f : A \rightarrow B$ and $g : C \rightarrow D$ are equal, denoted $f = g$, if and only if $A = C$, $B = D$, and $f = g$ as sets.

However, we have a more applicable result for functions.

Theorem 8.1.6: Equality of Functions, More Easily

Let A , B , C , and D be sets. Let the functions $f : A \rightarrow B$ and $g : C \rightarrow D$. Then $f = g$ if and only if $A = C$, $B = D$, and $f(x) = g(x)$ for all $x \in A$.

Proof. For the forward direction, suppose $f = g$ (as functions). Thus $A = C$, $B = D$, and $f = g$ as sets. Suppose $x \in A$ and $y \in B$ are arbitrary.

Note

$$\begin{aligned}
 y &= f(x) \\
 \iff (x, y) &\in f && \text{(Definition 8.1.2)} \\
 \iff (x, y) &\in g && \text{(as } f = g \text{ as sets)} \\
 \iff y &= g(x) && \text{(Definition 8.1.2)}
 \end{aligned}$$

which means $f(x) = g(x)$ for any $x \in A$.

For the reverse direction, suppose $A = C$, $B = D$, and $f(x) = g(x)$ for all $x \in A$. Then

$$\begin{aligned}
 (x, y) &\in f \\
 \iff y &= f(x) && \text{(Definition 8.1.2)} \\
 \iff y &= g(x) && \text{(as } f(x) = g(x) \forall x \in A) \\
 \iff (x, y) &\in g && \text{(Definition 8.1.2)}
 \end{aligned}$$

which means that $f = g$ as sets. Thus $f = g$ by definition of equality of functions. ■

There are three important sets related to a function.

Definition 8.1.7: Domain, Codomain, and Range (Functions)

Let X and Y be sets and let $f : X \rightarrow Y$ be a function.

- The *domain* of f is the set X .
- The *codomain* of f is the set Y .
- The *range* of f is the set $\{y \in Y \mid \exists x \in X \text{ s.t. } y = f(x)\}$.

Remark: Image of a Function

Some authors may call the range of a function the *image* of the function. However, this would likely lead to confusion with the image of an element under a function, so we will stick with “range”.

Example 8.1.8: The “Character of” Function

Let N , a subset of the natural numbers, be $\{1, 2, 3\}$ and C , a subset of all characters, be $\{+, a, b, c, d, e\}$. We define a function $f : N \rightarrow C$ (called the “character of” function) that maps a natural number to a character. In particular, let

$$f = \{(1, a), (2, b), (3, c)\}.$$

We can draw an arrow diagram for this function.

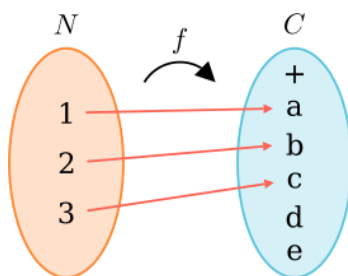


Figure 8.4. Arrow Diagram of f

For f , we see that

- its domain is N ;
- its codomain is C ; and
- its range is $\{a, b, c\}$.

Example 8.1.9: The Successor Function

Let's look at the successor function restricted to the natural numbers again, i.e. the function $S : \mathbb{N} \rightarrow \mathbb{N}$. We see that

- its domain is $\mathbb{N}$;
- its codomain is also $\mathbb{N}$; and
- its range is $\mathbb{N} \setminus \{0\}$.

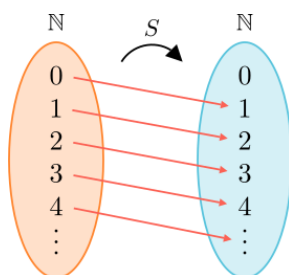
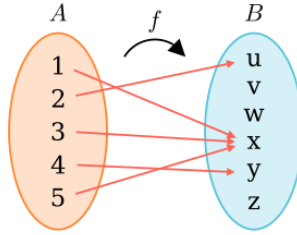


Figure 8.5. Arrow Diagram of S

Exercise 8.2: A Function's Properties

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{u, v, w, x, y, z\}$. Define the function $f : A \rightarrow B$ such that it has the following arrow diagram.



- (a) Find $f(1)$, $f(2)$, and $f(4)$.
- (b) What is the domain, codomain, and range of f ?
- (c) What is the set representation of f ?

8.2. Axiom Schema of Replacement

We now look at the axiom schema of replacement.

8.2.1. The Need for the Axiom Schema

We often define functions with a rule. For example, the successor function that is restricted to the natural numbers, i.e. $S : \mathbb{N} \rightarrow \mathbb{N}$, is defined by $S(n) = n \cup \{n\}$ for all $n \in \mathbb{N}$. Using relation notation, we can write this as

$$n S m \iff m = n \cup \{n\}.$$

Another example would be the "character of" function in [Example 8.1.8](#), where we defined a function $f : N \rightarrow C$ (where $N = \{1, 2, 3\}$ and $C = \{+, a, b, c, d, e\}$) such that $f(1) = a$, $f(2) = b$, and $f(3) = c$. We can write this as

$$x f y \iff (x = 1 \wedge y = a) \vee (x = 2 \wedge y = b) \vee (x = 3 \wedge y = c).$$

What these two examples show is that a function $f : X \rightarrow Y$ can be defined as

$$x f y \iff P(x, y)$$

where $P(x, y)$ is a binary predicate.

Essentially, what we want to ensure is that every function's image is indeed a set. Unfortunately, we are unable to quantify over definable functions in our language of axiomatic set theory. So, we are stuck with using an axiom schema that should achieve the same effect.

8.2.2. Forming the Axiom Schema

Let's start with our goal, expressed in a high-level manner: for every function, its range exists. As explained earlier, we are not able to quantify over functions, but we are able to quantify over WFFs (which include predicates).

Suppose we have a function $f : X \rightarrow Y$ such that

$$x f y \iff \phi(x, y, X)$$

where ϕ is a WFF. In essence, the WFF ϕ is the **rule** of f – it defines which elements in Y could be an image of an element from X . Note that X needs to be a free variable in ϕ because the function f may use it to determine which elements are within its range. Now recall that f is a function if and only if

$$\forall x \in X, \exists! y \in Y \text{ s.t. } x f y.$$

Thus

$$\forall x \in X, \exists! y \in Y \text{ s.t. } \phi(x, y, X)$$

is the condition that must be satisfied for f to be a function. In other words, using semi-formal language, we can write that condition above as

$$\forall x(x \in X \implies \exists! y(y \in Y \wedge \phi(x, y, X))).$$

Now, for our purposes, we actually want to make the condition a little weaker by removing the need for y to come from a known set Y . This is because the axiom itself will bring into existence a set that contains the images of all the elements in X , so y just needs to exist. Thus, the antecedent for our condition in the axiom schema of replacement is

$$\forall x(x \in X \implies \exists! y \phi(x, y, X)).$$

We turn our attention to the consequent of the axiom schema of replacement – that the range of the function f exists. In other words, there must exist a set containing all the images of the elements in X , and that any image of an element in X must be in that set. Note that if y is an image of an element in X , that means that there exists an $x \in X$ such that $y = f(x)$ (i.e., $x f y$, or $\phi(x, y, X)$ is true). We can express this idea as

$$\exists R \text{ s.t. } \forall y(y \in R \iff \exists x \in X \text{ s.t. } \phi(x, y, X))$$

where one can interpret R as the range of f . Formally, we can write this as

$$\exists R \forall y (y \in R \iff \exists x (x \in X \wedge \phi(x, y, X))).$$

There are a few more tweaks needed before we can finally arrive at the axiom schema of replacement. First, let us allow ϕ to have other free variables other than just x, y and X . In particular, let $w_1, w_2, \dots, w_n$ be free variables in ϕ . So we can write the antecedent and consequent of the axiom schema of replacement semi-formally as

$$\forall x (x \in X \implies \exists! y \phi(x, y, w_1, \dots, w_n, X))$$

and

$$\exists R \forall y (y \in R \iff \exists x (x \in X \wedge \phi(x, y, w_1, \dots, w_n, X)))$$

for all $X, w_1, w_2, \dots, w_n$. We then combine them into one statement, resulting in the semi-formal statement

$$\begin{aligned} &\forall X \forall w_1 \forall w_2 \dots \forall w_n (\\ &\quad \forall x (x \in X \implies \exists! y \phi(x, y, w_1, w_2, \dots, w_n, X)) \\ &\quad \implies \exists R \forall y (y \in R \iff \exists x (x \in X \wedge \phi(x, y, w_1, w_2, \dots, w_n, X))) \\ &). \end{aligned}$$

Lastly, recalling that the uniqueness quantifier is defined such that

$$\exists! x \psi(x) \equiv \exists x (\psi(x) \wedge \forall y (\psi(y) \implies y = x)),$$

we can finally write state the axiom schema of replacement formally.

Axiom 8.2.1: Axiom Schema of Replacement

Let ϕ be a WFF. Then

$$\begin{aligned} &\forall X \forall w_1 \forall w_2 \dots \forall w_n (\\ &\quad \forall x (x \in X \implies \exists y (\\ &\quad \quad \phi(x, y, w_1, \dots, w_n, X) \\ &\quad \quad \wedge \forall z (\phi(x, z, w_1, \dots, w_n, X) \implies z = y) \\ &\quad)) \\ &\quad \implies \exists R \forall y (y \in R \iff \exists x (x \in X \wedge \phi(x, y, w_1, \dots, w_n, X))) \\ &). \end{aligned}$$

where R is not free in ϕ .

This is undoubtedly one of the most involved axiom schema within ZFC, but also one of the most powerful. As we will soon see, functions play a crucial part when working with numbers.

8.3. Well-Defined Functions

Axiom 8.2.1 allows us to assert that the range of any function exists. However, care must be taken to ensure that the rule of the function indeed makes the function a function, and not just a relation. Specifically, referring to **Definition 8.1.1**, a ‘function’ that fails to satisfy either **F1** or **F2** (or both) is called a *ill-defined* function, and functions that satisfy both **F1** and **F2** are called *well-defined* functions (which are actual functions).

Example 8.3.1: A Possibly Ill-Defined Function

Let S_1 and S_2 be non-empty sets, and let $S = S_1 \cup S_2$. Let f be a relation from S to $\{1, 2\}$ such that

$$f = \{(s, 1) \mid s \in S_1\} \cup \{(s, 2) \mid s \in S_2\}.$$

In other words,

$$x f y \iff (x \in S_1 \wedge y = 1) \vee (x \in S_2 \wedge y = 2).$$

Is this a (well-defined) function?

Suppose $S_1 \cap S_2 = \emptyset$. This means that any element in S_1 would have an image of 1, and any element in S_2 would have an image of 2. In particular, every element from S will have one and only one image. Thus f is indeed a function. In fact, we can write it using a *piecewise definition* as

$$f(x) = \begin{cases} 1 & \text{if } x \in S_1 \\ 2 & \text{if } x \in S_2 \end{cases}$$

However, what happens when $S_1 \cap S_2 \neq \emptyset$? Suppose $S_1 = \{1, 2\}$ and $S_2 = \{2, 3\}$. Then we see

$$f = \{(1, 1), (2, 1), (2, 2), (3, 2)\}.$$

In particular, we see that $2 f 1$ and $2 f 2$ simultaneously. If we were to abuse some notation, this means that $f(2) = 1$ and $f(2) = 2$ simultaneously. This clearly violates **F2** in **Definition 8.1.1**, and thus f is not a (well-defined) function in this case (i.e., f is ill-defined as a function).

In conclusion, for $f : S \rightarrow \{1, 2\}$ to be a function such that

$$f(x) = \begin{cases} 1 & \text{if } x \in S_1 \\ 2 & \text{if } x \in S_2 \end{cases}$$

we must have $S_1 \cap S_2 = \emptyset$.

Exercise 8.3: Determining Function Validity

(Assume the addition and multiplication of natural numbers is defined.)

For each of the following relations, determine if they are an ill-defined function or a well-defined function.

- (a) The relation f from $\{1, 2, 3\}$ to $\{1, 8\}$ such that

$$x f y \iff y = x^3$$

for all $x \in \{1, 2, 3\}$ and $y \in \{1, 8\}$.

- (b) The relation g from $\mathbb{N}^2$ to $\mathbb{N}$ such that

$$x g y \iff \exists m, n \in \mathbb{N} \text{ s.t. } ((x = (m, n)) \wedge (y = m + n))$$

for all $x \in \mathbb{N}^2$ and $y \in \mathbb{N}$.

- (c) The relation h from $\mathbb{N}$ to $\mathbb{N}^2$ such that

$$x h y \iff \exists m, n \in \mathbb{N} \text{ s.t. } ((x = m + n) \wedge (y = (m, n)))$$

for all $x \in \mathbb{N}$ and $y \in \mathbb{N}^2$.

8.4. Injections, Surjections, Bijections

We now look at two essential properties that functions could satisfy – injectivity and surjectivity. Bijectivity is just the combination of both injectivity and surjectivity, and it results in the elements of the domain and codomain matching up exactly. Such a bijective function can result in a new function that undoes the original function, called the inverse function.

8.4.1. Injectivity

A function could send multiple elements in the domain to the same element in the codomain. The function's arrow diagram would have two arrows originating at different elements and terminating at the same element. Such a function is *non-injective*. It would be nice for a function to have no two distinct elements in the domain map to the same element in the codomain. Diagrammatically, this means that no two arrows originating at different elements terminate at the same element. For example, the “character of” function in [Example 8.1.8](#) satisfies this property. Such a function is *injective*.

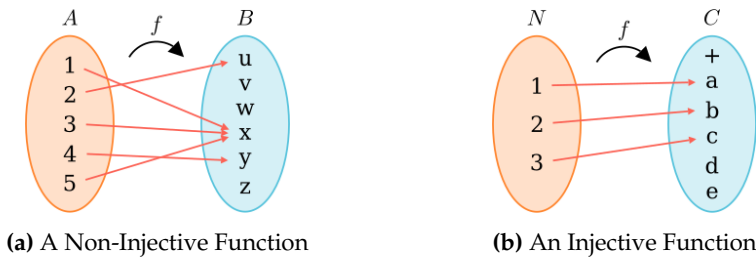


Figure 8.6. Injectivity and Non-injectivity

We can express the condition for a function to be injective formally.

Definition 8.4.1: Injectivity

Let X and Y be sets, and let $f : X \rightarrow Y$ be a function. Then f is *injective* (or *one-to-one*) if and only if

$$f(x_1) = f(x_2) \implies x_1 = x_2$$

for all $x_1, x_2 \in X$. Equivalently,

$$x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$$

for all $x_1, x_2 \in X$.

Let us look at some examples.

Example 8.4.2: Injectivity of Successor Function on Naturals

Consider the successor function restricted to the natural numbers $S : \mathbb{N} \rightarrow \mathbb{N}$ given by $S(n) = n \cup \{n\}$. We show that S is injective

using the contrapositive definition of injectivity.

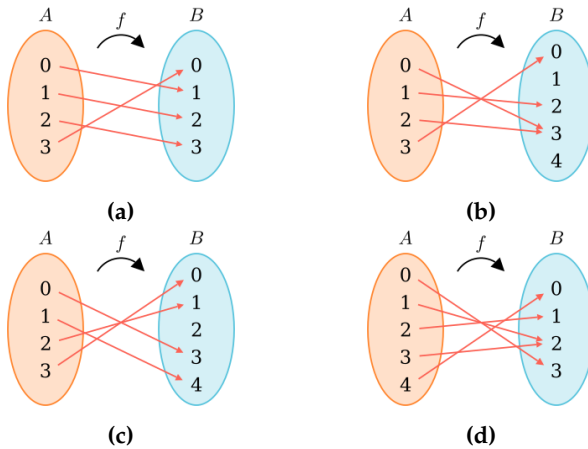
Proof. Suppose $n_1, n_2 \in \mathbb{N}$ such that $n_1 \neq n_2$. So the counter sets of n_1 and n_2 have differing number of elements, and hence $n_1 \cup \{n_1\} \neq n_2 \cup \{n_2\}$ because $n_1 \cup \{n_1\}$ and $n_2 \cup \{n_2\}$ have different numbers of elements. ■

Example 8.4.3: An Example of Non-Injectivity

Consider the function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(x) = (\text{last digit of } x)$. One clearly sees that f is not injective since $f(1) = f(11) = 1$.

Exercise 8.4: Injective Detective

For each function described by the arrow diagrams below, determine whether they are injective or not.



Exercise 8.5: Hash Functions

Let S be a set. Consider the function $H : S \rightarrow \mathbb{N}$ such that $H(x) = (\text{last two digits of } x)$. Suppose we have a table of 100 slots, each numbered 0 to 99, where we place the element $x \in S$ into the slot $H(x)$. A *collision* occurs if two elements from S map to the same slot in the table.

- (i) Describe the condition on H for no collisions to occur.
- (ii) Suppose all elements of S are placed into the table. Which of

the following possibilities of S results in no collisions?

- (a) $S = \{12, 34, 56\}$
- (b) $S = \{12, 34, 56, 78, 910, 1112, 1314\}$
- (c) $S = \{18323, 389892, 92349589, 8942894281\}$
- (d) $S = \{128, 7327, 83279, 8934289, 842892, 824379490\}$

Remark

Such a function as described above can be considered a *hash function* and the table a *hash table*.

8.4.2. Surjectivity

Recall that a relation from the set X to the set Y may not have its domain being X , but a function has to have its domain to be the entire set Y . Likewise, a function's range may not be the whole set Y , but it would be nice to have such a function with its range as the codomain. This is the appeal of *surjective* functions.

Definition 8.4.4: Surjectivity

Let X and Y be sets, and let $f : X \rightarrow Y$ be a function. Then f is *surjective* (or *onto*) if and only if

$$\forall y \in Y, \exists x \in X \text{ s.t. } y = f(x).$$

In other words, every $y \in Y$ has a corresponding $x \in X$, called its *preimage*, such that $y = f(x)$.

As mentioned, an important property of surjective functions is that their range is their codomain.

Proposition 8.4.5: Surjective Functions' Range is their Codomain

Let X and Y be sets, and let $f : X \rightarrow Y$ be a surjective function. Then the range of f is Y .

Proof. Recall that the range of f is the set

$$\{y \in Y \mid \exists x \in X \text{ s.t. } y = f(x)\}.$$

Since f is surjective, every $y \in Y$ has a corresponding preimage $x \in X$ such that $y = f(x)$. Thus the range of f is exactly Y . ■

Example 8.4.6: A Surjective Function

Let's modify the "character of" function in [Example 8.1.8](#) to make it surjective. Let N , a subset of the natural numbers, be $\{1, 2, 3, 4, 5\}$ and C , a subset of all characters, be $\{b, c, d\}$. Define the function $f : N \rightarrow C$ such that

$$f = \{(2, b), (3, c), (4, d)\}.$$

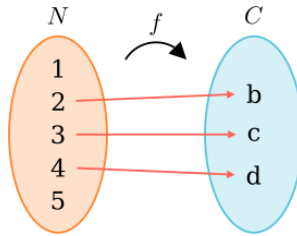


Figure 8.7. Arrow Diagram of f

We clearly see that every element in the codomain C has a corresponding preimage in N , and thus f is surjective.

Example 8.4.7: A Non-Surjective Function

Consider again the successor function restricted to the natural numbers again, i.e. the function $S : \mathbb{N} \rightarrow \mathbb{N}$.

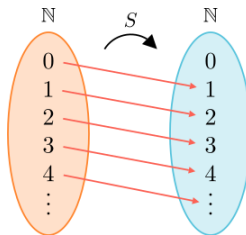
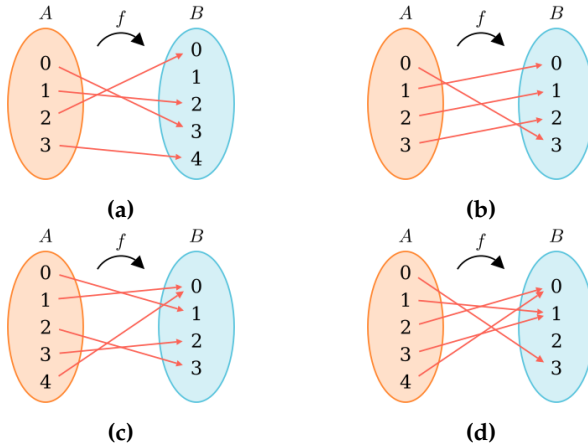


Figure 8.8. Arrow Diagram of S

We can see in the diagram that $0 \in \mathbb{N}$ has no preimage under S . Thus, this means that 0 is not in the range of S , which tells us that S is not surjective.

Exercise 8.6: Surjective Detective

For each function described by the arrow diagrams below, determine whether they are surjective or not.

**8.4.3. Bijectivity**

Now, let us consider a function $f : X \rightarrow Y$ which is both injective and bijective. Such functions are so important that we have a name for them – *bijections*.

Definition 8.4.8: Bijectivity

Let X and Y be sets, and let $f : X \rightarrow Y$ be a function. Then f is *bijective* (or that f is a *one-to-one correspondence*) if and only if it is both injective and surjective.

Why are they important? For a bijection $f : X \rightarrow Y$, any $x \in X$ has a unique corresponding element $y \in Y$ such that $y = f(x)$ (since f is a function). Also, any $y \in Y$ has a corresponding preimage $x \in X$ such that $y = f(x)$ (since f is surjective) and this x is that only one that results in y (since f is injective). In essence, a bijection creates a pairing between the elements of X and Y that matches each element of X exactly one element of Y (and vice versa).

Example 8.4.9: The “Decimal to Binary” Function

A binary string consists of only ones and zeros. Let N , a subset of all

natural numbers written in decimal (i.e., base 10), be $\{0, 1, 2, 3, 4, 5\}$. Let B , a subset of all binary strings, be $\{000, 001, 010, 011, 100, 101\}$. We may define the function $f : N \rightarrow B$ such that, in effect, it converts the decimal number in N into a binary string in B . That is,

$$f = \{(0, 000), (1, 001), (2, 010), (3, 011), (4, 100), (5, 101)\}.$$

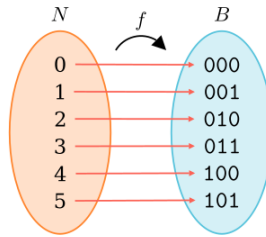


Figure 8.9. Arrow Diagram of f

We clearly see from the arrow diagram that

- each element of B has no more than one arrow pointing to it, which means f is injective; and
- every element of B has an arrow pointing to it, which means f is surjective,

and thus f is bijective (which means it is a bijection).

Example 8.4.10: The String Reversal Function

Suppose $\mathcal{L}$ is a formal language with alphabet Σ . Let Σ^* denote the set of all strings from using symbols from Σ . Define the function $r : \Sigma^* \rightarrow \Sigma^*$ where, for any string $s \in \Sigma^*$, $r(s)$ is the string obtained by reversing s . For example, for $\Sigma = \{a, b, c\}$,

- $r(\varepsilon) = \varepsilon$ (since we do not have any symbols to reverse);
- $r(a) = a$, and likewise $r(b) = b$ and $r(c) = c$;
- $r(ab) = ba$, and likewise $r(ac) = ca$ and $r(bc) = cb$;
- $r(abcaba) = abacba$; and
- $r(cacab) = bacac$.

We show that r is actually a bijection.

Proof. We are to show two things: r is injective and r is surjective.

- **Injective:** Suppose $s_1, s_2 \in \Sigma^*$ such that $r(s_1) = r(s_2)$. What does this mean? This means that the string obtained when writing the symbols in s_1 in reverse order is the same as the string obtained when writing the symbols in s_2 in reverse order. But if s_1 and s_2 are the same when written in reverse, they have to be the same to start with. In other words, we must have $s_1 = s_2$ and thus r is indeed injective.
- **Surjective:** Suppose s' is any string in Σ^* . By definition of r , we know that $r(s')$ is the string obtained by reversing s' . Note that when a reversed string is reversed again, we obtain the original string, i.e. $r(r(s')) = s'$. So let $s = r(s')$, noting that $s \in \Sigma^*$. Then $r(s) = r(r(s')) = s'$, and thus r is indeed surjective.

Since r is both injective and surjective, it is thus bijective. ■

Bijections create a function that undoes the effect of the original function. This function is called the *inverse function*, and only bijections have them.

Definition 8.4.11: Invertible Function and Inverse Function

Let X and Y be sets, and let $f : X \rightarrow Y$ be a function. Then f is an *invertible function* if and only if there exists a function $g : Y \rightarrow X$ such that

$$\begin{aligned} g(f(x)) &= x \text{ for all } x \in X \text{ and} \\ f(g(y)) &= y \text{ for all } y \in Y. \end{aligned}$$

In this case g is called the *inverse function of f* , and is usually denoted f^{-1} .

Theorem 8.4.12: Invertible If and Only If Bijective

Let X and Y be sets, and let $f : X \rightarrow Y$ be a function. Then f is invertible if and only if f is a bijection.

Proof. For the forward direction, suppose f is invertible. This means there exists $g : Y \rightarrow X$ such that $g(f(x)) = x$ for all $x \in X$ and $f(g(y)) = y$ for all $y \in Y$. We are to prove that f is both injective and surjective.

- **Injective:** Suppose $x_1, x_2 \in X$ such that $f(x_1) = f(x_2)$. Thus $g(f(x_1)) = g(f(x_2))$, which means $x_1 = x_2$ by definition of g .
- **Surjective:** Suppose $y \in Y$. We know that $g(y) \in X$ by definition of g . Furthermore we know that $f(g(y)) = y$ by definition of g . So the preimage of y under f is $g(y)$, which shows that f is surjective.

As f is both injective and surjective, it is thus bijective.

For the reverse direction, let $f : X \rightarrow Y$ be a bijection. Since a function is a relation, thus f is a subset of $X \times Y$. Considering f as a relation, let $g = f^{-1}$, i.e. g is the inverse relation of f . We show that g is the inverse function of f .

First, to prove that $g = f^{-1}$ is indeed a function. We need to prove **F1** and **F2** (see [Definition 8.1.1](#)). In the context of g , they can be written as follows.

F1. For all $y \in Y$, there exists $x \in X$ such that $y g x$.

F2. For all $y \in Y$, if $y g x_1$ and $y g x_2$ for some $x_1, x_2 \in X$, we must have $x_1 = x_2$.

We prove them one by one, using the properties of f .

- To prove **F1**, suppose $y \in Y$. Since f is surjective, we know that there exists a $x \in X$ such that $f(x) = y$ (i.e. $x f y$). In particular we know $y f^{-1} x$ by definition of inverse relation; since $g = f^{-1}$ we must have $y g x$. So there is an $x \in X$ such that $y g x$ (and in particular it is the $x \in X$ such that $f(x) = y$).
- To prove **F2**, suppose $y \in Y$ and $x_1, x_2 \in X$ such that $y g x_1$ and $y g x_2$. By our working above, we know that $x_1 f y$ and $x_2 f y$, i.e. $f(x_1) = y$ and $f(x_2) = y$. Thus $f(x_1) = f(x_2)$. Since f is injective, thus $x_1 = x_2$.

Thus $g = f^{-1}$ is a function.

Now we need to prove that $g(f(x)) = x$ for all $x \in X$. As above we know there exists a $y \in Y$ such that $x f y$ and $y g x$. Clearly this means $g(f(x)) = x$ for all $x \in X$.

Finally, we need to prove that $f(g(y)) = y$ for all $y \in Y$. As above we know there exists a $x \in X$ such that $y g x$ and $x f y$. Clearly this means $f(g(y)) = y$ for all $y \in Y$.

Hence we see f is invertible with g being the inverse function of f . ■

What the above theorem states is that the inverse function of $f : X \rightarrow Y$ is exactly the function induced by the inverse relation of f .

Example 8.4.13: The “Binary to Decimal” Function

Let us consider the “decimal to binary” function defined in [Example 8.4.9](#), where

$$f = \{(0, 000), (1, 001), (2, 010), (3, 011), (4, 100), (5, 101)\}.$$

So the inverse function of f is

$$f^{-1} = \{(000, 0), (001, 1), (010, 2), (011, 3), (100, 4), (101, 5)\}.$$

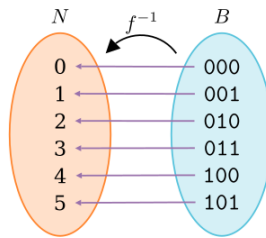


Figure 8.10. Arrow Diagram of f^{-1}

Example 8.4.14: The String Reversal Function

Consider the string reversal function $r : \Sigma^* \rightarrow \Sigma^*$ defined in [Example 8.4.10](#). One notes that it is its own inverse, since for any string $s \in \Sigma^*$,

$$\begin{aligned} r(r(s)) &= \text{the reversal of the reversal of } s \\ &= s. \end{aligned}$$

Exercise 8.7: Non-Trivial Bijection

Let the set $A = \{1, 2, 3, 4\}$. Find a function $f : A \rightarrow A$ such that f is a bijection but $f^{-1} \neq \text{id}_A$.

8.5. Composition of Functions

We may think of function composition as machines feeding into each other – the output of one machine is the input of the other.

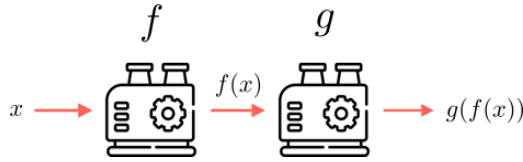


Figure 8.11. Composition of Functions f and g

In the above figure, we have two functions, namely f and g . Any input x that is fed into this machine will first pass through f , resulting in $f(x)$, and then pass through g , with the result being $g(f(x))$.

Definition 8.5.1: Composition of Functions

Let A , B , C , and D be sets. Let $f : A \rightarrow B$ and $g : C \rightarrow D$ be functions. The **composition of f and g** , denoted $g \circ f$, is the function $h : A \rightarrow D$ such that

$$h(x) = g(f(x))$$

for all $x \in A$, and is only defined if and only if C is a subset of the range of f (i.e., the domain of g is a subset of the range of f).

This definition can be displayed diagrammatically.

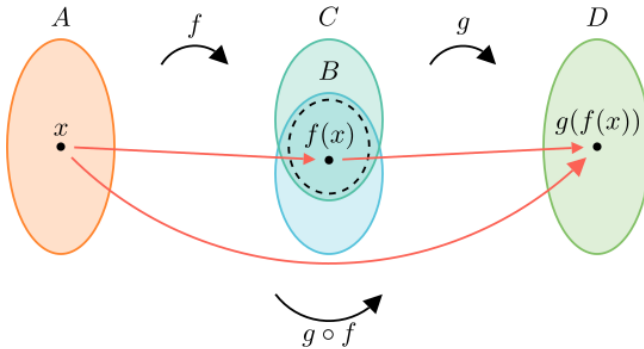


Figure 8.12. Composition of Functions f and g

In the diagram, any element in the domain of $g \circ f$ (i.e., any element in A) will first be mapped to the range of f (denoted by the dotted circle within the intersection of B and C), and then mapped to the final element in D (which is the codomain of $g \circ f$).

One might wonder whether the order of taking the compositions matter. It turns out that it does not.

Theorem 8.5.2: Associativity of Composition (Functions)

Let A, B, C, D, E , and F be sets. Let $f : A \rightarrow B$, $g : C \rightarrow D$, and $h : E \rightarrow F$ be functions such that $g \circ f$ and $h \circ g$ are defined. Then

$$(h \circ g) \circ f = h \circ (g \circ f).$$

Proof. Note that if either f , g , or h is the empty function (see [Exercise 8.1](#)), then $(h \circ g) \circ f = h \circ (g \circ f) = \emptyset$. So we may assume that f , g , and h are not empty functions.

Considering f and g as relations for a moment, the fact that $g \circ f$ is a function means that C is a subset of the range of f (which is a subset of B). Since f and g are not empty functions, thus we must have $B \cap C \neq \emptyset$. Likewise we see $D \cap E \neq \emptyset$ by using the same argument for g and h .

Therefore the claim is readily proven by using [Theorem 7.3.6](#) on f , g , and h when we consider them as relations. ■

Similar to relations, since there is no distinction between $(h \circ g) \circ f$ and $h \circ (g \circ f)$, we can write “ $h \circ g \circ f$ ” without any ambiguity.

The most trivial composition that we can consider is a composition with the identity function.

Proposition 8.5.3: Composition with Identity Function

Let X and Y be sets, and let $f : X \rightarrow Y$ be a function. Then

$$f \circ \text{id}_X = f \text{ and } \text{id}_Y \circ f = f.$$

Proof. Recall that $\text{id}_A(a) = a$ for all sets A and $a \in A$. So for any $x \in X$ we have

$$\begin{aligned} (f \circ \text{id}_X)(x) &= f(\text{id}_X(x)) && \text{(Definition of composition)} \\ &= f(x) && \text{(Definition of identity function)} \end{aligned}$$

which means $f \circ \text{id}_X = f$. Also, we have

$$\begin{aligned} (\text{id}_Y \circ f)(x) &= \text{id}_Y(f(x)) && \text{(Definition of composition)} \\ &= f(x) && \text{(Definition of identity function)} \end{aligned}$$

for any $x \in X$, meaning that $\text{id}_Y \circ f = f$. ■

A more interesting result can be found when we compose an invertible function with its inverse.

Example 8.5.4: Composing an Invertible Function with Its Inverse

Let $A = \{1, 2, 3\}$, $B = \{x, y, z\}$, and the function $f : A \rightarrow B$ such that

$$f = \{(1, z), (2, x), (3, y)\}.$$

Clearly f is a bijection, which means it has an inverse. In particular, its inverse is

$$f^{-1} = \{(x, 2), (y, 3), (z, 1)\}.$$

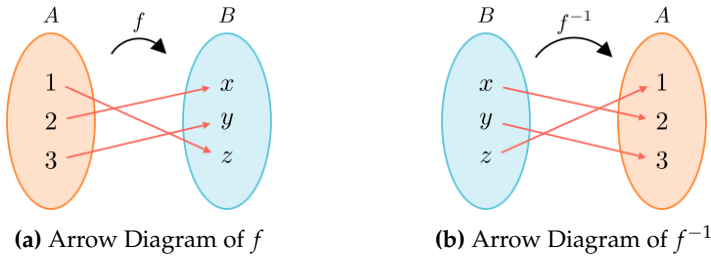


Figure 8.13. Arrow Diagrams of f and f^{-1}

Let us consider the composition $f^{-1} \circ f$. It can be found by following the arrows from A to B via f and back to A via f^{-1} . So we see that

$$\begin{aligned} (f^{-1} \circ f)(1) &= (f^{-1}(f(1))) = f^{-1}(z) = 1, \\ (f^{-1} \circ f)(2) &= (f^{-1}(f(2))) = f^{-1}(x) = 2, \text{ and} \\ (f^{-1} \circ f)(3) &= (f^{-1}(f(3))) = f^{-1}(y) = 3. \end{aligned}$$

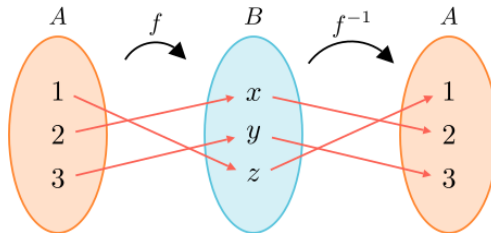


Figure 8.14. Arrow Diagram of $f^{-1} \circ f$

We can do the same for the composition $f \circ f^{-1}$. It can be found by

following the arrows from B to A via f^{-1} and back to B via f . Thus

$$\begin{aligned}(f \circ f^{-1})(x) &= (f(f^{-1}(x))) = f(2) = x, \\(f \circ f^{-1})(y) &= (f(f^{-1}(y))) = f(3) = y, \text{ and} \\(f \circ f^{-1})(z) &= (f(f^{-1}(z))) = f(1) = z.\end{aligned}$$

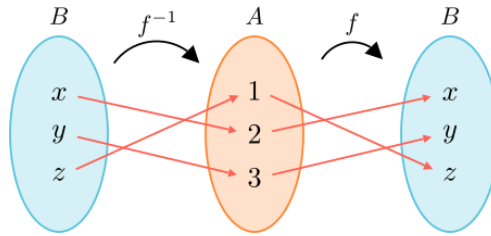


Figure 8.15. Arrow Diagram of $f \circ f^{-1}$

One sees from the above two computations that

$$f^{-1} \circ f = \text{id}_A \text{ and } f \circ f^{-1} = \text{id}_B.$$

The observation made above can be generalised.

Proposition 8.5.5: Composition of Function with Inverse

Let X and Y be sets, and let $f : X \rightarrow Y$ be an invertible function with inverse $f^{-1} : Y \rightarrow X$. Then

$$f^{-1} \circ f = \text{id}_X \text{ and } f \circ f^{-1} = \text{id}_Y.$$

Proof. For the first, let $x \in X$. Then

$$(f^{-1} \circ f)(x) = f^{-1}(f(x))$$

by definition of composition of functions. Letting $z = f^{-1}(f(x))$, we see that

$$f(z) = f(x)$$

by definition of the inverse function. As f is invertible, it is thus a bijection (**Theorem 8.4.12**), which means it is injective (by definition of bijection). Hence we see $z = x$, meaning $f^{-1}(f(x)) = x$. Since this holds for any $x \in X$, we must have $f^{-1} \circ f = \text{id}_X$.

For the second, let $y \in Y$. Then we see

$$(f \circ f^{-1})(y) = f(f^{-1}(y)).$$

Let $z = f(f^{-1}(y))$, which means that

$$f^{-1}(z) = f^{-1}(y)$$

by definition of the inverse function. Now note that f^{-1} clearly has an inverse of f , which thus means that f^{-1} is bijective ([Theorem 8.4.12](#)). So we must have $z = y$, meaning $f(f^{-1}(y)) = y$ for all $y \in Y$. Hence $f \circ f^{-1} = \text{id}_Y$. ■

The composition of two injective functions is also interesting.

Example 8.5.6: Composition of Injective Functions

Let $A = \{a, b, c\}$, $B = \{\text{I, II, III, IV}\}$, and $C = \{1, 2, 3, 4, 5\}$. Define the functions $f : A \rightarrow B$ and $g : B \rightarrow C$ such that

$$f = \{(a, \text{I}), (b, \text{II}), (c, \text{IV})\} \text{ and } g = \{(\text{I}, 1), (\text{II}, 3), (\text{III}, 4), (\text{IV}, 5)\}.$$

We can draw their individual effects as follows.

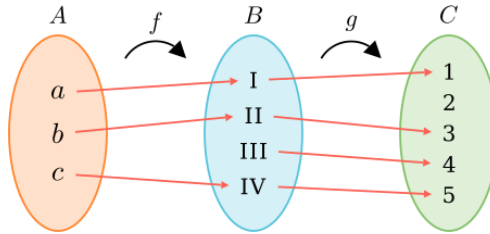


Figure 8.16. Arrow Diagrams of f and g , On The Same Diagram

The overall composition of f and g is

$$g \circ f = \{(a, 1), (b, 3), (c, 5)\}$$

which has an arrow diagram as follows.

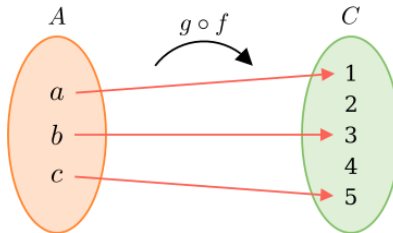


Figure 8.17. Arrow Diagram of $g \circ f$

One easily verifies that $g \circ f$ is also injective.

It is not a coincidence that the composition of injective functions is also injective.

Proposition 8.5.7: Injective Function Composition is Injective

Let A, B, C , and D be sets. Let $f : A \rightarrow B$ and $g : C \rightarrow D$ be injections such that $g \circ f$ is defined. Then $g \circ f$ is an injection.

Proof. Suppose $x_1, x_2 \in A$ such that $(g \circ f)(x_1) = (g \circ f)(x_2)$, i.e.

$$g(f(x_1)) = g(f(x_2))$$

by definition of composition of functions. Since g is injective, thus

$$f(x_1) = f(x_2).$$

But as f is also injective, thus $x_1 = x_2$. Therefore $g \circ f$ is an injection. ■

Exercise 8.8: Composition of an Injection and Non-Injection

Let X be a non-empty set, and let $f : X \rightarrow X$, $g : X \rightarrow X$, and $h : X \rightarrow X$ be functions. Suppose h is an injective function. Which of the following statements are always true?

(a) If $h \circ f = h \circ g$ then $f = g$.

(b) If $f \circ h = g \circ h$ then $f = g$.

(Hint: think of infinite sets. You may use concepts that are not yet explicitly defined in this book, e.g. odd and even numbers, and multiplication of natural numbers.)

One may notice a similar phenomenon occurring with the composition of surjective functions.

Example 8.5.8: Composition of Surjective Functions

Let $A = \{a, b, c, d, e\}$, $B = \{I, II, III, IV\}$, and $C = \{1, 2, 3\}$. Define the functions $f : A \rightarrow B$ and $g : B \rightarrow C$ such that

$$f = \{(a, II), (b, III), (c, I), (d, IV), (e, I)\}$$

and

$$g = \{(I, 2), (II, 1), (III, 1), (IV, 3)\}.$$

We can draw their individual effects as follows.

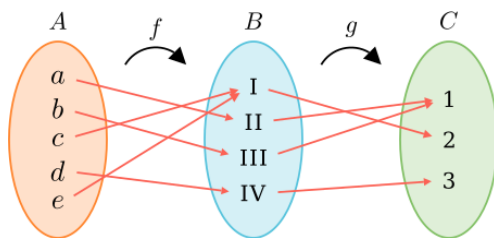


Figure 8.18. Arrow Diagrams of f and g , On The Same Diagram

The overall composition of f and g is

$$g \circ f = \{(a, 1), (b, 1), (c, 2), (d, 3), (e, 2)\}$$

which has an arrow diagram as follows.

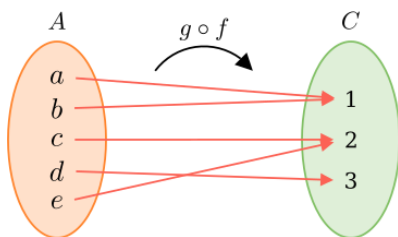


Figure 8.19. Arrow Diagram of $g \circ f$

One easily verifies that $g \circ f$ is also surjective.

Again, this is not a coincidence.

Proposition 8.5.9: Surjective Function Composition is Surjective

Let A , B , C , and D be sets. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be surjections such that $g \circ f$ is defined. Then $g \circ f$ is a surjection.

Proof. See [Exercise 8.9](#) (later). ■

The overall effect of [Proposition 8.5.7](#) and [Proposition 8.5.9](#) is the following.

Proposition 8.5.10: Bijective Function Composition is Bijective

Let A , B , C , and D be sets. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be bijections such that $g \circ f$ is defined. Then $g \circ f$ is a bijection.

Proof. Since f and g are bijections, they are both injective and surjective. **Proposition 8.5.7** thus tells us that $g \circ f$ is injective and **Proposition 8.5.9** tells us that $g \circ f$ is surjective, which means that $g \circ f$ is bijective. ■

Exercise 8.9: Composition of Surjective Functions is Surjective

Prove **Proposition 8.5.9**.

8.6. Axiom of Choice

We end this chapter by looking at the penultimate axiom of ZFC, and its namesake – the axiom of choice.

8.6.1. The Axiom

Unlike the other axioms, it is more enlightening to present its statement using the language of functions, and not directly using the formal language of axiomatic set theory.

Axiom 8.6.1: Choice

$$\forall X, \left((\emptyset \notin X) \implies \exists \left(f : X \rightarrow \bigcup_{A \in X} A \right) \text{ s.t. } \forall A \in X, f(A) \in A \right)$$

Let us unpack what this axiom says. For any non-empty set X , there exists a function f from X to the union of all the elements (which are sets) of X such that for any such element (set) in X , say A , the function maps A to an element in A . The function f is called a *choice function*.

Let us illustrate the process that **Axiom 8.6.1** performs. Suppose we have a set X , where

$$X = \{\{1, 2, 3\}, \{I, II, III, IV, V\}, \{a, b, c, d\}\}.$$

Clearly X is non-empty. So **Axiom 8.6.1** tells us there is a function from X to $\bigcup_{A \in X} A$ such that $f(A) \in A$ for all $A \in X$. Note that

$$\bigcup_{A \in X} A = \{1, 2, 3, I, II, III, IV, V, a, b, c, d\}.$$

So a possible choice function f is

$$f = \{(\{1, 2, 3\}, 2), (\{I, II, III, IV, V\}, III), (\{a, b, c, d\}, a)\}.$$

In effect, what we are doing is choosing one element from each of the sets in X .

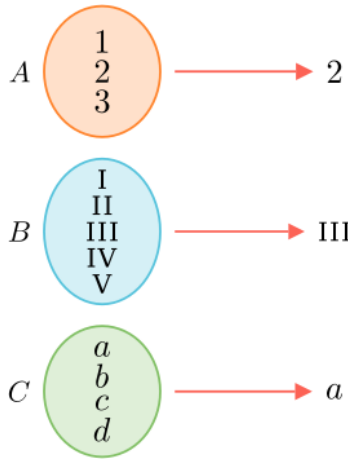


Figure 8.20. Action of the Choice Function f

8.6.2. The Necessity of the Axiom of Choice

One might wonder why we need to have the axiom of choice. Can't we always pick an element from each set within a collection of sets?

Suppose we have a set. It could be finite, it could be infinite, but we can definitely pick an element from it and place it in a singleton set. What if we have a finite collection of (finite or infinite) sets? We can still pick elements from them and place them in a finite set. The trouble comes when we are picking elements from an infinite number of sets. Our process of picking an elements from (possibly infinite) sets only work if we have a finite number of sets. If we have an infinite number of sets, we can never truly finish picking elements from them if we pick them 'manually'!

For some infinite collections of infinite sets, we still have a chance to pick elements using a certain rule.

Example 8.6.2

(Assume multiplication of natural numbers is defined, and that there is ordering within the natural numbers.)

Let N_k denote the set

$$\{x \in \mathbb{N} \mid \exists m \in \mathbb{N} \text{ s.t. } x = km\}.$$

Consider the set of sets

$$X = \{N_k \mid k \in \mathbb{N} \setminus \{0\}\}.$$

How might we define a procedure that picks an element from each of the sets in X , without using the axiom of choice? We could say “pick the smallest element from each of the sets”. This is okay – the element that we pick from each N_k is k itself, so the final set of chosen elements is

$$\{1, 2, 3, \dots, k, \dots\}$$

which is exactly $\mathbb{N}$.

Example 8.6.3

Here’s another simple example. Imagine you have infinitely many pairs of shoes. The fact is that, among each pair, we are able to distinguish the left shoe from the right shoe. So, if we were to define a function to choose a shoe from each of the infinitely many pairs of shoes, we can just say “pick the right shoe”.



Figure 8.21. Choosing a Shoe from Each Pair of Infinite Shoes

However, not every set has such a nice rule for us to use.

Example 8.6.4

Consider an infinite set of pairs of socks. Assuming that the left sock and right sock among each pair are identical, you cannot form an explicit rule for which sock to pick from each pair, and thus cannot create a function (without using the axiom of choice, that is) that picks a sock from each pair.

This is where the axiom of choice comes in. The axiom of choice asserts that such a selection is still possible, even without a specific rule. So it does not matter that the socks are indistinguishable from each other

within each pair, the axiom of choice simply tells us that there exists a way to select the socks.

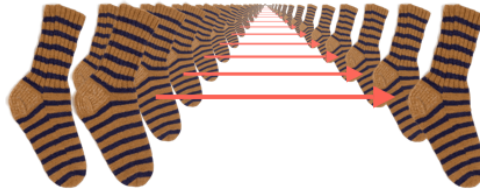


Figure 8.22. Choosing a Sock from Each Pair of Infinite Socks

Without the axiom of choice, many standard mathematical tools become unavailable or require much more complicated proofs. Whenever we invoke the idea of choosing, the axiom of choice comes to our aid.

8.6.3. An Alternate Formulation of the Axiom of Choice

The axiom of choice can be stated in many equivalent ways. We look at one of them here, relating to the idea of indexed family of sets.

Definition 8.6.5: Indexed Family of Sets

Let I be a non-empty set. A *family of sets indexed by I* is a set of sets $\mathcal{S} = \{S_i\}_{i \in I}$.

Example 8.6.6

Consider the set $S = \{a, b, c, d, e\}$. To create an indexed family of sets, we can use the set $I = \{1, 2, 3, 4, 5\}$ to index S to get $\mathcal{S} = \{S_1, S_2, S_3, S_4, S_5\}$ where $S_1 = a$, $S_2 = b$, $S_3 = c$, $S_4 = d$, and $S_5 = e$.

Theorem 8.6.7: Equivalent Statement of the Axiom of Choice

Axiom 8.6.1 holds if and only if any family of sets $\mathcal{S} = \{S_i\}_{i \in I}$ indexed by I has a corresponding family of sets $\mathfrak{S} = \{s_i\}_{i \in I}$ such that $s_i \in S_i$ for all $i \in I$.

Proof. We only prove the reverse direction here and leave the forward direction for [Exercise 8.10](#) (later).

For the reverse direction, suppose any family of sets $\mathcal{S} = \{S_i\}_{i \in I}$ indexed by I has a corresponding family of sets $\mathfrak{S} = \{s_i\}_{i \in I}$ such that

$s_i \in S_i$ for all $i \in I$. Let X be a non-empty set containing non-empty sets. We create a family of sets $\mathcal{X} = \{X_i\}_{i \in X}$ indexed by X where $X_i = i$ for all $i \in X$. In fact one sees that $X = \mathcal{X}$. By assumption we know there exists a corresponding family of sets $\mathfrak{X} = \{x_i\}_{i \in X}$ such that $x_i \in X_i$ for all $i \in X$.

One sees clearly that we have a bijection $f : \mathcal{X} \rightarrow \mathfrak{X}$ where $f(X_i) = x_i$ for all $i \in X$. Note $\mathfrak{X} \subseteq \bigcup_{S \in \mathcal{X}} S = \bigcup_{S \in X} S$. Thus the function $g : \mathcal{X} \rightarrow \bigcup_{S \in X} S$ where $g(X_i) = x_i$ for all $i \in X$ exists. Note also that $x_i \in X_i$ which means $g(X_i) \in X_i$ for all $i \in X$, and that $\mathcal{X} = X$. So the function g is a function from X to $\bigcup_{S \in X} S$ such that $g(X_i) \in X_i$ for all $i \in X$. Thus g is the choice function we are after, showing that [Axiom 8.6.1](#) holds. ■

Exercise 8.10: Equivalent Statement of the Axiom of Choice

Prove the forward direction of [Theorem 8.6.7](#).

CHAPTER 9

Counting and Cardinality

Ironically, it is often the case that rigorously defining something intuitive turns out to be exceedingly tricky. Such is the case of counting. Counting the number of things in a collection seems obvious, but creating a watertight protocol for counting is more challenging than it appears at first glance.

9.1. The Problem With 'Intuitive' Counting

One would be hard-pressed to disagree that counting is such an obvious and trivial task that it is not worth dedicating an entire chapter to it. The very idea of saying how many things exist is, to us, deeply ingrained in our biology. Why do we need to define counting?

The core issue with counting is ensuring everyone agrees on how many items are there. Surely this task is trivial! Simply tally how many things there are! How hard can it be to make everyone agree on the number of items?

In the spirit of Socratic dialogue, let us examine two characters attempting to agree on how many shapes were drawn on a field.

Achilles, the mightiest of Greek warriors, had been waiting on a barren field. With his mighty sword, he had been drawing shapes in the dirt to pass the time.



A certain Tortoise, fresh from his victory over a certain hare, came upon him thus engaged.

TORTOISE: You seem deep in contemplation.

ARCHILLES: Indeed I am. I find myself puzzling over the most elementary concept – the nature of counting itself.

TORTOISE: Counting! That must be the simplest thing anyone could do! I can demonstrate it to you with the shapes you've drawn. There're six of them, you see.

Achilles was shaken out of his reverie upon hearing that comment.

ARCHILLES: *Six?* My dear friend, that's quite impossible! How do you arrive at such a count?

TORTOISE: Nothing simpler. I'll demonstrate by labelling them.

With great satisfaction, the Tortoise, upon producing a stick, proceeded to mark several of the shapes multiple times until he indeed reached the number six.



ARCHILLES: Preposterous! You can't count the same shape more than once!

TORTOISE: Ah, my mistake. In that case, there are three shapes.

And he proceeds to label only some of the shapes, leaving others unmarked entirely.



Achilles, who started to show signs of the same despair that would later plague him in matters of logic, protested.

ARCHILLES: No, no – every shape must have at least one label!

TORTOISE: Oh, I didn't know that was a requirement. Well, then—

He proceeded to label all shapes but the star, which he crossed out with particular vigour.



TORTOISE: That spiky thing hardly counts as a shape, so we have exactly four shapes.

Achilles, with what patience he could still muster, attempted to query the Tortoise again.

ARCHILLES: Let us suppose that 'spiky thing' – which, I might add, is called a star – is a shape. I hope you grant me at least this much.

TORTOISE: Of course.

ARCHILLES: How many shapes would you count then?

The Tortoise considered this for a moment, then brightened, and declared thus–

TORTOISE: Seven!

–and proceeded to demonstrate this absurdity with yet another creative labelling system.



At this point, Achilles – who would later spend countless hours discussing the logistics of logical propositions with this same Tortoise – decided that discretion was the better part of valour and walked away without another word.

Of course, this is a contrived example, but it is important to ask where it went wrong. The two main issues faced above were that either

- not all the shapes were counted; or
- a shape was counted more than once.

With this in mind, how do determine how many elements exist in a set?

9.2. An Introduction to Cardinality

Definition 9.2.1: Cardinality

Cardinality describes a relationship between sets which compares their relative size. The cardinality of a set A is denoted $|A|$.

It turns out that the cardinality of a finite set is the number of elements inside the set. Of course, defining the notion of cardinality only for finite sets defeats the purpose of defining counting – how do we know how many elements are inside the set in the first place? To that end, let us **not** define cardinality of a set directly; instead, let us define the relationships between cardinalities, an approach adopted by the originator of set theory, Georg Ferdinand Cantor, around 1880.

9.2.1. Bijections and Cardinality

Definition 9.2.2: Equal Cardinality

Let A and B be sets. Then A and B have equal cardinality (denoted $|A| = |B|$) if and only if there exists a bijection $f : A \rightarrow B$. Such sets are called *equinumerous*.

This should be intuitive for finite sets – if we can pair the elements of two sets up exactly, they should have the same number of elements (which is analogous to their cardinalities).

Example 9.2.3: Equinumerous Sets

Let $A = \{a, b, c, d\}$ and $B = \{w, x, y, z\}$. We can define a bijection $f : A \rightarrow B$ where

$$f = \{(a, w), (b, x), (c, y), (d, z)\}.$$

So $|A| = |B|$, i.e. A and B are equinumerous.

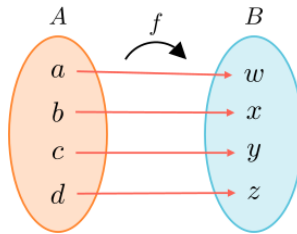


Figure 9.1. Arrow Diagram of the Bijection f

Of course, there are other possible bijections. For example, the function $g : A \rightarrow B$ where

$$g = \{(a, z), (b, x), (c, w), (d, y)\}$$

is also a bijection.

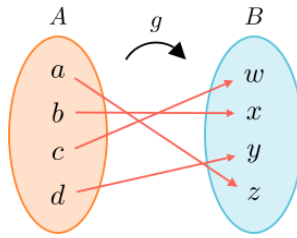


Figure 9.2. Arrow Diagram of the Bijection g

The key thing that we need to show is that there exists at least one bijection. As long as we can find one such bijection, we can conclude that two sets are equinumerous.

This way of defining equal cardinality addresses the two issues that we noted when trying to count the shapes earlier. Every element must be accounted for, and every element must be counted at most once.

We note one important equivalence of cardinalities here.

Proposition 9.2.4: Cardinality of Image of Injection

Let A and B be sets and let $f : A \rightarrow B$ be an injective function. Then $|A| = |f(A)|$, where

$$f(A) = \{f(a) \mid a \in A\}.$$

Proof. Let $\tilde{f} : A \rightarrow f(A)$ where $\tilde{f}(a) = f(a)$. We show that $\tilde{f}$ is a bijection.

- **Injective:** Suppose $\tilde{f}(a_1) = \tilde{f}(a_2)$ for some $a_1, a_2 \in A$. Thus $f(a_1) = f(a_2)$ by definition of $\tilde{f}$. Since f is assumed to be injective, thus $a_1 = a_2$, which means $\tilde{f}$ is injective.
- **Surjective:** Suppose $y \in f(A)$. Note $y = f(a)$ for some $a \in A$ by definition of $f(A)$. Thus a is the preimage of y because $\tilde{f}(a) = f(a) = y$. Hence $\tilde{f}$ is surjective.

Therefore $\tilde{f}$ is a bijection, which means $|A| = |f(A)|$. ■

9.2.2. Injections/Surjections and Cardinality

What happens when we try and measure the cardinality of two sets that do not have a bijection between them?

Example 9.2.5: Non-Equinumerous Sets

Consider the sets $A = \{a, b, c, d\}$ and $B = \{v, w, x, y, z\}$. No matter how hard we try, we cannot find a bijection between these two sets. There seems to always be at least one element left unmapped from A to B .

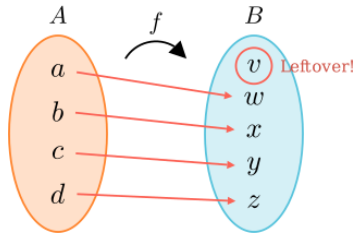


Figure 9.3. Arrow Diagram by Some Function f

Okay, let us try weakening the condition for the function f in **Definition 9.2.1**. What happens if we don't require f to be surjective (i.e., f is injective only). Of course, if f is surjective we can still conclude that $|A| = |B|$. What happens when we can only find injective functions from A to B ? Just like the previous example, we can see that the set B is 'larger' than the set A . This motivates the definition of smaller cardinality.

Definition 9.2.6: Smaller (or Equal) Cardinality

Let A and B be sets. Then A has a cardinality smaller than or equal to B (denoted $|A| \leq |B|$) if and only if there exists an injection $f : A \rightarrow B$.

We say that A has a cardinality smaller than B (denoted $|A| < |B|$) if and only if $|A| \leq |B|$ and there does not exist a bijection from A to B .

What if a surjection from A to B exists? What can we conclude?

Definition 9.2.7: Larger (or Equal) Cardinality

Let A and B be sets. Then A has a cardinality larger than or equal to B (denoted $|A| \geq |B|$) if and only if there exists a surjection $f : A \rightarrow B$.

We say that A has a cardinality larger than B (denoted $|A| > |B|$) if and only if $|A| \geq |B|$ and there does not exist a bijection from A to B .

Wait a minute, does **Definition 9.2.6** agree with **Definition 9.2.7**? That is, if we have two sets A and B , if $|A| \leq |B|$, do we also have $|B| \geq |A|$? It turns out that we do.

Theorem 9.2.8: Agreement of **Definition 9.2.6** and **Definition 9.2.7**

Let A and B be sets. Then $|A| \geq |B|$ if and only if there exists an injection from B to A .

Proof. We only prove the forward direction here and leave the reverse direction for **Exercise 9.1** (later).

Suppose $|A| \geq |B|$. By **Definition 9.2.7** this means that there exists a surjection from A to B , say f . Our goal is to construct an injection, say g , from B to A .

First, because f is surjective, which means that for any $b \in B$ there exists at least one $a \in A$ such that $f(a) = b$. So define a function $g : B \rightarrow A$ such that $g(b) = a$ for one specific a where $f(a) = b$ for all $b \in B$. Note that we specifically choose only one a per b for the definition of g , which makes it a valid function. We are able to choose this a due to the axiom of choice (**Axiom 8.6.1**).

Now we need to prove that g is injective. Suppose $g(b_1) = g(b_2)$ for some $b_1, b_2 \in B$. Let $g(b_1) = a$ (which also means $g(b_2) = a$ too). By definition of g , we must have $f(a) = b_1$ and $f(a) = b_2$, which means $b_1 = b_2$. Hence g is injective, completing the forward direction. ■

Exercise 9.1: Complete the Proof!

Complete the proof of **Theorem 9.2.8**.

9.2.3. Cantor-Schröder-Bernstein

The symbols “ $\leq$ ” and “ $\geq$ ” used for comparing cardinalities hint at an interesting idea: if $|A| \leq |B|$ and $|B| \leq |A|$, is $|A| = |B|$? In fact, this is a corollary of the Cantor-Schröder-Bernstein Theorem. The proof of this theorem is complex, so we will leave it to the end of this chapter.

Theorem 9.2.9: Cantor-Schröder-Bernstein

Let A and B be sets. If there are injections $f : A \rightarrow B$ and $g : B \rightarrow A$, then there exists a bijection $h : A \rightarrow B$.

Cantor-Schröder-Bernstein allows us to prove two important results.

Corollary 9.2.10

Let A and B be sets. If $|A| \leq |B|$ and $|B| \leq |A|$ then $|A| = |B|$.

Proof. By definition we know that $|A| \leq |B|$ means that there exists an injection from A to B and $|B| \leq |A|$ means that there exists an injection from B to A . Thus [Theorem 9.2.9](#) tells us that there is a bijection from A to B , which hence means $|A| = |B|$. ■

Corollary 9.2.11: Injection Into Subset Means Equinumerous

Let A be a set and B be a subset of A . If there is an injection from A to B , then $|A| = |B|$.

Proof. Our goal is to find injections from A to B and from B to A and then use Cantor-Schröder-Bernstein ([Theorem 9.2.9](#)) to finish the proof.

The injection $f : A \rightarrow B$ is assumed to exist by our antecedent. The injection $g : B \rightarrow A$ can be found simply by defining $f(x) = x$ for all $x \in B$. One clearly sees that if $g(x_1) = g(x_2)$ then $x_1 = x_2$ for all $x_1, x_2 \in B$.

Thus Cantor-Schröder-Bernstein tells us that there exists a bijection from A to B , which means $|A| = |B|$. ■

There are more interesting examples of using the Cantor-Schröder-Bernstein theorem, but we will cover them when the time comes.

9.3. Finite Counting

With an introduction of cardinality out of the way, we can finally define a way to count the number of items in a finite set.

Example 9.3.1: A Set of 5

Consider the set $A = \{a, b, c, d, e\}$. One clearly sees that the set $B = \{0, 1, 2, 3, 4\}$ is equinumerous with the set A by using the bijection

$$f = \{(a, 0), (b, 1), (c, 2), (d, 3), (e, 4)\}.$$

But notice that the set B is exactly the counter set for 5. So, in a sense, we can write

$$|A| = |5|.$$

Example 9.3.2: A Set of 7

Consider the set $A = \{73, 75, 93, 103, 848, 3742, 3999\}$. One clearly sees that the set $B = \{0, 1, 2, 3, 4, 5, 6\}$ is equinumerous with the set A by using the bijection

$$f = \{(73, 0), (75, 1), (93, 2), (103, 3), (848, 4), (3742, 5), (3999, 6)\}.$$

But notice that the set B is exactly the counter set for 7. So, in a sense, we can write

$$|A| = |7|.$$

Example 9.3.3: A Set of 0

Consider the empty set $\emptyset$. This is exactly the counter set for 0, and thus

$$|\emptyset| = |0|.$$

It is surely inconvenient (and unintuitive) to use the cardinality of the counter set of a natural number. Thus, we may abuse the notation for cardinality slightly.

Definition 9.3.4: A Slight Abuse of Cardinality of Counter Sets

Define $|n| = n$ for all $n \in \mathbb{N}$. That is, the cardinality of the counter set of n is exactly n .

With this definition we can write the cardinality of finite sets more intuitively.

Example 9.3.5: Cardinalities of Finite Sets

- For $A = \{a, b, c, d, e\}$, since $|A| = |5|$, we can write $|A| = 5$. This is read as “the cardinality of A is 5”.
- For $A = \{73, 75, 93, 103, 848, 3742, 3999\}$ we can write $|A| = 7$, which is read “the cardinality of A is 7”.
- For the empty set we can write $|\emptyset| = 0$. This can be read as “the empty set has cardinality 0”.

Exercise 9.2: Finite Cardinalities

- (a) What is the cardinality of the counter set of 7125?
- (b) What is $|373267|$?
- (c) Write down a set of cardinality 1.
- (d) What is the cardinality of $13 \cup 14$?
(Hint: think of them as counter sets.)

We have an interesting result regarding maps between finite sets with the same cardinality.

Theorem 9.3.6: Map Between Finite Sets of Equal Cardinality

Let A and B be finite sets such that $|A| = |B|$. Suppose $f : A \rightarrow B$ is a function. Then f is a bijection if and only if f is an injection.

Proof. For the forward direction, if f is a bijection then f must be injective by definition of bijections.

For the reverse direction, suppose f is an injection. By [Proposition 9.2.4](#) we know that $|A| = |f(A)|$ where

$$f(A) = \{f(a) \mid a \in A\}.$$

As $|A| = |B|$, we have $|f(A)| = |B|$. Note that $f(A) \subseteq B$. Thus $f(A)$ is a subset of B with the same number of elements as B , which means $f(A) = B$. But clearly this means that any element in B can be expressed in the form $f(a)$ where $a \in A$, and thus under f any $b = f(a)$ has a preimage of a . Hence f is also surjective, which means f is a bijection. ■

9.4. Infinite Sets and Infinite Counting

We have an intuition of what a finite set is – there is a limit to the number of items in the set. With our language of cardinality we can formally state this, using the definition found in [\[Boo98, p. 262\]](#).

Definition 9.4.1: Finite Set

A set S is *finite* if and only if there exists a bijection $f : S \rightarrow n$ for some $n \in \mathbb{N}$ (here n represents the counter set of n). In other words, S is finite if and only if $|S| = n$ for some $n \in \mathbb{N}$.

Example 9.4.2: Examples of Finite Sets

- The set $\{2, 3, 5\}$ is finite since $|\{2, 3, 5\}| = 3$.
- The set $\{a, b, 1, 2, !\}$ is finite since $|\{a, b, 1, 2, !\}| = 5$.
- The empty set is finite since $|\emptyset| = 0$.

Infinite sets can thus be defined as follows.

Definition 9.4.3: Infinite Set

A set S is *infinite* if and only if S is not finite. In other words, there does not exist a bijection $f : S \rightarrow n$ for all $n \in \mathbb{N}$.

We have already seen some examples of infinite sets.

Example 9.4.4: $\mathbb{N}$

Based on the definition above, it should be the case that $\mathbb{N}$ given by the axiom of infinity ([Axiom 6.4.1](#)) is an infinite set. In fact, it is.

Proof. We consider a proof by contradiction. Suppose $\mathbb{N}$ is finite. Consider the successor function $S : \mathbb{N} \rightarrow \mathbb{N}$ where $S(n) = n \cup \{n\}$ for all $n \in \mathbb{N}$. By [Example 8.4.2](#) we know that S is injective. As $\mathbb{N}$ is (assumed to be) finite, we know by [Theorem 9.3.6](#) this means that S is bijective, which means S is surjective. But one sees that 0 has no preimage under S , which thus means S cannot be surjective. This is a contradiction. Hence $\mathbb{N}$ is not finite. ■

Example 9.4.5: $\mathbb{N} \setminus \{0\}$

We show that $\mathbb{N} \setminus \{0\}$ is also not a finite set by showing that $|\mathbb{N} \setminus \{0\}| = |\mathbb{N}|$.

Proof. First consider the function $S : \mathbb{N} \rightarrow \mathbb{N} \setminus \{0\}$ where $S(n) = n \cup \{n\}$ for all $n \in \mathbb{N}$. Using a similar argument as [Example 8.4.2](#) we can show that S is injective.

Now consider the function $S' : \mathbb{N} \setminus \{0\} \rightarrow \mathbb{N}$ where $S'(n) = n \cup \{n\}$ for all $n \in \mathbb{N} \setminus \{0\}$. Again using a similar argument as [Example 8.4.2](#) we can show that S' is injective.

So Cantor-Schröder-Bernstein ([Theorem 9.2.9](#)) tells us that there is a bijection from $\mathbb{N} \setminus \{0\}$ to $\mathbb{N}$, i.e. $|\mathbb{N} \setminus \{0\}| = |\mathbb{N}|$. ■

But this is weird. Clearly $\mathbb{N} \setminus \{0\}$ contains one fewer element than $\mathbb{N}$, and yet they have the same cardinality! This is the trap when we think of cardinalities as ‘sizes’ of sets – an infinite set with ‘more’ elements can have the same ‘size’ as an infinite set with ‘less’ elements. Thus, cardinalities for infinite sets should be considered its own beast, and not the same as the ‘sizes’ for finite sets.

9.5. Axiom of Power Set

To create sets with a larger cardinality than $\mathbb{N}$, we need the power set – the set of all subsets of a set. This set is explicitly instantiated for all sets in ZFC via the axiom of power set, the last axiom in ZFC.

Axiom 9.5.1: Power Set

$$\forall A \exists P \forall S (S \in P \iff \forall x (x \in S \implies x \in A))$$

In other words, for any set A there is a set P (the power set of A) such that for all sets S in P , any element in S is also in A . This clearly must mean that S is a subset of A and thus P , being the collection of all such subsets, is the power set $\mathcal{P}(A)$.

Example 9.5.2: Examples of Power Sets

(Assume exponentiation is defined for the natural numbers.)

- $\mathcal{P}(\emptyset) = \{\emptyset\}$, since the empty set is always a subset of any set, including itself. We see that $|\mathcal{P}(\emptyset)| = 1 = 2^0$.
- $\mathcal{P}(\{1\}) = \{\emptyset, \{1\}\}$. We see that $|\mathcal{P}(\{1\})| = 2 = 2^1$.
- $\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$. We see that $|\mathcal{P}(\{1, 2\})| = 4 = 2^2$.
- $\mathcal{P}(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$. So $|\mathcal{P}(\{1, 2, 3\})| = 8 = 2^3$.

From the above example we can come up with the following claim.

Proposition 9.5.3: Cardinality of Power Set For Finite Sets

Let A be a finite set with cardinality n . Then $|\mathcal{P}(A)| = 2^n$.

For the proof, we will assume that addition and exponentiation is defined for the natural numbers.

Proof. We induct on n .

For the base case, assume $n = 0$. So $|A| = 0$ which means that $A = \emptyset$. Thus $\mathcal{P}(A) = \mathcal{P}(\emptyset) = \{\emptyset\}$. So $|\mathcal{P}(A)| = 1 = 2^0$, proving the base case.

Now let's suppose that $|\mathcal{P}(A)| = 2^k$ for some finite set A with cardinality k . We are to show that the power set of a set with cardinality $k + 1$ has cardinality 2^{k+1} .

Let B be a set of cardinality of $k + 1$. By the axiom of choice ([Axiom 8.6.1](#)), pick an element $x \in B$. Let $A = B \setminus \{x\}$; notice $|A| = k$. Now, there are two kinds of subsets of B .

- The first kind are subsets that do not include x . These subsets are exactly the subsets of A , i.e. they come from $\mathcal{P}(A)$.
- The second kind are the subsets that include x . In particular, a subset of this kind takes the form $S \cup \{x\}$ where $S \in \mathcal{P}(A)$.

All in all, we see that

$$\mathcal{P}(B) = \{S \mid S \in \mathcal{P}(A)\} \cup \{S \cup \{x\} \mid S \in \mathcal{P}(A)\}.$$

Notice that each of the two sets that we are unioning together have the same cardinality as $\mathcal{P}(A)$. By the induction hypothesis we know that $|\mathcal{P}(A)| = 2^k$. Thus,

$$|\mathcal{P}(B)| = |\mathcal{P}(A)| + |\mathcal{P}(A)| = 2^k + 2^k = 2^{k+1}$$

which proves the case for $k + 1$.

Thus, by mathematical induction, we see that $|\mathcal{P}(A)| = 2^n$ for any finite set A with cardinality n . ■

This fact about the power set is useful when we want to define larger infinities.

Exercise 9.3: A Power Set

Find $\mathcal{P}(\mathcal{P}(\{a, b\}))$, and state its cardinality.

9.6. Cantor's Theorem and Larger Infinities

So far, all the examples of infinite sets that we have seen so far seem to be of the same cardinality. For example, the set of natural numbers $\mathbb{N}$ has

the same cardinality as $\mathcal{S}(\mathbb{N})$. So is it that all infinite sets have the same cardinality? No. This surprising result is called Cantor's Theorem, and it involves the power set.

Theorem 9.6.1: Cantor

Let A be a set. Then any function $f : A \rightarrow \mathcal{P}(A)$ is not surjective. Consequently $|A| < |\mathcal{P}(A)|$ for any set A .

Notice that this theorem does not just apply to finite sets – this works for all sets, including infinite sets.

Proof. Let $f : A \rightarrow \mathcal{P}(A)$ be a function. Seeking a contradiction, suppose f is surjective.

Note that the set

$$B = \{x \in A \mid x \notin f(x)\}$$

exists by the axiom schema of specification ([Axiom 5.6.1](#)). Since $B \subseteq A$, thus $B \in \mathcal{P}(A)$. As f is (assumed to be) surjective, there must be a $\xi \in A$ such that $f(\xi) = B$.

Let us determine whether ξ is in B or not.

- Suppose $\xi \in B$. Then, by definition of B , we know $\xi \notin f(\xi)$. But $f(\xi) = B$, which means $\xi \notin B$, a clear contradiction to the assumption that $\xi \in B$.
- However, if instead $\xi \notin B$, we see $\xi \notin f(\xi)$ since $B = f(\xi)$. But this must mean that $\xi \in B$ by definition of B , another contradiction.

So, in either case, we have a contradiction. What did we contradict? We contradicted the assumption that f is surjective. Thus there is no surjective mapping from A to $\mathcal{P}(A)$, and hence no bijection from A to $\mathcal{P}(A)$.

To prove that $|A| < |\mathcal{P}(A)|$ we simply need to find an injection from A to $\mathcal{P}(A)$. This is left for [Exercise 9.4](#) (later). ■

This surprising result uses the set B , sometimes called the *Cantor diagonal set of f* (see [Das14, Theorem 1210, p. 362]). We will see the use of the set later in this book.

Exercise 9.4: The Last Part of Cantor's Theorem

Find an injection from A to $\mathcal{P}(A)$ for any set A .

In light of [Theorem 9.6.1](#), we can start defining larger infinities. The ‘smallest’ infinity is the cardinality of the real numbers, which is implied by the definition of infinite sets. We give notation for the cardinality of infinite sets.

Definition 9.6.2: Beth Numbers

Define $\beth_0$ (read “beth zero” or “beth null”) to be the cardinality of $\mathbb{N}$, i.e. $\beth_0 = |\mathbb{N}|$. Subsequently (assuming addition of natural numbers is defined), for any $n \in \mathbb{N}$, define

$$\beth_{n+1} = |\mathcal{P}(A)|$$

where $|A| = \beth_n$.

With Cantor’s theorem ([Theorem 9.6.1](#)) we know that

$$\beth_0 < \beth_1 < \beth_2 < \dots < \beth_n < \dots$$

Example 9.6.3: The First Few Beth Numbers

- $\beth_0 = |\mathbb{N}|$.
- $\beth_1 = |\mathcal{P}(\mathbb{N})|$.
- $\beth_2 = |\mathcal{P}(\mathcal{P}(\mathbb{N}))|$.
- $\beth_3 = |\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathbb{N})))|$.

The more common notation used for denoting the cardinality of an infinite set are the aleph numbers.

Definition 9.6.4: Aleph Numbers

Define $\aleph_0$ (read as “aleph zero” or “aleph null”) to be $\beth_0$. Subsequently (assuming addition of natural numbers is defined), for any $n \in \mathbb{N}$, define $\aleph_{n+1}$ as the next cardinality such that $\aleph_n < \aleph_{n+1}$ and there does not exist a cardinality κ such that $\aleph_n < \kappa < \aleph_{n+1}$.

Unfortunately, determining the exact ‘form’ of $\aleph_1$ is tricky. A well known question in mathematics is the *Continuum hypothesis*, which, in essence, asks whether

$$\aleph_1 = \beth_1.$$

The surprising thing about this question is that **this is not decidable using ZFC** – we can neither prove nor disprove this statement within ZFC.

To decide on the truth or falsity of this statement requires us to create a new axiom, which is not needed for this book. We shall just leave this as a mathematical curiosity.

9.7. Proof of Cantor-Schröder-Bernstein

In this final section, we will prove [Theorem 9.2.9](#). However, to do so, we will need a few preliminaries.

- Assume that addition and subtraction of natural numbers is defined.
- Let the notation

$$\bigcup_{n \in \mathbb{N}} A_n$$

denote

$$A_0 \cup A_1 \cup A_2 \cup A_3 \cup A_4 \cup \dots$$

where each A_i is a set.

- Suppose $f : A \rightarrow A$ is a function from the set A to the set A . Then define $f^2 = f \circ f$, $f^3 = f \circ f \circ f$, $f^4 = f \circ f \circ f \circ f$, etc. More formally, define

$$f^n = \begin{cases} \text{id}_A & \text{if } n = 0 \\ f \circ f^{n-1} & \text{otherwise} \end{cases}$$

for all $n \in \mathbb{N}$.

- Suppose $f : A \rightarrow B$ is a function from the set A to the set B . Define

$$f(A) = \{f(a) \mid a \in A\},$$

i.e. $f(A)$ is the range of f .

Alright. Recall that [Theorem 9.2.9](#) essentially states that, for any two sets A and B , if there are injections $f : A \rightarrow B$ and $g : B \rightarrow A$, then there is a bijection $h : A \rightarrow B$. Before we present the proof of this theorem, let us use diagrams to see what needs to be proven.

We are given two sets A and B , with the injections $f : A \rightarrow B$ and $g : B \rightarrow A$.

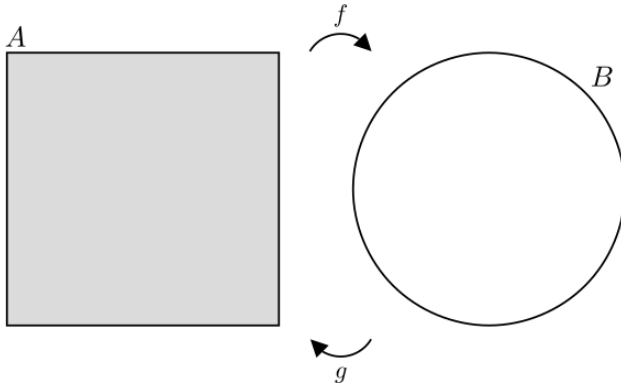


Figure 9.4. Sets A and B with Injections f and g

Let's suppose we map the entirety of A to B using the injection f . That would place a "copy" of A in B (in particular, this "copy" is $f(A)$). Likewise, let's map the entirety of B to A using the injection g , placing a "copy" of B (i.e., $g(B)$) in A . But after mapping A to B (resulting in $f(A)$ in B), this second mapping from B to A creates a "nested" copy of A inside B inside A (in fact, this is $(g \circ f)(A)$). We can do this again, creating a copy of B inside A inside B (which is $(f \circ g)(B)$). So, repeating this process indefinitely we will get this nice fractal-like diagram.

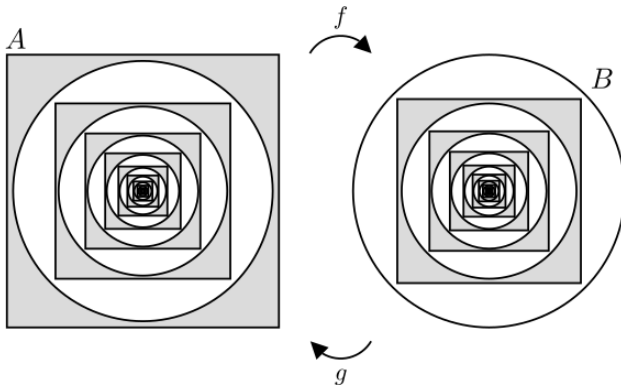


Figure 9.5. Infinite Nesting of A and B Within Themselves

We can trace how each of these mappings occurred and label each (nested) set accordingly. Of course, the figure can't show all the labels, but you get the point.

the set

$$\mathfrak{U} = A \setminus \mathfrak{S}.$$

Note that the set $\mathfrak{S}$ does indeed exist by the axiom of union ([Axiom 5.5.1](#)), so the set $\mathfrak{U}$ also exists.

Now we can start defining a possible bijection. We know that any element in a shaded region of A can be mapped to a shaded region in B under f . But what about an unshaded region in B ? Well, since the unshaded regions of A are images of elements in the unshaded regions in B under g , we can just ‘invert’ g to get a mapping from A to B for the unshaded regions in A .

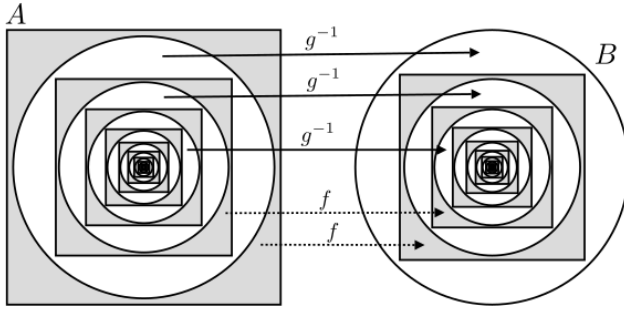


Figure 9.7. Bijection Mapping from A to B

Let us be clear – g^{-1} does not exist since g may not be bijective. However, restricting g to map from B to the entirety of the outermost circle of A (i.e., the range of g) gives us a bijective map, and so g^{-1} does exist for that part. In other words, if we define a new function $\tilde{g} : B \rightarrow g(B)$ such that $\tilde{g}(x) = g(x)$ for all $x \in B$, this is bijective.

We are now ready to prove the Cantor-Schröder-Bernstein theorem ([Theorem 9.2.9](#)).

Proof (see [[Ham18](#), Theorem 14.10, pp. 286–287]). Suppose A and B are sets with injections $f : A \rightarrow B$ and $g : B \rightarrow A$. Henceforth restrict g to have a codomain of $g(B)$, making g a bijection.

Consider the set

$$\mathfrak{S} = \bigcup_{n \in \mathbb{N}} S_n$$

where

$$S_n = (g \circ f)^n(A \setminus g(B))$$

for all $n \in \mathbb{N}$, and let the set

$$\mathfrak{U} = A \setminus \mathfrak{S}.$$

Notice that $A = \mathfrak{S} \cup \mathfrak{U}$ and that $\mathfrak{S} \cap \mathfrak{U} = \emptyset$. This means that any element in A must be in either $\mathfrak{S}$ or $\mathfrak{U}$, but not both.

Define the function $h : A \rightarrow B$ where

$$h(x) = \begin{cases} f(x) & \text{if } x \in \mathfrak{S} \\ g^{-1}(x) & \text{if } x \in \mathfrak{U} \end{cases}$$

for all $x \in A$. We show that h is a bijection.

- **Injective:** Suppose $h(x_1) = h(x_2)$ for some $x_1, x_2 \in A$. There are four cases.

1. $x_1, x_2 \in \mathfrak{S}$. Thus, by definition of h , we have $f(x_1) = f(x_2)$. Since f is injective, we must have $x_1 = x_2$.
2. $x_1, x_2 \in \mathfrak{U}$. This means $g^{-1}(x_1) = g^{-1}(x_2)$ by definition of h . Hence $x_1 = x_2$ by applying g on both sides.
3. $x_1 \in \mathfrak{S}$ and $x_2 \in \mathfrak{U}$. Since $x_1 \in \mathfrak{S}$, it must belong in a S_k , i.e. $x_1 = (g \circ f)^k(z)$ for some $k \in \mathbb{N}$ and $z \in A \setminus g(B)$. So, by definition of h , we must have

$$f((g \circ f)^k(z)) = g^{-1}(x_2),$$

which means $g(f((g \circ f)^k(z))) = x_2$, i.e.

$$(g \circ f)^{k+1}(z) = x_2.$$

But, since $z \in A \setminus g(B)$, this shows that $x_2 \in S_{k+1}$ and thus $x_2 \in \mathfrak{S}$, which is a contradiction since no element is simultaneously in $\mathfrak{S}$ and $\mathfrak{U}$ at the same time.

4. $x_1 \in \mathfrak{U}$ and $x_2 \in \mathfrak{S}$. Similar argument as case 3 shows that this is not possible.

Thus, in all (possible) cases, we have $x_1 = x_2$, meaning h is injective.

- **Surjective:** Suppose $y \in B$. Note $g(y) \in A$, which means $g(y) \in \mathfrak{S}$ or $g(y) \in \mathfrak{U}$.

1. $g(y) \in \mathfrak{S}$. Like we argued above, we see that $g(y) \in S_k$ for some $k \in \mathbb{N}$, i.e. $g(y) = (g \circ f)^k(z)$ for some $k \in \mathbb{N}$ and $z \in A \setminus g(B)$. Note that $k \neq 0$ because, otherwise, we will have

$$g(y) = z \in A \setminus g(B),$$

where $g(y) \in g(B)$ is equal to z which is not in $g(B)$, a contradiction. So we may write

$$\begin{aligned} g(y) &= (g \circ f)^k(z) \\ &= ((g \circ f) \circ (g \circ f)^{k-1})(z) \\ &= g(f((g \circ f)^{k-1}(z))). \end{aligned}$$

Since g is injective, we have

$$y = f((g \circ f)^{k-1}(z)).$$

Let $x = f((g \circ f)^{k-1}(z))$. By definition of $\mathfrak{S}$ we see that $x \in \mathfrak{S}$. So one sees

$$\begin{aligned} h(x) &= f(x) && \text{(Definition of } h) \\ &= y \end{aligned}$$

which means a preimage of y under h is $x = f((g \circ f)^{k-1}(z))$.

2. $g(y) \in \mathfrak{U}$. Let $x = g(y)$, noting that $x \in \mathfrak{U} \subseteq A$. Then one sees

$$\begin{aligned} h(x) &= g^{-1}(x) && \text{(Definition of } h) \\ &= g^{-1}(g(y)) && \text{(Definition of } x) \\ &= y \end{aligned}$$

which means a preimage of y under h is $x = g(y)$.

Thus, in either case, we have found a preimage of y under h , which means that h is surjective.

Thus h is a bijection as required. ■

CHAPTER 10

Operations Involving the Natural Numbers

What is the point of having numbers but with no way to interact with them? Numbers without any relationships defined between them are useless to us; we need operations between them to create relationships between them. This chapter will introduce three well-defined operations: addition, multiplication, and exponentiation.

10.1. Binary Operations

To define the operations on the natural numbers, we first need to understand what “operation” means. In particular, we are only looking at binary operations.

Definition 10.1.1: Binary Operation

A *binary operation on a set S* is a function $f : S \times S \rightarrow S$ (i.e., $f : S^2 \rightarrow S$).

Remark: n -ary Operation

We can generalize the definition of an operation to be an *n -ary operation on a set S* , which is a function $f : S^n \rightarrow S$ where $n \in \mathbb{N}$. We will stick to the case when $n = 2$ for this chapter.

It should be noted that we usually use infix notation to write binary operations.

Example 10.1.2: Union Operation

Let A be a set. We can define the set union operation on the set A by the function $u : \mathcal{P}(A) \times \mathcal{P}(A) \rightarrow \mathcal{P}(A)$ where

$$u(S, T) = \{x \in A \mid x \in S \vee x \in T\}$$

for all $S, T \in \mathcal{P}(A)$. Note that $u(S, T)$ would be written as $S \cup T$ using infix notation.

For example, for the set $A = \{1, 2, 3, 4, 5\}$, $S = \{1, 2\}$ and $T =$

$\{2, 3\}$, we see that

$$u(S, T) = S \cup T = \{1, 2, 3\}.$$

If instead $S = \{1, 4, 5\}$ and $T = \{2, 3, 5\}$ one notes that

$$S \cup T = \{1, 2, 3, 4, 5\} = A.$$

Example 10.1.3: Intersection Operation

Likewise, we can define the set intersection operation on a set A by the function $n : \mathcal{P}(A) \times \mathcal{P}(A) \rightarrow \mathcal{P}(A)$ where

$$S \cap T = \{x \in A \mid x \in S \wedge x \in T\}$$

for all $S, T \in \mathcal{P}(A)$.

For example, for the set $A = \{1, 2, 3, 4, 5\}$, we see that

$$\{1, 2\} \cap \{2, 3\} = \{2\}.$$

and

$$\{1, 3, 5\} \cap \{2, 4\} = \emptyset.$$

Example 10.1.4: Function Composition Operation

Let S be a set, and let $\mathcal{F}$ denote the set of all functions f from S to S . We can then define an operation $\circ : \mathcal{F}^2 \rightarrow \mathcal{F}$ such that

$$o(f, g) = f \circ g$$

for all $f, g \in \mathcal{F}$, i.e. o acts like function composition.

Exercise 10.1: Set Difference

Let A be a set. Write down a definition for the set difference operation on the set A .

10.2. Axiom of Induction

Earlier, in [Section 3.5](#), we covered the basics of mathematical induction. However, so far, there has not been an explicit axiom proving that mathematical induction works as a proof technique.

Why does this matter?

Let $P(n)$ be a predicate that is defined for all natural numbers. Suppose $P(0)$ is true, and that if $P(k)$ is true then $P(k+1)$ (which really should be written as $P(S(k))$ for now) is true for all $k \in \mathbb{N}$. One clearly sees that $P(0)$ being true implies $P(1)$ being true, which thus means $P(2)$ is true, which thus means $P(3)$ is true, and so on. But at no point can we assert that $P(n)$ is true for all $n \in \mathbb{N}$! Each $P(n)$ can be proven using a finite number of steps, but to prove all of them requires an infinite number of steps, which we cannot perform.

But clearly mathematical induction (as stated above) clearly works! It is a very useful (and clever) proof technique that allows us to evade proving each case of $P(n)$ manually. And yet we cannot prove it using the tools that we have so far. Thus, we need an additional axiom.

The *axiom of induction* is **not** an axiom of ZFC. Rather, it can be seen as an axiom of logic when working with the natural numbers. However, to state it, we will appeal to the notation used in ZFC, although it would not be in the formal language of axiomatic set theory.

Axiom 10.2.1: Induction

$$\forall A, ((0 \in A) \wedge (\forall k \in \mathbb{N}, (k \in A \implies S(k) \in A))) \implies \mathbb{N} \subseteq A$$

Essentially, this axiom is the same as what we have covered in [Section 3.5](#). In the axiom, the set A can be thought of as the set containing elements satisfying our desired predicate $P(n)$. For the consequent of the axiom to hold, it requires two conditions.

- **Base Case:** $0 \in A$, i.e. the statement $P(0)$ must be true.
- **Induction Step:** For all natural numbers k , if $k \in A$ then $S(k) \in A$, i.e. if $P(k)$ is true for some k then $P(S(k))$ (which is the same as $P(k+1)$) is also true.

If both these conditions hold, then we know that all of the natural numbers belong in A (i.e., $P(n)$ is true for all $n \in \mathbb{N}$).

The axiom of induction is needed to prove the properties of the operations on the natural numbers.

10.3. Addition

Intuitively, addition is the simplest operation on the natural numbers. How hard can it be to add two numbers together? However, as we will see, creating a proper definition of addition will take some work.

10.3.1. Basic Definitions

Let us first define what addition is. We know that addition can be thought of as a binary operation on $\mathbb{N}$, but that isn't that helpful. What we want to do is to find properties that this operation must satisfy.

We can probably all agree that adding zero to any natural number should just return the number. It should be the case that $3 + 0 = 3$, just like how $42 + 0 = 42$, $101 + 0 = 101$, and $7125 + 0 = 7125$. So that should be the first property that addition must satisfy: adding zero to any natural number is just that natural number.

A less obvious property that addition has to satisfy is adding with successors. Consider $3 + S(4)$. Intuitively, we know that $S(4) = 4 + 1 = 5$, and so $3 + S(4) = 3 + 5 = 8$. Notice, however, that we can also write $3 + S(4)$ as $S(3 + 4) = (3 + 4) + 1 = 8$. This is the second thing that addition has to satisfy: adding a successor of a number to another number is the successor of the sum of the two numbers.

With these two properties, we can create a definition for addition.

Definition 10.3.1: Addition (Natural Numbers)

The **addition operation** is the function $+: \mathbb{N}^2 \rightarrow \mathbb{N}$ such that

- A1. for all $n \in \mathbb{N}$ we have $n + 0 = n$; and
- A2. for all $n, d \in \mathbb{N}$ we have $n + S(d) = S(n + d)$.

Surprisingly, these two conditions on the addition operation are all we need to define the rest of the properties of addition.

10.3.2. Basic Properties of Addition

The first result should be very obvious.

Proposition 10.3.2: Successor of n

$S(n) = n + 1$ for all $n \in \mathbb{N}$.

Proof. See [Exercise 10.2](#) (later). ■

Exercise 10.2: Successor of n

Prove [Proposition 10.3.2](#).

(Note: recall that we define $S(0) = 1$.)

The next result should also be obvious, but its proof is more involved.

Lemma 10.3.3: Sum of 0 and n

$0 + n = n$ for all $n \in \mathbb{N}$.

Notice that **A1** only works for $n + 0$, not $0 + n$. That is why we need to prove this as a separate lemma.

Proof. We induct on n .

For the base case $n = 0$, one clearly sees that $0 + 0 = 0$ by **A1**.

Now suppose the statement holds for some natural number k , i.e. $0 + k = k$. We are to prove that the statement holds for the successor of k , i.e. $0 + \mathcal{S}(k) = \mathcal{S}(k)$.

Let's start with $0 + \mathcal{S}(k)$. By **A2**, we know that it is the same as $\mathcal{S}(0 + k)$. Using the induction hypothesis we can immediately rewrite this as $\mathcal{S}(k)$, which is what we wanted to show. Thus the statement also holds for $\mathcal{S}(k)$.

By induction (via the axiom of induction (**Axiom 10.2.1**)), we have proven this proposition. ■

The following result is the 'reversed' version of **A2**.

Lemma 10.3.4: Sum of $\mathcal{S}(d)$ and n

$\mathcal{S}(d) + n = \mathcal{S}(d + n)$ for all $n, d \in \mathbb{N}$.

Proof. See **Exercise 10.3** (later). ■

We are now ready to prove the commutativity of addition.

Theorem 10.3.5: Commutativity of Addition (Naturals)

$m + n = n + m$ for all $m, n \in \mathbb{N}$.

Proof. Take m as an arbitrary natural number. We will induct on n .

For the base case, the left-hand side is $m + 0$ which is m by **A1**. The right-hand side is $0 + m$ which is also m by **Lemma 10.3.3**. Thus the two sides are equal, proving the base case.

Suppose that $m + k = k + m$ for some $k \in \mathbb{N}$. We will prove that $m + \mathcal{S}(k) = \mathcal{S}(k) + m$.

One sees that

$$\begin{aligned}
 m + \mathcal{S}(k) &= \mathcal{S}(m + k) && \text{(by A2)} \\
 &= \mathcal{S}(k + m) && \text{(induction hypothesis)} \\
 &= \mathcal{S}(k) + m && \text{(by Lemma 10.3.4)}
 \end{aligned}$$

which proves the case for $\mathcal{S}(k)$. ■

Exercise 10.3: Sum of $\mathcal{S}(d)$ and n

Prove **Lemma 10.3.4**.

We can now also prove the associativity of addition.

Theorem 10.3.6: Associativity of Addition (Naturals)

$(a + b) + c = a + (b + c)$ for all $a, b, c \in \mathbb{N}$.

Proof. Let a and b be arbitrary natural numbers. We induct on c .

For the base case $c = 0$, the left-hand side is $(a + b) + 0$. **A1** allows us to rewrite this as $a + b$. On the other hand, the right-hand side is $a + (b + 0)$ which **A1** again allows us to rewrite as $a + b$. Thus the base case is proven.

Suppose that $(a + b) + k = a + (b + k)$ for some $k \in \mathbb{N}$. We show that $(a + b) + \mathcal{S}(k) = a + (b + \mathcal{S}(k))$.

One sees clearly that

$$\begin{aligned}
 (a + b) + \mathcal{S}(k) &= \mathcal{S}((a + b) + k) && \text{(by A2)} \\
 &= \mathcal{S}(a + (b + k)) && \text{(induction hypothesis)} \\
 &= a + \mathcal{S}(b + k) && \text{(by A2)} \\
 &= a + (b + \mathcal{S}(k)) && \text{(by A2)}
 \end{aligned}$$

which proves the case for $\mathcal{S}(k)$. ■

Since addition is associative, we can just write expressions like $a + b + c$ without any ambiguity.

There is a useful corollary from these two theorems.

Corollary 10.3.7

$(a + b) + c = (a + c) + b$ for all $a, b, c \in \mathbb{N}$.

Proof. See **Exercise 10.4** (later). ■

Exercise 10.4: A Commutative Result

Prove **Corollary 10.3.7**.

(Hint: you do not need induction.)

10.3.3. Implications of Addition

With these basic properties of addition defined, we can look at some consequences with regards to addition.

Proposition 10.3.8: Cancellation Property of Addition

Let $a, b, n \in \mathbb{N}$. If $a + n = b + n$ then $a = b$. Likewise if $n + a = n + b$ then $a = b$.

Proof. We will only prove that if $a + n = b + n$ then $a = b$ since the other case is clearly taken care of by commutativity of addition (**Theorem 10.3.5**). We will do this by inducting on n .

When $n = 0$, the left-hand side is $a + 0 = b + 0$ which clearly simplifies to $a = b$ by **A1**. Thus the base case holds.

Suppose $a + k = b + k$ means $a = b$ for some $k \in \mathbb{N}$. We will show that $a + \mathcal{S}(k) = b + \mathcal{S}(k)$ means $a = b$.

We see that

$$\begin{aligned}
 a + \mathcal{S}(k) &= b + \mathcal{S}(k) \\
 \iff \mathcal{S}(a + k) &= \mathcal{S}(b + k) && \text{(by A2)} \\
 \iff a + k &= b + k && \text{(by Example 8.4.2)} \\
 \iff a &= b && \text{(induction hypothesis)}
 \end{aligned}$$

so the case for $\mathcal{S}(k)$ is also true. ■

Proposition 10.3.9: Addition that Equals Self

Let $a, b \in \mathbb{N}$. If $a + b = b$ then $a = 0$. Likewise if $a + b = a$ then $b = 0$.

Proof. See **Exercise 10.5** (later). ■

Proposition 10.3.10: Addition that Equals Zero

Let $a, b \in \mathbb{N}$. If $a + b = 0$ then $a = 0$ and $b = 0$.

Proof. We only prove the case that if $a + b = 0$ then $a = 0$, since the other is easily implied by commutativity of addition ([Theorem 10.3.5](#)).

There are two cases to consider. The first is if $b = 0$. Then one clearly sees that the left-hand side is $a + 0$ which is a by **A1**, and the right-hand side is 0. Thus $a = 0$ as required.

The other case is that $b \neq 0$. So b is a successor of some other natural number, say $d \in \mathbb{N}$. Thus we have $a + \mathcal{S}(d) = 0$, which by **A2** is the same as $\mathcal{S}(a + d) = 0$. So

$$\mathcal{S}(\underbrace{a + d}_{\text{in } \mathbb{N}}) = 0$$

which is a contradiction since 0 is not the successor of any natural number.

Thus we have $a = 0$ (and $b = 0$). ■

Exercise 10.5: Addition that Equals Self

Prove [Proposition 10.3.9](#).

10.4. Multiplication

With addition defined, we can now turn our attention to defining the next most important operation: multiplication.

10.4.1. Basic Definitions

Like with addition, it is of no use to us to just say that multiplication is a binary operation on $\mathbb{N}$. We need to find properties that multiplication satisfy in order to build a meaningful definition for it. Just like with addition, we will focus our attention on the interaction of multiplication with 0 and the successor of a number.

What happens when we multiply a number by zero? That is easy – the answer is just zero. But what happens when we multiply a number by the successor of another number? Well, by [Proposition 10.3.2](#), we know that $\mathcal{S}(n) = n + 1$, and we know by algebra that $x(y + 1) = xy + x$. So we can adopt this property into our definition for multiplication.

Definition 10.4.1: Multiplication (Natural Numbers)

The *multiplication operation* is the function $\times : \mathbb{N}^2 \rightarrow \mathbb{N}$ such that

M1. for all $n \in \mathbb{N}$ we have $n \times 0 = 0$; and

M2. for all $n, d \in \mathbb{N}$ we have $n \times \mathcal{S}(d) = (n \times d) + n$.

Like with addition, these two conditions will be all we need to prove the most essential properties of multiplication.

For the ease of notation, we may choose to omit the “ $\times$ ” when writing the multiplication of two algebraic symbols, i.e. ab is the same as $a \times b$ for $a, b \in \mathbb{N}$. However, if it serves to improve readability, we will write the “ $\times$ ” in between the symbols.

10.4.2. Commutativity

First, an easy warm-up property.

Proposition 10.4.2: Multiplication by One

$n \times 1 = n$ for all $n \in \mathbb{N}$.

Proof. One sees

$$\begin{aligned}
 n \times 1 &= n \times \mathcal{S}(0) && \text{(by definition of 1)} \\
 &= n \times 0 + n && \text{(by M2)} \\
 &= 0 + n && \text{(by M1)} \\
 &= n && \text{(by Lemma 10.3.3)}
 \end{aligned}$$

so $n \times 1 = n$ as required. ■

The next few results are analogous to those covered in the earlier section on addition. Again, note that **M1** works for $n \times 0$, not $0 \times n$, so we need this lemma.

Lemma 10.4.3: Product of 0 and n

$0 \times n = 0$ for all $n \in \mathbb{N}$.

Proof. See [Exercise 10.6](#) (later). ■

Lemma 10.4.4: Product of $\mathcal{S}(d)$ and n

$\mathcal{S}(d) \times n = (d \times n) + n$ for all $n, d \in \mathbb{N}$.

Proof. We consider induction on n .

When $n = 0$, we have $\mathcal{S}(d) \times 0$ on the left-hand side, which easily simplifies to 0 by **M1**. The right-hand side, upon substituting $n = 0$,

becomes $d \times 0 + 0$ which is $0 + 0$ by **M1** and becomes 0 upon using **A1**. Since the left-hand side and right-hand side both equal 0, thus the base case is proven.

Suppose that $\mathcal{S}(d) \times k = (d \times k) + k$ for some $k \in \mathbb{N}$. We show that $\mathcal{S}(d) \times \mathcal{S}(k) = (d \times \mathcal{S}(k)) + \mathcal{S}(k)$.

One sees that

$$\begin{aligned}
 \mathcal{S}(d) \times \mathcal{S}(k) &= (\mathcal{S}(d) \times k) + \mathcal{S}(d) && \text{(by M2)} \\
 &= ((d \times k) + k) + \mathcal{S}(d) && \text{(induction hypothesis)} \\
 &= (d \times k) + (k + \mathcal{S}(d)) && \text{(by associativity of addition)} \\
 &= (d \times k) + (k + (d + 1)) && \text{(by Proposition 10.3.2)} \\
 &= (d \times k) + ((k + d) + 1) && \text{(by associativity of addition)} \\
 &= (d \times k) + ((k + 1) + d) && \text{(by Corollary 10.3.7)} \\
 &= (d \times k) + (\mathcal{S}(k) + d) && \text{(by Proposition 10.3.2)} \\
 &= (d \times k) + (d + \mathcal{S}(k)) && \text{(by commutativity of addition)} \\
 &= ((d \times k) + d) + \mathcal{S}(k) && \text{(by associativity of addition)} \\
 &= (d \times \mathcal{S}(k)) + \mathcal{S}(k) && \text{(by M2)}
 \end{aligned}$$

which proves the case for $\mathcal{S}(k)$. ■

We are now ready to prove the commutativity of multiplication.

Theorem 10.4.5: Commutativity of Multiplication (Naturals)

$m \times n = n \times m$ for all $m, n \in \mathbb{N}$.

Proof. See **Exercise 10.6** (later). ■

Exercise 10.6: Proof of Commutativity

- (i) Prove **Lemma 10.4.3**.
- (ii) Prove **Theorem 10.4.5**.

With the commutativity of multiplication proven, we can prove some intuitive results.

Corollary 10.4.6: Multiplication by One (on the Left)

$1 \times n = n$ for all $n \in \mathbb{N}$.

Proof. One can use **Theorem 10.4.5** to see that $1 \times n = n \times 1$, which clearly equals n by **Proposition 10.4.2**. ■

10.4.3. Distributivity

Recall that one of the core interactions of multiplication with addition is distributivity. For example, we can evaluate $3 \times (4 + 5)$ in two ways. The first is to first calculate $4 + 5 = 9$, and then calculate $3 \times 9 = 27$. The second is to ‘expand’ the expression to become $3 \times 4 + 3 \times 5$, and then evaluating $3 \times 4 = 12$ and $3 \times 5 = 15$, and finally summing them both to get $12 + 15 = 27$.

This is summarised in the following theorem.

Theorem 10.4.7: Distributivity of $\times$ over $+$ (Naturals)

For all $a, b, c \in \mathbb{N}$ we have

$$\begin{aligned} a \times (b + c) &= (a \times b) + (a \times c) \text{ and} \\ (a + b) \times c &= (a \times c) + (b \times c). \end{aligned}$$

Proof. We prove the first statement via induction on a .

When $a = 0$, the left-hand side is $0 \times (b + c) = 0$ by [Lemma 10.4.3](#). The right-hand side is $0 \times b + 0 \times c = 0 + 0$ by [Lemma 10.4.3](#) again, which clearly equals 0 by **A1**. So the left-hand side equals the right-hand side, proving the base case.

Suppose $k \times (b + c) = (k \times b) + (k \times c)$ for some $k \in \mathbb{N}$. We show that $\mathcal{S}(k) \times (b + c) = (\mathcal{S}(k) \times b) + (\mathcal{S}(k) \times c)$.

Observe that

$$\begin{aligned} \mathcal{S}(k) \times (b + c) &= (k \times (b + c)) + (b + c) && \text{(by Lemma 10.4.4)} \\ &= ((k \times b) + (k \times c)) + (b + c) && \text{(induction hypothesis)} \\ &= ((k \times b) + b) + ((k \times c) + c) && \text{(by properties of addition)} \\ &= (\mathcal{S}(k) \times b) + (\mathcal{S}(k) \times c) && \text{(by Lemma 10.4.4)} \end{aligned}$$

which proves the case for $\mathcal{S}(k)$.

The second statement is readily proven by using [Theorem 10.4.5](#) on the first statement to get

$$(b + c) \times a = (a \times b) + (a \times c)$$

Then by ‘renaming’ b to a , c to b , and a to c , we will get that

$$(a + b) \times c = (c \times a) + (c \times b).$$

Using [Theorem 10.4.5](#) on the two products on the right-hand side readily proves the second theorem. ■

10.4.4. Associativity

With distributivity proven, we are ready to prove the associativity of multiplication.

Theorem 10.4.8: Associativity of Multiplication (Naturals)

$(a \times b) \times c = a \times (b \times c)$ for all $a, b, c \in \mathbb{N}$.

Proof. We induct on c .

When $c = 0$, the left-hand side is $(a \times b) \times 0$ which is 0 by **M1**. The right-hand side is

$$\begin{aligned} a \times (b \times 0) &= a \times 0 && \text{(by M1)} \\ &= 0 && \text{(by M1 again)} \end{aligned}$$

so the left-hand side and the right-hand side are the same, proving the base case.

Suppose that $(a \times b) \times k = a \times (b \times k)$ for some $k \in \mathbb{N}$. We will demonstrate that $(a \times b) \times \mathcal{S}(k) = a \times (b \times \mathcal{S}(k))$.

One can see that

$$\begin{aligned} &(a \times b) \times \mathcal{S}(k) \\ &= (a \times b) \times (k + 1) && \text{(by Proposition 10.3.2)} \\ &= ((a \times b) \times k) + ((a \times b) \times 1) && \text{(by Theorem 10.4.7)} \\ &= ((a \times b) \times k) + (a \times b) && \text{(by Corollary 10.4.6)} \\ &= (a \times (b \times k)) + (a \times b) && \text{(induction hypothesis)} \\ &= a \times ((b \times k) + b) && \text{(by Theorem 10.4.7)} \\ &= a \times ((b \times k) + (b \times 1)) && \text{(by Corollary 10.4.6)} \\ &= a \times (b \times (k + 1)) && \text{(by Theorem 10.4.7)} \\ &= a \times (b \times \mathcal{S}(k)) && \text{(by Proposition 10.3.2)} \end{aligned}$$

which proves the case for $\mathcal{S}(k)$. ■

10.4.5. Implications of Multiplication

Let us now look at some consequences with regards to multiplication.

Proposition 10.4.9: Product that Equals Zero

Let $a, b \in \mathbb{N}$. If $a \times b = 0$ then $a = 0$ or $b = 0$.

Proof. Seeking a contradiction, suppose $a \times b = 0$ and $a \neq 0$ and $b \neq 0$. Since $b \neq 0$, we can write $b = \mathcal{S}(p)$ for some $p \in \mathbb{N}$. So we see

$$\begin{aligned} 0 &= a \times \mathcal{S}(p) \\ &= (a \times p) + a \end{aligned}$$

which must mean $a \times p = 0$ and $a = 0$ by [Proposition 10.3.10](#), a contradiction to the assumption that $a \neq 0$. ■

Proposition 10.4.10: Cancellation Property of Multiplication

Let $a, b, n \in \mathbb{N} \setminus \{0\}$. If $a \times n = b \times n$ then $a = b$. Likewise if $n \times a = n \times b$ then $a = b$.

Proof. Like with [Proposition 10.3.8](#) we will prove $a \times n = b \times n$ means $a = b$ only. Unlike that proposition, however, we will induct on a .

Suppose $a = 1$. Seeking a contradiction, suppose $b \neq 1$. Since $b \neq 0$ also, we can write $b = \mathcal{S}(p)$ for some $p \in \mathbb{N}$. So we see

$$\begin{aligned} 1 \times n &= \mathcal{S}(p) \times n \\ &= (p \times n) + n && \text{(by [Lemma 10.4.4](#))} \\ &= (p \times n) + 1 \times n. && \text{(by [Corollary 10.4.6](#))} \end{aligned}$$

Adding 1 on both sides, we get

$$1 + (1 \times n) = 1 + (p \times n) + (1 \times n)$$

which means that

$$1 = 1 + p \times n$$

by cancellation property of addition ([Proposition 10.3.8](#)). But note that $1 + p \times n = (p \times n) + 1 = \mathcal{S}(p \times n)$, which means $1 = \mathcal{S}(p \times n) = \mathcal{S}(0)$. So $p \times n = 0$ since the successor function is injective, which thus means $p = 0$ or $n = 0$ by [Proposition 10.4.9](#). If $p = 0$ then $b = 1$, contradicting that $b \neq 1$; if $n = 0$ this contradicts $n \neq 0$. Thus we must have $b = 1$, i.e. $a = b = 1$. This proves the base case.

Now suppose $k \times n = b \times n$ means $k = b$ for some $k \in \mathbb{N} \setminus \{0\}$ and for all $b \in \mathbb{N} \setminus \{0\}$. We show that if $\mathcal{S}(k) \times n = b \times n$ then $\mathcal{S}(k) = b$.

Since $k \neq 0$, thus $\mathcal{S}(k) \neq 1$, which clearly means $b \neq 1$. So write $b = \mathcal{S}(m)$ for some $m \in \mathbb{N} \setminus \{0\}$; we get

$$\mathcal{S}(k) \times n = \mathcal{S}(m) \times n$$

which is equivalent to

$$(k \times n) + n = (m \times n) + n$$

by [Lemma 10.4.4](#). Thus $k \times n = m \times n$ by cancellation property of addition; using the induction hypothesis here means $k = m$, and thus $\mathcal{S}(k) = \mathcal{S}(m) = b$. This proves the case for $\mathcal{S}(k)$. ■

Proposition 10.4.11: Product that Equals Self

Let $a, b \in \mathbb{N}$. If $b \neq 0$ and $a \times b = b$ then $a = 1$. Likewise if $a \neq 0$ and $a \times b = a$ then $b = 1$.

Proof. See [Exercise 10.7](#) (later). ■

Proposition 10.4.12: Product that Equals 1

Let $a, b \in \mathbb{N}$. If $a \times b = 1$ then $a = 1$ and $b = 1$.

Proof. First, observe that $a \neq 0$ and $b \neq 0$; otherwise, the left-hand side will be 0 (by either [Corollary 10.4.6](#) or [Proposition 10.4.2](#)) while the right-hand side is 1. So we can write $a = \mathcal{S}(p)$ and $b = \mathcal{S}(q)$ for some $p, q \in \mathbb{N}$. Thus we have

$$\begin{aligned} \mathcal{S}(p) \times \mathcal{S}(q) &= 1 \\ \implies \mathcal{S}(p) \times q + \mathcal{S}(p) &= 1 && \text{(by M2)} \\ \implies (p \times q + q) + \mathcal{S}(p) &= 1 && \text{(by Lemma 10.4.4)} \\ \implies (p \times q + q) + (p + 1) &= 1 && \text{(by Proposition 10.3.2)} \\ \implies (p \times q) + p + q + 1 &= 1 && \text{(by properties of addition)} \\ \implies p \times q + p + q &= 0 && \text{(by Proposition 10.3.8)} \\ \implies (p \times q) + (p + q) &= 0. && \text{(by associativity of addition)} \end{aligned}$$

Hence we have $p \times q = 0$ and $p + q = 0$ by [Proposition 10.3.10](#); applying it again on $p + q = 0$ means $p = 0$ and $q = 0$, thus showing that $a = \mathcal{S}(p) = \mathcal{S}(0) = 1$ and $b = \mathcal{S}(q) = \mathcal{S}(0) = 1$. ■

Exercise 10.7: Product that Equals Self

Prove [Proposition 10.4.11](#).

10.5. Exponentiation

This last section will introduce exponentiation in the natural numbers. Just like how multiplication can be thought of as repeated addition, exponentiation can be thought of as repeated multiplication.

10.5.1. Basic Definitions

As before, we investigate the interactions of exponentiation with 0 and the successor of a number.

When we raise any non-zero number to the power of 0, what do we get? Before that, what does it mean to raise a number to the power of 0? Well, that means that we do not multiply anything at all. But what should this value be? Just like how multiplying by 1 does not change the product of a sequence of numbers, if we don't multiply by anything at all we should just get 1.

Remark: 0^0

The elephant in the room is, of course, the value of 0^0 . Some authors define this to be 1, some say 0, and others say that this is undefined. To make our lives easier for proving the properties of exponentiation later, let us just say that $0^0 = 1$. We will see this proved later.

Now what about if we raise any number to a successor of another number? Well, if that other number is a , then we know by [Proposition 10.3.2](#) that its successor is $a + 1$. Logically this should mean that we multiply one more copy of the base number.

We can thus define exponentiation as follows.

Definition 10.5.1: Exponentiation (Natural Numbers)

The *exponentiation operation* is the function $\wedge : \mathbb{N}^2 \rightarrow \mathbb{N}$ such that

- E1. for all $n \in \mathbb{N}$ we have $n \wedge 0 = 1$; and
- E2. for all $n, d \in \mathbb{N}$ we have $n \wedge \mathcal{S}(d) = (n \wedge d) \times n$.

Of course, we do not usually write “ $\wedge$ ” for exponentiation; we just use a superscript. Thus $(2 \wedge 3) \wedge 4 = (2^3)^4$. However, if it serves to improve readability, we will write the “ $\wedge$ ”.

Exercise 10.8: Exponentiation is Unlike the Rest

Which of the following statement(s) are true?

- (a) For all $a, b \in \mathbb{N}$ we have $a \wedge b = b \wedge a$.
- (b) For all $a, b, c \in \mathbb{N}$ we have $(a \wedge b) \wedge c = a \wedge (b \wedge c)$.

10.5.2. Basic Properties

Let us look at some basic properties of exponentiation.

Proposition 10.5.2: Zero to the Power of a Natural Number

$$0^n = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

for all $n \in \mathbb{N}$.

Proof. The case when $n = 0$ is obvious by using **E1**. Otherwise, we know that n is the successor of some other natural number, say m . Thus

$$\begin{aligned} 0^n &= 0^{S(m)} \\ &= 0^m \times 0 && \text{(by E2)} \\ &= 0, && \text{(by M1)} \end{aligned}$$

proving the second case. ■

Proposition 10.5.3: Natural Number to the Power of One

$n^1 = n$ for all $n \in \mathbb{N}$.

Proof. See [Exercise 10.9](#) (later). ■

Proposition 10.5.4: One to the Power of a Natural Number

$1^n = 1$ for all $n \in \mathbb{N}$.

Proof. We induct on n .

The base case of $n = 0$ is easily proven by using **E1**.

Suppose $1^k = 1$ for some $k \in \mathbb{N}$. We show $1^{S(k)} = 1$.

Note

$$\begin{aligned}
 1^{\mathcal{S}(k)} &= (1^k) \times 1 && \text{(by E2)} \\
 &= 1 \times 1 && \text{(induction hypothesis)} \\
 &= 1 && \text{(by Corollary 10.4.6)}
 \end{aligned}$$

which proves the case for $\mathcal{S}(k)$. ■

Proposition 10.5.5: Square

$n^2 = n \times n$ for all $n \in \mathbb{N}$.

Proof. We see that

$$\begin{aligned}
 n^2 &= n^{\mathcal{S}(1)} \\
 &= n^1 \times n && \text{(by E2)} \\
 &= n \times n && \text{(by Proposition 10.5.3)}
 \end{aligned}$$

which proves the proposition. ■

Exercise 10.9: Power of One

Prove Proposition 10.5.3.

10.5.3. Laws of Exponents

We end this chapter by looking at the laws of exponents.

Theorem 10.5.6: Product With Same Base

$a^m \times a^n = a^{m+n}$ for all $a, m, n \in \mathbb{N}$.

Proof. We induct on n .

For the base case $n = 0$, we see that

$$\begin{aligned}
 a^m \times a^0 &= a^m \times 1 && \text{(by E1)} \\
 &= a^m && \text{(by Proposition 10.4.2)} \\
 &= a^{m+0} && \text{(by A1)}
 \end{aligned}$$

so the base case is true.

Suppose $a^m \times a^k = a^{m+k}$ for some $k \in \mathbb{N}$. We will demonstrate that $a^m \times a^{\mathcal{S}(k)} = a^{m+\mathcal{S}(k)}$.

One sees that

$$\begin{aligned}
 a^m \times a^{\mathcal{S}(k)} &= a^m \times (a^k \times a) && \text{(by E2)} \\
 &= (a^m \times a^k) \times a && \text{(by Theorem 10.4.8)} \\
 &= a^{m+k} \times a && \text{(induction hypothesis)} \\
 &= a^{\mathcal{S}(m+k)} && \text{(by E1)} \\
 &= a^{m+\mathcal{S}(k)} && \text{(by A2)}
 \end{aligned}$$

which proves the case for $\mathcal{S}(k)$. ■

Theorem 10.5.7: Product to a Power

$a^n \times b^n = (a \times b)^n$ for all $a, b, n \in \mathbb{N}$.

Proof. See [Exercise 10.10](#) (later). ■

Theorem 10.5.8: Power Raised to a Power

$(a^m)^n = a^{m \times n}$ for all $a, m, n \in \mathbb{N}$.

Proof. We induct on n .

For the base case $n = 0$, one sees clearly that

$$\begin{aligned}
 (a^m)^0 &= 1 && \text{(by E1)} \\
 &= a^0 && \text{(by E1 again)} \\
 &= a^{m \times 0} && \text{(by M1)}
 \end{aligned}$$

which proves the base case.

Assume $(a^m)^k = a^{m \times k}$ for some $k \in \mathbb{N}$; we are to show that $(a^m)^{\mathcal{S}(k)} = a^{m \times \mathcal{S}(k)}$.

We observe that

$$\begin{aligned}
 (a^m)^{\mathcal{S}(k)} &= (a^m)^k \times a^m && \text{(by E2)} \\
 &= a^{m \times k} \times a^m && \text{(induction hypothesis)} \\
 &= a^{m \times k + m} && \text{(by Theorem 10.5.6)} \\
 &= a^{m \times \mathcal{S}(k)} && \text{(by M2)}
 \end{aligned}$$

which proves the case for $\mathcal{S}(k)$. ■

Exercise 10.10: Product to a Power

Prove **Theorem 10.5.7**.

PART II
Zahlen



Die ganzen Zahlen hat der liebe Gott gemacht, alles andere ist Menschenwerk.

(God made the integers, all else is the work of man.)

— Leopold Kronecker, 1893

([Web93](#), p. 15)

CHAPTER 11

Relations II

Earlier, we looked at the basics of relations. We looked at their definition, some basic properties of relations, and their composition. In this chapter, we will continue exploring relations, particularly relations related to equality and relations that create an ordering of elements in a set.

11.1. Equivalence Relations

Before we can define what an equivalence relation is, we define three properties that a relation could possess.

Definition 11.1.1: Reflexivity

A relation R on a set A is *reflexive* if and only if

$$\forall x \in A, x R x.$$

Definition 11.1.2: Symmetry

A relation R on a set A is *symmetric* if and only if

$$\forall x, y \in A, (x R y \implies y R x).$$

Definition 11.1.3: Transitivity

A relation R on a set A is *transitive* if and only if

$$\forall x, y, z \in A, (x R y \wedge y R z \implies x R z).$$

That was certainly a lot. Let us look at some examples to hammer in these definitions.

Example 11.1.4

Let the set $A = \{0, 1, 2, 3\}$. Define the relation

$$R = \{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (2, 2), (3, 0), (3, 3)\}.$$

It has the following directed graph.

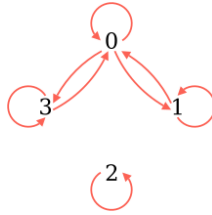


Figure 11.1. Directed Graph of R

We note the following.

- R is reflexive since $0 R 0$, $1 R 1$, and $2 R 2$. This can be seen in the figure by the loops from each element to itself.
- R is symmetric, since
 - given $0 R 1$ we also see $1 R 0$; and
 - given $0 R 3$ we also see $3 R 0$.

These are shown by the double arrows in the figure. For example, there is an arrow from 0 to 1 and an arrow from 1 to 0. There is also an arrow from 0 to 3 and an arrow from 3 to 0.

- R is not transitive, since $1 R 0$ and $0 R 3$ but $1 \not R 3$. This can be seen by the absence of an arrow from 1 to 3.

Example 11.1.5

Let the set $A = \{0, 1, 2, 3\}$. Define the relation

$$S = \{(0, 0), (0, 2), (0, 3), (2, 3)\}$$

on A , which has the following directed graph.

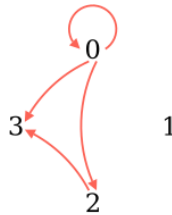


Figure 11.2. Directed Graph of S

We note the following.

- S is not reflexive since $1 \not S 1$. This can be seen by the absence of a loop from 1 to 1.
- S is not symmetric since, while $0 S 2$, we see $2 \not S 0$. This can be seen in the diagram by an arrow from 0 to 2 but the absence of an arrow from 2 to 0.
- S is transitive since $0 S 2$ and $2 S 3$ means $0 S 3$. This is shown in the diagram by an arrow from 0 to 3 after there were arrows from 0 to 2 and 2 to 3.

Exercise 11.1: A Relation on a Set

Let the set $A = \{0, 1, 2, 3\}$. Is the relation $T = \{(0, 1), (2, 3)\}$ on A

- (a) reflexive? (b) symmetric? (c) transitive?

With these three definitions, we can now define equivalence relations.

Definition 11.1.6: Equivalence Relation

An *equivalence relation on a set A* , usually denoted $\sim$, is a relation on A that is reflexive, symmetric, and transitive.

Example 11.1.7: An Equivalence Relation

Consider $A = \{0, 1, 2, 3\}$ and define a relation $\sim$ on A by

$$\sim = \{(0, 0), (1, 1), (2, 2), (3, 3)\}$$

which has the following directed graph.

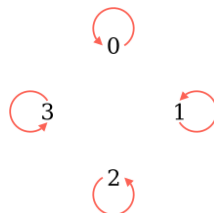


Figure 11.3. Directed Graph of $\sim$

We note that $\sim$ is

- reflexive since $0 \sim 0$, $1 \sim 1$, $2 \sim 2$, and $3 \sim 3$;
- symmetric since the symmetric requirements are implied by reflexivity; and
- transitive by vacuous definition of transitivity.

Therefore $\sim$ is an equivalence relation.

Example 11.1.8: Another Equivalence Relation

Consider $A = \{0, 1, 2, 3\}$ and define a relation $\sim$ on A by

$$\sim = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2), (3, 3)\}$$

which has the following directed graph.

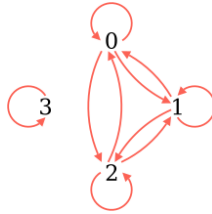


Figure 11.4. Directed Graph of $\sim$

We note that $\sim$ is

- reflexive since $0 \sim 0$, $1 \sim 1$, $2 \sim 2$, and $3 \sim 3$;
- symmetric since
 - given $0 \sim 1$ we have $1 \sim 0$;
 - given $0 \sim 2$ we have $2 \sim 0$; and
 - given $1 \sim 2$ we have $2 \sim 1$.

All other symmetric requirements are implied by reflexivity; and

- transitive since
 - given $0 \sim 1$ and $1 \sim 2$ we have $0 \sim 2$;
 - given $1 \sim 2$ and $2 \sim 0$ we have $1 \sim 0$; and

– given $2 \sim 0$ and $0 \sim 1$ we have $2 \sim 1$.

All other transitive requirements are implied by prior properties.

Therefore $\sim$ is an equivalence relation.

Example 11.1.9: An Equivalence Relation on a Formal Language

Let Σ be the set of English letters, i.e. $\Sigma = \{a, b, c, \dots, x, y, z\}$. Let $\mathcal{L}$ be the formal language using the alphabet Σ such that it contains strings formed from Σ .

Define a function $\text{Start}_n : \mathcal{L} \rightarrow \mathcal{L}$, where $n \in \mathbb{N} \setminus \{0\}$, such that

$$\text{Start}_n(s) = \begin{cases} s & \text{if length of } s \text{ is less than } n \\ \text{the first } n \text{ characters of } s & \text{otherwise} \end{cases}$$

for all $s \in \mathcal{L}$. For example, $\text{Start}_1(abc) = a$, $\text{Start}_2(abc) = ab$, $\text{Start}_3(abc) = abc$, and $\text{Start}_4(abc) = abc$.

We can now define a relation $\sim$ on $\mathcal{L}$ by

$$s \sim t \iff \text{Start}_3(s) = \text{Start}_3(t)$$

for all $s, t \in \mathcal{L}$. We show that $\sim$ is an equivalence relation.

Proof. We need to show that $\sim$ is reflexive, symmetric, and transitive.

- **Reflexive:** Let $s \in \mathcal{L}$. Clearly s has the same first 3 letters as itself, i.e. $\text{Start}_3(s) = \text{Start}_3(s)$. Thus $s \sim s$, proving that $\sim$ is reflexive.
- **Symmetric:** Let $s, t \in \mathcal{L}$ such that $s \sim t$. This means that $\text{Start}_3(s) = \text{Start}_3(t)$. But clearly this means that $\text{Start}_3(t) = \text{Start}_3(s)$, i.e. $t \sim s$, proving that $\sim$ is symmetric.
- **Transitive:** Let $s, t, u \in \mathcal{L}$ such that $s \sim t$ and $t \sim u$. So we have $\text{Start}_3(s) = \text{Start}_3(t)$ and $\text{Start}_3(t) = \text{Start}_3(u)$. But clearly this means that $\text{Start}_3(s) = \text{Start}_3(u)$, i.e. $s \sim u$, proving that $\sim$ is transitive.

Thus we see that $\sim$ is an equivalence relation. ■

Exercise 11.2: Equivalence Relation Detective

Let $S = \{1, 2, 3\}$. Which of the following relations are equivalence relations on S ?

- (a) $A = \emptyset$
- (b) $B = \{(1, 1), (2, 2), (3, 3)\}$
- (c) $C = \{(1, 1), (1, 2), (2, 2), (3, 3)\}$
- (d) $D = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3)\}$

With an equivalence relation $\sim$ on a set A , one might want to find the set of all elements that is related to a fixed element in a . This motivates the definition of equivalence classes.

Definition 11.1.10: Equivalence Class

Let $\sim$ be an equivalence relation on a set A , and suppose $a \in A$. The *equivalence class* of a under $\sim$ is

$$[a]_{\sim} = \{x \in A \mid x \sim a\}.$$

If the equivalence relation involved in the equivalence class is clear, we may just write $[a]$ for the equivalence class of a .

In other words,

$$x \in [a]_{\sim} \iff x \sim a.$$

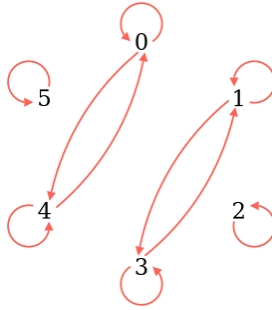
One clearly sees that $[x]$ is non-empty. This is because $x \sim x$ by reflexivity, which hence means $x \in [x]$.

Example 11.1.11: Equivalence Classes of an Equivalence Relation

Consider the equivalence relation

$$\begin{aligned} \sim = \{ & (0, 0), (0, 4), (1, 1), (1, 3), (2, 2), \\ & (3, 1), (3, 3), (4, 0), (4, 4), (5, 5) \} \end{aligned}$$

defined on the set $A = \{0, 1, 2, 3, 4, 5\}$, which has the following directed graph.

Figure 11.5. Directed Graph of $\sim$

Let us look at the equivalence class of each element in A .

- $[0] = \{x \in A \mid x \sim 0\} = \{0, 4\}$.
- $[1] = \{x \in A \mid x \sim 1\} = \{1, 3\}$.
- $[2] = \{x \in A \mid x \sim 2\} = \{2\}$.
- $[3] = \{x \in A \mid x \sim 3\} = \{3, 1\}$.
- $[4] = \{x \in A \mid x \sim 4\} = \{4, 0\}$.
- $[5] = \{x \in A \mid x \sim 5\} = \{5\}$.

Note that $[0] = [4]$ and $[1] = [3]$. Thus the distinct equivalence classes of $\sim$ are $\{0, 4\}$, $\{1, 3\}$, $\{2\}$, and $\{5\}$.

Notice in the above example that $0 \sim 4$ and $[0] = [4]$. Similarly we have $1 \sim 3$ and $[1] = [3]$. This observation forms part of a more important result regarding equivalence relations and equivalence classes.

Theorem 11.1.12: Equivalence Class Equivalence

Let $\sim$ be an equivalence relation on a set A , and let $x, y \in A$. Then the following statements are equivalent.

1. $x \sim y$
2. $[x] = [y]$
3. $[x] \cap [y] \neq \emptyset$

Proof. We will prove $(1) \implies (2) \implies (3) \implies (1)$.

- $(1) \implies (2)$ We are given $x \sim y$. Suppose $a \in [x]$. By definition of equivalence classes, this means $a \sim x$. But since $x \sim y$, thus $a \sim y$ by transitivity. So we see $a \in [y]$. Repeating this argument with x and y swapped means that everything in $[x]$ is in $[y]$ and vice versa, i.e. $[x] = [y]$.
- $(2) \implies (3)$ We are given that $[x] = [y]$. Pick any $a \in [x]$. Clearly $a \in [y]$. Thus $a \in [x] \cap [y]$, which means $[x] \cap [y] \neq \emptyset$.
- $(3) \implies (1)$ See [Exercise 11.4](#) (later).

This proves the theorem. ■

Exercise 11.3: Equal Cardinality Relation

Let X be a set. Define a relation $\sim$ on $\mathcal{P}(X)$ such that

$$A \sim B \iff |A| = |B|$$

for all $A, B \in \mathcal{P}(X)$.

- (a) Prove that $\sim$ is an equivalence relation.
- (b) Suppose $X = \{1, 2, 3\}$. Find the distinct equivalence classes of $\sim$.

Exercise 11.4: Complete the Proof!

Let $\sim$ be an equivalence relation on a set A , and let $x, y \in A$. Prove that if $[x] \cap [y]$ is non-empty, then $x \sim y$.

11.2. Partial Orders

Partial orders give us a way to compare elements of a set. However, not all pairs of elements need to be comparable – hence the term “partial”. Before we can properly look at them, we need to define a few more properties of relations.

11.2.1. More Properties of Relations

Earlier, we looked at reflexivity, symmetry, and transitivity. Let us introduce more of these properties, starting with antisymmetry.

Definition 11.2.1: Antisymmetry

A relation R on a set A is *antisymmetric* if and only if

$$\forall x, y \in A, (x R y \wedge y R x \implies x = y).$$

Remark: Antisymmetry Versus Not Symmetric

It is very important to realise that antisymmetry is **not** the opposite of symmetry. The negation of the symmetric condition is

$$\exists x, y \in A \text{ s.t } x R y \wedge y \not R x,$$

which is distinct from antisymmetry.

Example 11.2.2: The Subset Relation

Let X be the power set of any set. Define a relation R on X such that

$$A R B \iff A \subseteq B$$

for all $A, B \in X$. We see that R is antisymmetric – if $A \subseteq B$ and $B \subseteq A$, then $A = B$.

Exercise 11.5: Symmetric and Antisymmetric?

Does there exist a relation R on a set A that is both symmetric and antisymmetric?

Related to antisymmetry is the idea of asymmetry.

Definition 11.2.3: Asymmetry

A relation R on a set A is *asymmetric* if and only if

$$\forall x, y \in A, x R y \implies y \not R x.$$

Remark: Asymmetry Versus Not Symmetric

Notice that asymmetry is also **not** the opposite of symmetry.

Example 11.2.4: The Strict Subset Relation

Let X be the power set of any set. Define a relation R on X such that

$$A R B \iff A \subset B$$

for all $A, B \in X$. We see that R is asymmetric – if $A \subset B$, then $B \not\subset A$.

The last new property that we will look is irreflexivity.

Definition 11.2.5: Irreflexive

A relation R on a set A is *irreflexive* if and only if

$$\forall x \in A, x \not R x.$$

Remark: Irreflexive Versus Not Reflexive

Like with the other properties explored in this section, irreflexivity is **not** the negation of reflexivity.

Example 11.2.6: The “Element of” Relation

Let S be a set. Define a relation R on S such that

$$a R b \iff a \in b$$

for all $a, b \in S$. By the axiom of regularity ([Axiom 5.7.1](#)), we know that $x \notin x$ for all $x \in S$, so R is irreflexive.

There is an interesting relationship between asymmetry and irreflexivity.

Proposition 11.2.7: Asymmetry Implies Irreflexivity

Any asymmetric relation is irreflexive.

Proof. See [Exercise 11.6](#) (later). ■

There is a stronger result for transitive relations.

Proposition 11.2.8: Asymmetric $\iff$ Irreflexive

A transitive relation is asymmetric if and only if it is irreflexive.

Proof. See [Exercise 11.6](#) (later). ■

Exercise 11.6: Asymmetry and Irreflexivity

Let R be a relation on a set A .

- (a) Prove that if R is asymmetric, then R is irreflexive.
- (b) Prove that if R is transitive and irreflexive, then R is asymmetric.

11.2.2. Partial Order Relations

We are now in a position to introduce partial order relations.

Definition 11.2.9: Partial Order Relation

A relation $\preccurlyeq$ on a set A is a *partial order relation* (also known as a *partial order*) if and only if $\preccurlyeq$ is reflexive, antisymmetric, and transitive.

Example 11.2.10: The Subset Relation

Let $\mathcal{A}$ be a collection of sets. Define the relation $\preccurlyeq$ on $\mathcal{A}$ such that

$$A \preccurlyeq B \iff A \subseteq B$$

for all $A, B \in \mathcal{A}$. We show that $\preccurlyeq$ is a partial order relation.

Proof. We need to show $\preccurlyeq$ is reflexive, antisymmetric, and transitive.

- **Reflexive:** Clearly $A \subseteq A$ for any $A \in \mathcal{A}$, so $A \preccurlyeq A$ for all $A \in \mathcal{A}$.
- **Antisymmetric:** Suppose $A, B \in \mathcal{A}$ such that $A \preccurlyeq B$ and $B \preccurlyeq A$. This means that $A \subseteq B$ and $B \subseteq A$, which means we must have $A = B$.
- **Transitive:** Suppose $A, B, C \in \mathcal{A}$ such that $A \preccurlyeq B$ and $B \preccurlyeq C$. This means that $A \subseteq B$ and $B \subseteq C$. But clearly this means that $A \subseteq C$, so $A \preccurlyeq C$.

Therefore $\preccurlyeq$ is a partial order relation. ■

We can now define the “less than or equal to” relation for the natural numbers.

Definition 11.2.11: Less Than or Equal To Relation (Naturals)

The *less than or equal to relation* on the natural numbers is the relation $\leq$ on $\mathbb{N}$ where

$$m \leq n \iff m \subseteq n$$

for any $m, n \in \mathbb{N}$.

Let's unpack why this definition makes sense. Consider the natural number 4. We see that the counter set of 4, which is $\{0, 1, 2, 3\}$. Any number that is smaller than 4, say 2 (which has a counter set of $\{0, 1\}$), necessarily has to have a counter set that is a subset of $\{0, 1, 2, 3\}$. Extending this logic, any natural number whose counter set is a subset of the counter set of a natural number n is smaller than n .

Proposition 11.2.12: $\leq$ is a Partial Order

The less than or equal to relation, $\leq$, is a partial order on $\mathbb{N}$.

Proof. Use [Example 11.2.10](#). ■

Another relation that can be defined is the “divides” relation.

Definition 11.2.13: Divides Relation

The *divides relation* on the natural numbers is the relation $|$ on $\mathbb{N}$ where

$$d | n \iff \exists q \in \mathbb{N} \setminus \{0\} \text{ s.t. } n = dq$$

for all $n \in \mathbb{N}$.

For a fixed $n \in \mathbb{N}$, any $d \in \mathbb{N}$ where $d | n$ is called a *divisor of n* (or a *factor of n*), and the set of all such d are called the *divisors of n* (or *factors of n*).

Notice that we did not need to define division in order to define divisibility – the only thing needed to define divisibility is multiplication.

Proposition 11.2.14: $|$ is a Partial Order

The divides relation, $|$, is a partial order on $\mathbb{N}$.

Proof. See [Exercise 11.7](#) (later). ■

Example 11.2.15: $\mid$ on a Subset of $\mathbb{N}$

Consider $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. One sees that

- 1 divides everybody since $n = 1 \times n$ for all $n \in S$ ([Corollary 10.4.6](#));
- 2 divides 2, 4, 6, and 8 (where the q -values are 1, 2, 3, and 4 respectively);
- 3 divides 3, 6, and 9 (where the q -values are 1, 2, and 3 respectively);
- 4 divides 4 and 8 (where the q -values are 1 and 2 respectively); and
- 5, 6, 7, 8, and 9 divides only themselves in S .

Exercise 11.7: Divides Relation

Prove [Proposition 11.2.14](#).

11.2.3. Strict Partial Order Relations

Like how we can remove the “or equal to” condition for the “less than or equal to relation”, we can remove the requirement that any element must be related to itself to obtain a strict partial order.

Definition 11.2.16: Strict Partial Order Relation

A relation $\prec$ on a set A is a **strict partial order relation** (also known as a **strict partial order**) if and only if $\prec$ is irreflexive, asymmetric, and transitive.

Remark: Redundancy in the Definition

Note that we only require $\prec$ to be either

- irreflexive and transitive; or
- asymmetric and transitive

due to [Proposition 11.2.8](#). However, we explicitly state the three conditions in the definition for clarity.

Example 11.2.17: The Strict Subset Relation

Let $\mathcal{A}$ be a collection of sets. Define the relation $\prec$ on $\mathcal{A}$ such that

$$A \prec B \iff A \subset B$$

for all $A, B \in \mathcal{A}$. We show that $\prec$ is a strict partial order relation. We do this by proving irreflexivity and transitivity, as asymmetry would be implied by [Proposition 11.2.8](#).

- **Irreflexive:** Clearly $A \not\subset A$ since A does not contain an element that is not an element of A ; thus $A \not\prec A$ for all $A \in \mathcal{A}$.
- **Transitive:** Suppose $A \prec B$ and $B \prec C$. This means that $A \subset B$ and $B \subset C$. But clearly this means that $A \subset C$, so $A \prec C$.

Thus $\prec$ is a strict partial order relation.

Like how we defined the “less than or equal to” relation, we define the “less than” relation for the natural numbers.

Definition 11.2.18: Less Than Relation (Naturals)

The *less than relation* on the natural numbers is the relation $<$ on $\mathbb{N}$ where

$$m < n \iff m \subset n$$

for any $m, n \in \mathbb{N}$.

Proposition 11.2.19: $<$ is a Strict Partial Order

The less than relation, $<$, is a strict partial order on $\mathbb{N}$.

Proof. Use [Example 11.2.17](#). ■

To end this section, we note one interesting property that a (strict) partial order’s inverse has.

Proposition 11.2.20: Inverse of (Strict) Partial Order

Let A be a set and R be a relation on A . If R is a (strict) partial order on A , then R^{-1} is also a (strict) partial order on A .

Remark

For a partial order $\preceq$, we denote its inverse by $\succ$. Similarly, for a strict partial order $\prec$, we denote its inverse by $\succ$.

Proof. See [Exercise 11.8](#) (later). ■

With [Proposition 11.2.20](#) we can define the “greater than or equal to” partial order relation and “greater than” strict partial order relation on the natural numbers.

Definition 11.2.21: Greater Than or Equal To Relation (Naturals)

The *greater than or equal relation* on the natural numbers is the relation $\geq$ on $\mathbb{N}$ where $\geq = \leq^{-1}$, i.e. $m \geq n$ if and only if $n \leq m$ for any $m, n \in \mathbb{N}$.

Definition 11.2.22: Greater Than Relation (Naturals)

The *greater than relation* on the natural numbers is the relation $<$ on $\mathbb{N}$ where $< = >^{-1}$, i.e. $m > n$ if and only if $n < m$ for any $m, n \in \mathbb{N}$.

Exercise 11.8: Inverse of (Strict) Partial Order

Let A be a set.

- (a) If $\preceq$ is a partial order on A , prove that $\succ$ is also a partial order on A .
- (b) If $\prec$ is a strict partial order on A , prove that $\succ$ is also a strict partial order on A .

11.3. Hasse Diagrams

To motivate Hasse diagrams, consider the divides relation on the set $A = \{1, 2, 3, 9, 18\}$ (call it $\preceq$). One sees that

$$R = \{(1, 1), (1, 2), (1, 3), (1, 9), (1, 18), (2, 2), (2, 18), \\ (3, 3), (3, 9), (3, 18), (9, 9), (9, 18), (18, 18)\}$$

and has the following directed graph.

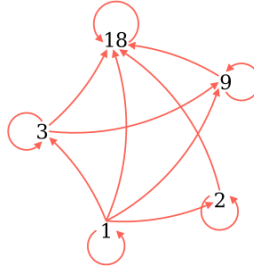


Figure 11.6. Directed Graph of $\preceq$

The directed graph exhibits a couple of redundancies for partial orders: we know that every element is related to itself, and transitivity holds. We can draw the directed graph of any partial order $\preceq$ in a simpler way, called a Hasse diagram, which is named after Helmut Hasse.

Definition 11.3.1: Hasse Diagram

Let $\preceq$ be a partial order on a set A . The *Hasse diagram* of $\preceq$ is obtained from the directed graph of $\preceq$ by performing the following steps.

1. Draw the directed graph of $\preceq$ such that all arrows (except self-loops) point upwards.
2. Remove any self-loops.
3. Remove all arrows that are implied by transitivity.
4. Remove the direction indicators on the arrows.

For the relation $\preceq$ above, the Hasse diagram is as follows (after some shifting of the elements).

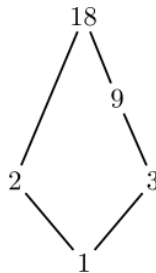


Figure 11.7. Hasse Diagram of $\preceq$

Example 11.3.2: The Subset Relation

Let $X = \mathcal{P}(\{0, 1, 2\})$. Consider the partial order relation $\preceq$ where

$$A \preceq B \iff A \subseteq B$$

for all $A, B \in X$. The Hasse diagram is as follows.

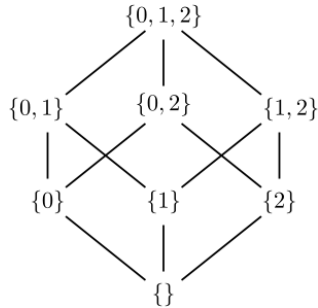


Figure 11.8. Hasse Diagram of $\preceq$

It is arguably easier to see the structure of the partial order from the Hasse diagram than from a directed graph.

Example 11.3.3: $|$ on a Subset of $\mathbb{N}$

Consider the divides relation on the set $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ as in [Example 11.2.15](#). The Hasse diagram is as follows.

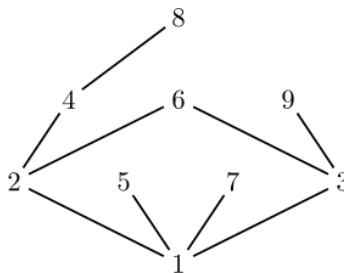


Figure 11.9. Hasse Diagram of $|$ on S

Remark: Primes in the Hasse Diagram

One might notice that the second level of the Hasse diagram (i.e., 2,

3, 5, and 7) are prime numbers. In fact, the lowest level are the numbers with no prime factors, the second highest level are the numbers with two prime factors, and so on.

The Hasse diagram does not necessarily need to have one “root” element.

Example 11.3.4: $|$ on Another Subset of $\mathbb{N}$

The divides relation on the set $S = \{2, 3, 4, 5, 7, 8, 9, 10, 16, 20, 21, 27\}$ has the following Hasse diagram.

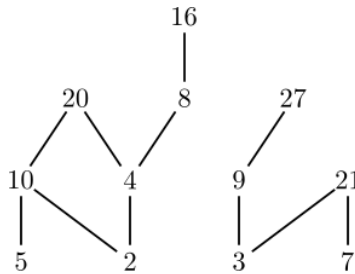
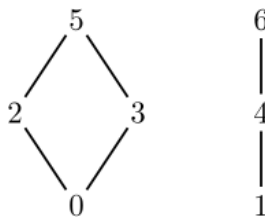


Figure 11.10. Hasse Diagram of $|$ on S

Notice now that there are 4 “root” elements (namely the primes 2, 3, 5, and 7). There are also two distinct “parts” to the Hasse diagram – the part with the “roots” of 2 and 5 and the part with the “roots” of 3 and 7.

Exercise 11.9: Relation Recovery

Let $\preceq$ be the partial order on $A = \{0, 1, 2, 3, 4, 5\}$ with the following Hasse diagram.



List the elements in $\preceq$.

11.4. Maximal, Minimal, Greatest, and Least Elements

One might notice that, in a Hasse diagram, some elements are at the “bottom” of the diagram with no edge pointing into them. There could also be some elements at the “top” of the diagram with no edge coming out of them. The way that the elements are positioned from bottom to top creates a sort-of ‘ordering’ of the elements; these “bottom” and “top” elements have special names.

Definition 11.4.1: Maximal and Minimal Elements

Let $\preceq$ be a partial order relation on a set A .

- An element $m \in A$ is called a *maximal element of A* if and only if, for all $x \in A$, if $m \preceq x$ then $x = m$.
- An element $m \in A$ is called a *minimal element of A* if and only if, for all $x \in A$, if $x \preceq m$ then $x = m$.

A related construct is the idea of greatest and least elements.

Definition 11.4.2: Greatest and Least Elements

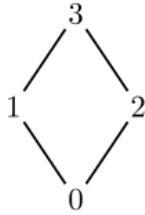
Let $\preceq$ be a partial order relation on a set A .

- An element $m \in A$ is called a *greatest element of A* if and only if, for all $x \in A$ we have $x \preceq m$.
- An element $m \in A$ is called a *least element of A* if and only if, for all $x \in A$ we have $m \preceq x$.

Note the difference between maximal and greatest elements. Greatest elements require that **all** elements be related to the greatest element, whereas maximal elements have the antecedent that x is related to m , so not all elements need to be related to m . Similarly, least elements require that **all** elements be related to the least element, whereas minimal elements have the antecedent that x is related to m .

Example 11.4.3

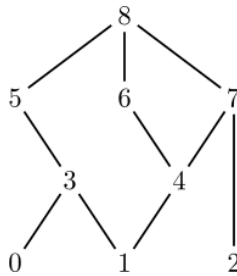
Let $\preceq$ be the partial order on $A = \{0, 1, 2, 3\}$ with the following Hasse diagram.



One sees that $\preccurlyeq$ has minimal (and least) element 0, and has a maximal (and greatest) element 3.

Example 11.4.4

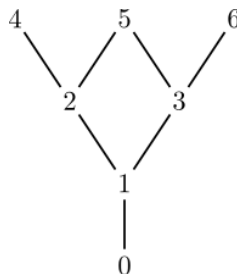
Let $\preccurlyeq$ be the partial order on $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$ with the following Hasse diagram.



It has minimal elements 0, 1, and 2 (and no least element), and has a maximal (and greatest) element of 8.

Example 11.4.5

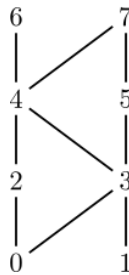
Let $\preccurlyeq$ be the partial order on $A = \{0, 1, 2, 3, 4, 5, 6\}$ with the following Hasse diagram.



That partial order has minimal (and least) element 0, and has a maximal elements 4, 5, and 6 (and no greatest element).

Example 11.4.6

Let $\preceq$ be the partial order on $A = \{0, 1, 2, 3, 4, 5, 6, 7\}$ with the following Hasse diagram.



This partial order has minimal elements 0 and 1 (and thus no least element), and has maximal elements 6 and 7 (and thus no greatest element).

Exercise 11.10: Element Order Properties

Let $\preceq$ be a partial order relation on a set A . Prove or disprove the following statements.

- (a) If there is a greatest element, it is unique.
- (b) If there is a least element, it is unique.
- (c) If there is more than one maximal element, then there is no greatest element.
- (d) If there is more than one minimal element, then there is no least element.

11.5. Total Orders

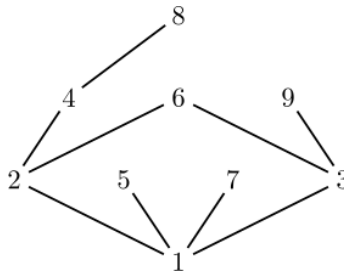
Recall in [Example 11.3.4](#) that the Hasse diagram has two distinct parts. Elements in one part does not divide any element in the other part, and vice versa. This motivates the idea of comparability.

Definition 11.5.1: Comparability

Let $\preceq$ be a partial order relation on a set A . Then $a, b \in A$ are said to be **comparable** if and only if either $a \preceq b$ or $b \preceq a$. Otherwise they are said to be **noncomparable**.

Example 11.5.2

Consider the divides relation in [Example 11.2.15](#), which has the following Hasse diagram.



From the Hasse diagram, we can see that $2 \mid 4$, so 2 and 4 are comparable. On the other hand, notice that $5 \nmid 8$ and $8 \nmid 5$, so 5 and 8 are noncomparable.

What happens when all elements in a set are comparable by a partial order relation? That results in a total order.

Definition 11.5.3: Total Order

A partial order $\preceq$ on a set A is a **total order relation** (or a **total order**) if and only if a and b are comparable for all $a, b \in A$.

What does it mean for a partial order to be a total order? In terms of the Hasse diagram, this means that all the elements are in a straight line, from the bottom to the top. This is why some authors call total orders **linear orders**.

Example 11.5.4: $\mid$ on Yet Another Subset of $\mathbb{N}$

Consider the divides relation on the subset $A = \{1, 2, 4, 8\}$ of $\mathbb{N}$. One sees that, for any $a, b \in A$, they are comparable. Thus the divides relation on A is a total order. One can see this by looking at its Hasse diagram.



Figure 11.11. $|$ is a Total Order on A

The Hasse diagram is a straight line – a clear sign that $|$ is a total order on A . We can also see that, under $|$, the minimal (and least) element of A is 1, and the maximal (and greatest) element of A is 8.

Example 11.5.5: The Subset Relation

In general, the subset relation is **not** a total order. One can look at [Example 11.3.2](#) to see why.

There is one essential total order that we will look at – the “less than or equal to” relation on that natural numbers.



Figure 11.12. $\leq$ is a Total Order on $\mathbb{N}$

One observes from the Hasse diagram that “vertical” property indicative of a total order. We note that the minimal (and least) element of $\mathbb{N}$ under $\leq$ is 0, and there is no maximal (and greatest) element.

Proposition 11.5.6: $\leq$ is a Total Order on $\mathbb{N}$

$\leq$ is a total order on $\mathbb{N}$.

Proof. We are to show that, for any $m, n \in \mathbb{N}$, either $m \leq n$ or $n \leq m$.

Suppose otherwise, i.e. there is $m, n \in \mathbb{N}$ such that neither $m \leq n$ nor $n \leq m$. This means that neither $m \subseteq n$ nor $n \subseteq m$, i.e. $m \not\subseteq n$ and $n \not\subseteq m$.

However, note how $\mathbb{N}$ is constructed. If $m \not\subseteq n$, then it has to be the case that m has at least one more element than n . In particular, by the way that $\mathbb{N}$ is defined, we know that m must contain the elements of n and n itself, i.e. $\mathcal{S}(n) \subseteq m$. But note $n \subseteq \mathcal{S}(n)$, and thus $n \subseteq m$, contradicting that $n \not\subseteq m$.

Thus we see that $\leq$ is a total order on $\mathbb{N}$. ■

CHAPTER 12

Defining the Integers

Constructing more general numbers requires the numbers we defined before; we will use natural numbers and equivalence relations to create a definition for the integers. We will also define relationships between integers, just like we did with the natural numbers.

12.1. The Insufficiency of the Natural Numbers

Consider the action of adding two natural numbers, say 2 and 3. We know that $3 + 4 = 7$ by definition, but how do we express this diagrammatically?

Let us consider a *number line*. We place the natural numbers on the number line, and draw a line between each pair of adjacent numbers.

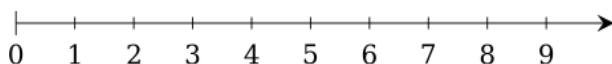


Figure 12.1. The Number Line

When we add 4 to 3, we first move right by 3 (see the first arrow below) and then move along by 4 afterwards (see the second arrow below).



Figure 12.2. Adding 4 to 3

Of course, the result is the same if we first move right by 4 and then move along by 3, due to commutativity of addition ([Theorem 10.3.5](#)).

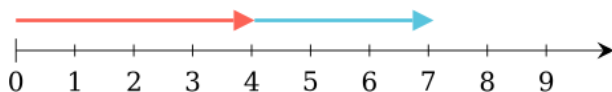


Figure 12.3. Adding 3 to 4

What if there was a way to “undo” addition? Suppose we have the result of 7, and we knew that one of the numbers is 3. What is the other number? Diagrammatically, this looks like moving left by 3 from 7.

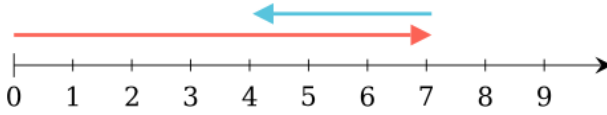


Figure 12.4. Undoing Adding 3 From 7

For some numbers, we are able to find a natural number if we perform this “undo” operation. For others, we can’t. For example, what if we move left by 4 from 1? There simply is no natural number that is there.

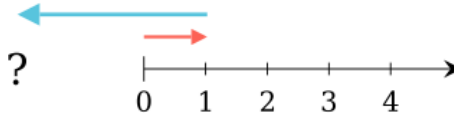


Figure 12.5. Undoing Adding 3 From 1

This is the insufficiency of the natural numbers that we wish to address by defining the integers – a way to move an arbitrary number of units left or right on the number line.

12.2. Leftward and Rightward “Shifts” of Natural Numbers

Let us use these shifts to define numbers. The natural numbers can easily be defined this way – for example, the natural number 4 can be thought of as the number obtained by performing a rightward shift of 7 followed by a leftward shift of 3.

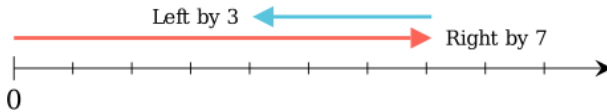


Figure 12.6. Creating 4 By Using Leftward and Rightward “Shifts”

We may denote this pair of shifts by $(7, 3)$, signifying that we shift rightward by 7 and then shift leftward by 3. Of course, this is not the only combination of shifts that result in 4. We could have, for instance, shifted rightward by 5 and then leftward by 1, or shifted rightward by 6 and then leftward by 2, or shifted rightward by 8 and then leftward by 4, and so on.

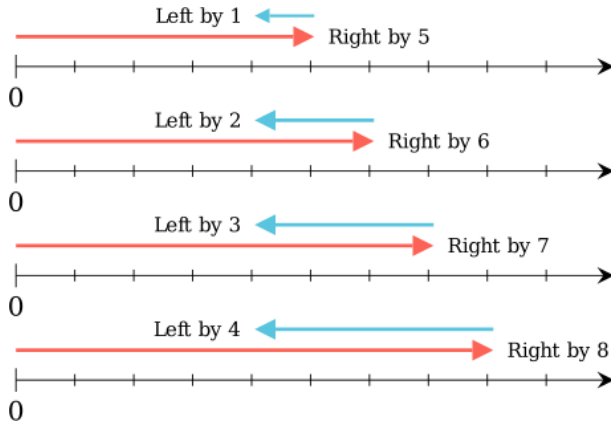


Figure 12.7. Other Ways to Create 4

But clearly all these shifts just result in 4! We want a way to say that these different combinations of shifts are the same. To put it more concretely, we want an equivalence relation $\sim$ on the set $\mathbb{N}^2$ such that $(a, b) \sim (c, d)$ if and only if a rightward shift of a followed by a leftward shift of b is the same as a rightward shift of c followed by a leftward shift of d , for all $a, b, c, d \in \mathbb{N}$. That way, we can say that $(5, 1) \sim (6, 2)$, $(6, 2) \sim (7, 3)$, $(7, 3) \sim (8, 4)$, and so on.

Exercise 12.1: Defining the “Shift Equivalence” Relation

Create a definition for $\sim$ as mentioned above, proving that it is an equivalence relation.

(Note: you are restricted to using the operations defined in [Chapter 10](#).)

Of course, there’s nothing stopping us from creating a resultant shift that is not within the natural numbers. Consider the shift defined by $(2, 5)$. This means that we first shift rightward by 2, and then shift leftwards by 5. But clearly the result (marked by the question mark) is not a natural number, and yet it does not matter.

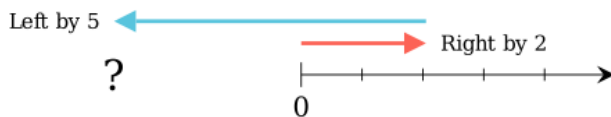


Figure 12.8. A Non-Natural Number Combination of “Shifts”

Again, there are other combinations of shifts that result in ‘?’, and they

are all equivalent via $\sim$. For example, one sees $(1, 4) \sim (2, 5)$, $(2, 5) \sim (3, 6)$, $(3, 6) \sim (4, 7)$, and so on.

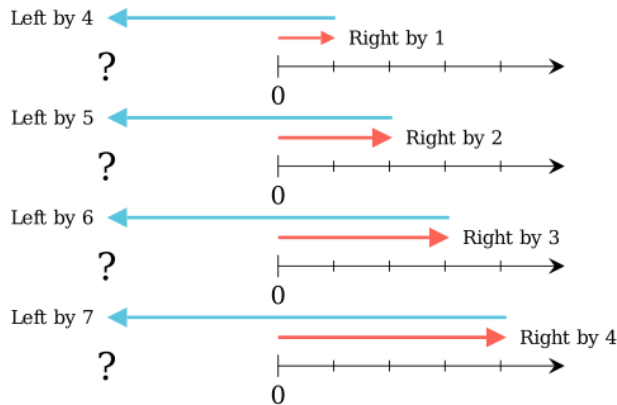


Figure 12.9. Other Ways to Create ‘’

With the equivalence relation $\sim$, we are ready to define the integers.

12.3. The Set of Integers

Notice that the shifts of $(5, 1)$, $(6, 2)$, $(7, 3)$, and $(8, 4)$ all represent the natural number 4. We can thus think of all these shifts as the same thing by considering their equivalence class under $\sim$. Since they are all related to each other, thus $[(5, 1)] = [(6, 2)] = [(7, 3)] = [(8, 4)] = \dots$ (**Theorem 11.1.12**). One notices that $[(4, 0)] = [(5, 1)]$, and it is precisely the natural number 4 that the shifts in the equivalence class $[(4, 0)]$ represent. Thus, we can define the integer 4 to be $[(4, 0)]$. We can do this for the rest of the natural numbers.

Definition 12.3.1

Let n be a natural number. The integer n is the equivalence class $[(n, 0)]$.

What about the shifts $(1, 4)$, $(2, 5)$, $(3, 6)$, $(4, 7)$, and so on? These are not equivalent to any natural number, but are numbers found on the ‘left’ of 0. In particular, these shifts all result in an overall leftward shift of 3, so they all belong in the equivalence class $[(0, 3)]$. We can define an integer to represent this equivalence class.

Definition 12.3.2

Let n be a non-zero natural number. The integer $-n$, read as “negative n ”, is the equivalence class $[(0, n)]$.

We can visualize these equivalence classes on a two-dimensional grid. A shift (m, n) is represented by a point that is m along the horizontal axis and n along the vertical axis, and shifts belonging in the same equivalence class lie on the same dotted line.

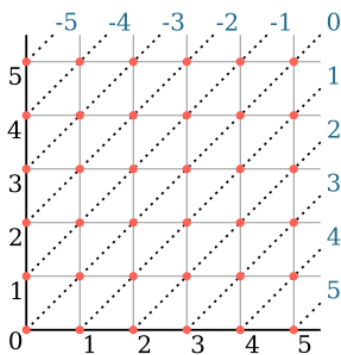


Figure 12.10. Integers as Equivalence Classes

With these definitions out of the way, we are now ready to define the set of integers.

Definition 12.3.3: The Set of Integers

The *set of integers*, denoted $\mathbb{Z}$, is the set of equivalence classes under $\sim$. That is,

$$\mathbb{Z} = \{[(m, n)] \mid m, n \in \mathbb{N}\}$$

where $[(m, n)]$ is the equivalence class of (m, n) under $\sim$.

Remark: $\mathbb{N}$ as a Subset of $\mathbb{Z}$

One notes that the equivalence classes are not exactly natural numbers. However, we will think of the natural numbers as being ‘embedded’ in the integers. This means that when we write $\mathbb{N} \subseteq \mathbb{Z}$, we think of the representative equivalence classes of the natural numbers. More concretely, think about $\mathbb{N}$ within $\mathbb{Z}$ is that $\mathbb{N}$, in this

context, is the range of the function $\mathcal{E}_{\mathbb{N}} : \mathbb{N} \rightarrow \mathbb{Z}$ where

$$\mathcal{E}_{\mathbb{N}}(n) = [(n, 0)].$$

Let us now define the negation of an integer.

Definition 12.3.4: Negation

The **negation operation** is a unary operation given by the function $\text{neg} : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$\text{neg}([(m, n)]) = [(n, m)].$$

We denote the negation of an integer x by $-x$. That is, for any integer $x = [(m, n)]$, where $m, n \in \mathbb{N}$, we have $-x = [(n, m)]$.

Example 12.3.5

- $1 = [(1, 0)]$ and $-1 = [(0, 1)]$.
- $5 = [(5, 0)]$ and $-5 = [(0, 5)]$.
- $0 = [(0, 0)]$ and $-0 = [(0, 0)]$, which is the same as 0.
- $-3 = [(0, 3)]$ and $-(-3) = [(3, 0)]$, which is the same as 3.

The observation that $-(-3)$ is the same as 3 leads us to define a more general result.

Proposition 12.3.6: Negation of Negative

$-(-x) = x$ for all $x \in \mathbb{Z}$.

Proof. Suppose $x = [(m, n)]$ where $m, n \in \mathbb{N}$. Note that $-x = [(n, m)]$ by definition of negation, and so $-(-x) = [(m, n)] = x$. ■

With negation defined, we can define the absolute value operation.

Definition 12.3.7: Absolute Value

The **absolute value operation** is a unary operation given by the function $\text{abs} : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$\text{abs}([(m, n)]) = \begin{cases} [(m, n)] & \text{if } n \leq m \\ [(n, m)] & \text{if } m < n \end{cases}$$

We denote the absolute value of an integer x by $|x|$. That is, for any integer x we have

$$|x| = \begin{cases} x & \text{if } x \in \mathbb{N} \\ -x & \text{if } x \notin \mathbb{N} \end{cases}$$

Remark: $|\cdot|$ As Cardinality Versus $|\cdot|$ As Absolute Value

Recall that $|A|$ denotes the cardinality of a set A , which can be confusing when $|n|$ denotes the absolute value of an integer n . If there is any chance of confusion, we will use $\text{abs}(n)$ to denote the absolute value of an integer instead of $|n|$.

Example 12.3.8

- $|1| = \text{abs}([(1,0)]) = [(1,0)] = 1.$
- $|0| = \text{abs}([(0,0)]) = [(0,0)] = 0.$
- $|-3| = \text{abs}([(0,3)]) = [(3,0)] = 3.$

12.4. Ordering of Integers

Earlier, we defined the “less than or equal to” relation ($\leq$) on the natural numbers, which is a total order on the naturals ([Proposition 11.5.6](#)) and thus gives us a way to order the natural numbers. We can do the same for the integers.

Definition 12.4.1: Less Than or Equal To Relation (Integers)

The *less than or equal to relation* on the integers is the relation $\leq$ on $\mathbb{Z}$ where

$$[(a,b)] \leq [(c,d)] \iff a + d \leq b + c$$

for any $[(a,b)], [(c,d)] \in \mathbb{Z}$.

Note that the “ $\leq$ ” on the right-hand side of the definition is the “less than or equal to” partial order on the natural numbers.

We can similarly define the “less than” relation on the integers.

Definition 12.4.2: Less Than Relation (Integers)

The *less than relation* on the integers is the relation $<$ on $\mathbb{Z}$ where

$$[(a, b)] < [(c, d)] \iff a + d < b + c$$

for any $[(a, b)], [(c, d)] \in \mathbb{Z}$.

Exercise 12.2: Proving Partial Orders

Prove that

- (a) $\leq$ on $\mathbb{Z}$ is a partial order; and
- (b) $<$ on $\mathbb{Z}$ is a strict partial order.

Using [Proposition 11.2.20](#) we can also define the “greater than or equal to” partial order relation and “greater than” strict partial order relation on the integers.

Definition 12.4.3: Greater Than or Equal To Relation (Integers)

The *greater than or equal relation* on the integers is the relation $\geq$ on $\mathbb{Z}$ where $\geq = \leq^{-1}$, i.e. $m \geq n$ if and only if $n \leq m$ for any $m, n \in \mathbb{Z}$.

Definition 12.4.4: Greater Than Relation (Integers)

The *greater than relation* on the integers is the relation $>$ on $\mathbb{Z}$ where $> = <^{-1}$, i.e. $m > n$ if and only if $n < m$ for any $m, n \in \mathbb{Z}$.

It should come as no surprise that, like with the natural numbers, the “less than or equal to” relation ($\leq$) on the integers is a total order.

Proposition 12.4.5: $\leq$ is a Total Order on $\mathbb{Z}$

$\leq$ on $\mathbb{Z}$ is a total order.

Proof. We are to show that, for any $m, n \in \mathbb{Z}$, either $m \leq n$ or $n \leq m$.

Let $m, n \in \mathbb{Z}$. Write

$$m = [(a_m, b_m)]$$

$$n = [(a_n, b_n)]$$

where $a_m, b_m, a_n, b_n \in \mathbb{N}$. Note that $a_m + b_n \in \mathbb{N}$ and $b_m + a_n \in \mathbb{N}$. Since $\leq$ on $\mathbb{N}$ is a total order, thus we must have either $a_m + b_n \leq b_m + a_n$ or $b_m + a_n \leq a_m + b_n$. The first case leads to $[(a_m, b_m)] \leq [(a_n, b_n)]$ and the second leads to $[(a_n, b_n)] \leq [(a_m, b_m)]$, i.e. either $m \leq n$ or $n \leq m$. This proves that $\leq$ on $\mathbb{Z}$ is a total order. ■

We can draw a Hasse diagram for $\leq$ on $\mathbb{Z}$.

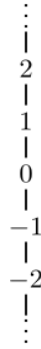


Figure 12.11. $\leq$ is a Total Order on $\mathbb{Z}$

One clearly sees that $\mathbb{Z}$ has no maximal, minimal, greatest, and least element.

12.5. Cardinality of the Integers

To end this chapter, let us examine the size (that is, cardinality) of the set of integers. We know that the natural numbers is a subset of the integers, so we know that the set of integers is also an infinite set. This observation is generalizable by the following proposition.

Proposition 12.5.1: Set Containing Infinite Subset is Infinite

Let S be a set. If T is an infinite subset of S , then S is infinite.

Proof. We prove this by contrapositive. The proposition's statement is that "if S has an infinite subset T , then S is an infinite set". Thus the contrapositive is "if S is a finite set, then S has a finite subset T ".

So suppose S is finite. Choose $T = S$. Clearly T is a subset of S , and that it is finite. This proves the proposition. ■

Corollary 12.5.2: $\mathbb{Z}$ is Infinite

$\mathbb{Z}$ is an infinite set.

Proof. Since $\mathbb{N}$ is a subset of $\mathbb{Z}$ that is infinite, the above proposition tells us that $\mathbb{Z}$ is infinite. ■

The more interesting question is whether the cardinality of the set $\mathbb{Z}$ is more than $\mathbb{N}$. Clearly $-1 \in \mathbb{Z}$ but $-1 \notin \mathbb{N}$, so $\mathbb{Z}$ clearly has “more elements”, but we’ve seen how cardinalities of infinite sets are not intuitive. In fact, we will prove that the cardinality of the integers is the same as the cardinality of the natural numbers.

12.5.1. Odd and Even Numbers

To prove this fact, we will first introduce a categorization of the natural numbers – the odd and even numbers.

Definition 12.5.3: Odd and Even Numbers

The natural number n is said to be *even* if and only if there exists a natural number k such that $n = 2k$. A natural number that is not even is said to be *odd*.

Proposition 12.5.4: Formula For An Odd Number

Any odd number takes the form $2k + 1$ where $k \in \mathbb{N}$.

Proof. Let n be an odd natural number. We can write

$$n = 0 + \underbrace{1 + 1 + 1 + 1 + 1 + \cdots + 1}_{n \text{ times}}$$

by definition of the natural numbers. By associativity of addition on natural numbers, we can each pair of 1s as follows.

$$n = 0 + (1 + 1) + (1 + 1) + (1 + 1) + \cdots$$

Since $1 + 1 = 2$, we can replace all these groups of “ $1 + 1$ ” with 2. So we

must have

$$\begin{aligned}
 n &= 0 + 2 + 2 + \cdots + 2, \text{ or} \\
 n &= 0 + 2 + 2 + \cdots + 2 + 1, \text{ or} \\
 n &= 0 + 2 + 2 + \cdots + 2 + 1 + 1, \text{ or} \\
 n &= 0 + 2 + 2 + \cdots + 2 + 1 + 1 + 1, \text{ or} \\
 n &= 0 + 2 + 2 + \cdots + 2 + 1 + 1 + 1 + 1, \text{ or} \\
 &\vdots \quad \vdots
 \end{aligned}$$

But the first case is not possible since that simplifies to a form of $n = 2k$ where $k \in \mathbb{N}$, and the third, fourth, fifth etc. cases are not possible since we can just group additional “1 + 1”s into another group of 2. The last remaining case is the second case which takes the form $n = 2k + 1$ for some $k \in \mathbb{N}$. ■

12.5.2. Equinumerosity of the Integers and the Naturals

With odd and even numbers defined, let’s state the theorem that we are to prove.

Theorem 12.5.5: Integers Are Equinumerous With Naturals

$$|\mathbb{Z}| = |\mathbb{N}|.$$

How do we approach this? Let us use Cantor-Schröder-Bernstein ([Theorem 9.2.9](#)) to make our lives easier. Recall that $\mathbb{N}$ is (thought of as) a subset of $\mathbb{Z}$, so all we need to find is an injection from $\mathbb{Z}$ to $\mathbb{N}$ and we are done (since the heavy lifting is given by [Corollary 9.2.11](#)).

For this exploration step, let us ignore the fact that we haven’t define operations on the integers. How can we create an injection from $\mathbb{Z}$ to $\mathbb{N}$? With the result of odd and even numbers, one might think of a way to map all the non-negative integers to the even numbers and the negative integers to all the odd numbers. That way, we will have a injective map from $\mathbb{Z}$ to $\mathbb{N}$.

To map the non-negative integers to the even natural numbers is easy – just double every non-negative integer. 0 becomes 0, 2 becomes 4, 4 becomes 8, 6 becomes 12, and so on. How about the negative integers? We need -1 to become 1, -2 to become 3, -3 to become 5, -4 to become 7, and so on. One might notice that, for any odd integer n , the resulting natural number is $-2n - 1$.

In summary, the injection that we came up with is

$$f(n) = \begin{cases} 2n & \text{if } n \geq 0 \\ -2n - 1 & \text{if } n < 0 \end{cases}$$

for all $n \in \mathbb{Z}$. This function's effect can be visualized by the following diagram.

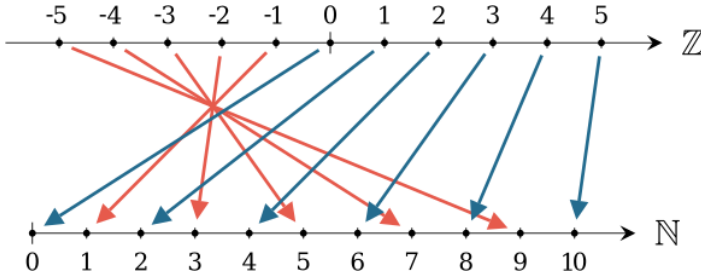


Figure 12.12. Effect of the Injection $f : \mathbb{Z} \rightarrow \mathbb{N}$

This is all well and good, but we haven't defined any operations on the integers yet! This means that, unfortunately, we have to appeal to the equivalence class definition of integers to define the injection f .

Suppose $n = [(x, y)]$.

- Doubling n (i.e., $2n$) is the same as moving twice as far to the right as compared to the shift (x, y) , and moving twice as far to the left as compared to the shift (x, y) , i.e. $2n = [(2x, 2y)]$.
- The expression $-2n - 1$ can be rewritten as $-(2n + 1)$ using simple algebra. Adding 1 to (x, y) simply means an additional shift of 1 unit to the left, so $2n + 1 = [(2x + 1, 2y)]$. Since negation swaps the order of the elements in the representative of the equivalence class, thus $-(2n + 1) = [(2y, 2x + 1)]$.

So, in the proof, we will appeal to these equivalence classes to define the injection $f : \mathbb{Z} \rightarrow \mathbb{N}$. However, since we are now invoking equivalence classes, we have to worry about the function being well-defined. This will be proven later.

With this set up, we can now formally prove the equinumerosity of $\mathbb{N}$ and $\mathbb{Z}$.

Proof. We will find an injection from $\mathbb{Z}$ to $\mathbb{N}$ and then use a corollary of the Cantor-Schröder-Bernstein theorem ([Corollary 9.2.11](#)) to finish the proof.

Consider the function $f : \mathbb{Z} \rightarrow \mathbb{N}$ where

$$f([(m, n)]) = \begin{cases} [(2m, 2n)] & \text{if } [(m, n)] \in \mathbb{N} \\ [(2n, 2m + 1)] & \text{if } [(m, n)] \notin \mathbb{N} \end{cases}$$

We need to prove two things: that f is a well-defined function, and that f is an injection.

First, to prove that f is well-defined. This comprises two things – that $f([(m, n)])$ is indeed in $\mathbb{N}$ for any $[(m, n)] \in \mathbb{Z}$, and that it is unique.

- **In $\mathbb{N}$ (i.e., F1):** Suppose $[(m, n)] \in \mathbb{Z}$. There are two cases to consider
 - If $[(m, n)] \in \mathbb{N}$, say $k = [(m, n)] \in \mathbb{N}$, then clearly $2k = [(2m, 2n)] \in \mathbb{N}$.
 - Otherwise, since $[(m, n)] \notin \mathbb{N}$, one sees that $n > m$. In particular, one notes that $n \geq m + 1$. Hence $2n \geq 2m + 2 > 2m + 1$, and thus $[(2n, 2m + 1)] \in \mathbb{N}$.

Thus in either case, we have $f([(m, n)]) \in \mathbb{N}$.

- **Uniqueness (i.e., F2):** Suppose we have $[(m_1, n_1)], [(m_2, n_2)] \in \mathbb{Z}$ such that $[(m_1, n_1)] = [(m_2, n_2)]$. Note that, by [Theorem 11.1.12](#), this means that $(m_1, n_1) \sim (m_2, n_2)$, i.e. $m_1 + n_2 = n_1 + m_2$. Again, there are two cases to consider.

- If $[(m_1, n_1)] = [(m_2, n_2)] \in \mathbb{N}$, then we have

$$\begin{aligned} f([(m_1, n_1)]) &= [(2m_1, 2n_1)] \text{ and} \\ f([(m_2, n_2)]) &= [(2m_2, 2n_2)]. \end{aligned}$$

Since $m_1 + n_2 = n_1 + m_2$, thus $2m_1 + 2n_2 = 2n_1 + 2m_2$. By [Theorem 11.1.12](#) again, this means $(2m_1, 2n_1) \sim (2m_2, 2n_2)$, so $f([(m_1, n_1)]) = [(2m_1, 2n_1)] = [(2m_2, 2n_2)] = f([(m_2, n_2)])$.

- Otherwise, we have $[(m_1, n_1)] = [(m_2, n_2)] \notin \mathbb{N}$. One sees that

$$\begin{aligned} f([(m_1, n_1)]) &= [(2n_1, 2m_1 + 1)] \text{ and} \\ f([(m_2, n_2)]) &= [(2n_2, 2m_2 + 1)]. \end{aligned}$$

Now, using $m_1 + n_2 = n_1 + m_2$ again, and by using various properties of addition, one obtains

$$2n_2 + (2m_1 + 1) = (2m_2 + 1) + 2n_1,$$

and so we see that $[(2n_2, 2m_2 + 1)] = [(2n_1, 2m_1 + 1)]$. Hence one sees that $f([(m_1, n_1)]) = [(2n_1, 2m_1 + 1)] = [(2n_2, 2m_2 + 1)] = f([(m_2, n_2)])$.

Thus, in either case, we have $f([(m_1, n_1)]) = f([(m_2, n_2)])$, which proves uniqueness.

Now we need to prove that f is injective. Let $[(m_1, n_1)], [(m_2, n_2)] \in \mathbb{Z}$ be such that $f([(m_1, n_1)]) = f([(m_2, n_2)])$. There are four cases to consider.

- $[(m_1, n_1)] \in \mathbb{N}$ and $[(m_2, n_2)] \in \mathbb{N}$ For this case we have

$$[(2m_1, 2n_1)] = [(2m_2, 2n_2)]$$

which means $2m_1 + 2n_2 = 2n_1 + 2m_2$. We can rewrite this slightly and turn it into

$$2(m_1 + n_2) = 2(n_1 + m_2).$$

By the cancellation property of multiplication for natural numbers ([Proposition 10.4.10](#)), and since $2 \neq 0$, we obtain

$$m_1 + n_2 = n_1 + m_2$$

which hence must mean that $[(m_1, n_1)] = [(m_2, n_2)]$.

- $[(m_1, n_1)] \notin \mathbb{N}$ and $[(m_2, n_2)] \in \mathbb{N}$ For this case, one notes that we will obtain

$$[(2n_1, 2m_1 + 1)] = [(2m_2, 2n_2)]$$

which means that $2n_1 + 2n_2 = (2m_1 + 1) + 2m_2$. We can move some things around to obtain

$$2(\underbrace{n_1 + n_2}_{\text{In } \mathbb{N}}) = 2(\underbrace{m_1 + m_2}_{\text{In } \mathbb{N}}) + 1.$$

But this is an issue – the left-hand side is an even natural number (by definition) while the right-hand side is an odd natural number (by [Proposition 12.5.4](#)). This is impossible, since an even number cannot be an odd number by definition.

- $[(m_1, n_1)] \in \mathbb{N}$ and $[(m_2, n_2)] \notin \mathbb{N}$ Same as case 2.
- $[(m_1, n_1)] \notin \mathbb{N}$ and $[(m_2, n_2)] \notin \mathbb{N}$ For this last case we have

$$[(2n_1, 2m_1 + 1)] = [(2n_2, 2m_2 + 1)]$$

which means $2n_1 + (2m_2 + 1) = (2m_1 + 1) + 2n_2$. Rearranging some terms, we get

$$2m_2 + 2n_1 + 1 = 2n_1 + 2m_2 + 1.$$

By using the cancellation property of addition for natural numbers (**Proposition 10.3.8**), we can turn that equation into the same equation in case 1, which thus means that $[(m_1, n_1)] = [(m_2, n_2)]$.

Therefore f is an injection from $\mathbb{Z}$ to $\mathbb{N}$. By the aforementioned corollary, this means that $|\mathbb{Z}| = |\mathbb{N}|$. ■

Appendices

APPENDIX A

Solutions to Exercises

A.1. Mathematical Logic

1.
 - (a) Not a statement; it is neither true nor false, it is just a sentence.
 - (b) A statement; it is definitely false.
 - (c) A statement; it can (eventually) be shown to be true or false.
 - (d) Not a statement; the truth or falsity of the statement depends on the value of n .
 - (e) A statement; it is definitely true.
 - (f) A statement; we can find such a number n , for instance $n = 3$, so we can confirm that this is true.
 - (g) Not a statement. This is tricky since, algebraically, it is true for all x . However, the fact remains that we can only evaluate the truth of this statement by substituting values of x since x is an unknown.
2. We work from the inner-most bracket outwards. We note p is true, q is false, and r is true.
 - $p \vee q$ can be understood to be “1 is a positive number **or** 1 is an even number”. This is true since 1 is indeed a positive number.
 - $(p \vee q) \wedge r$ is “(1 is a positive number **or** 1 is an even number) **and** 1 is an odd number”. Since $p \vee q$ is true and 1 is, indeed, an odd number, thus this statement is true.
 - $\neg((p \vee q) \wedge r)$ is false since $(p \vee q) \wedge r$ is true.Hence, the overall statement “ $\neg((p \vee q) \wedge r)$ is false” is true.
3.
 - (i) s is the statement “20 is a multiple of 5”.
 - (ii) The atomic statements are “20 ends in a 0” and “20 ends in a 5”.
 - (iii) t is $(p \vee q) \implies s$.
 - (iv) “20 is a multiple of 5 if and only if 20 ends in a 0 or 5”, which is true.

4. Again we consider a truth table.

p	q	$\neg p$	$\neg q$	$p \implies q$	$(\neg q) \implies (\neg p)$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Therefore we establish the required result.

5. The first statement is proven by considering the truth table below.

p	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg(p \wedge q)$	$(\neg p) \vee (\neg q)$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

The second statement is proven by considering the truth table below.

p	q	$\neg p$	$\neg q$	$p \vee q$	$\neg(p \vee q)$	$(\neg p) \wedge (\neg q)$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

6. One sees that

$$\begin{aligned}
 & ((p \vee \neg q) \wedge \neg r) \vee p \\
 & \equiv p \vee ((p \vee \neg q) \wedge \neg r) && \text{(Commutativity)} \\
 & \equiv (p \vee (p \vee \neg q)) \wedge (p \vee \neg r) && \text{(Distributivity)} \\
 & \equiv ((p \vee p) \vee \neg q) \wedge (p \vee \neg r) && \text{(Associativity)} \\
 & \equiv (p \vee \neg q) \wedge (p \vee \neg r) && \text{(Clearly } p \vee p \equiv p) \\
 & \equiv p \vee (\neg q \wedge \neg r) && \text{(Distributivity)} \\
 & \equiv p \vee \neg(q \vee r). && \text{(De Morgan's Law)}
 \end{aligned}$$

7. One formulation is " $\forall n, ((n > 2) \implies (\forall x, y, z, (x^n + y^n \neq z^n)))$ ".

8. We use a truth table.

p	q	$\neg q$	$p \implies q$	$\neg(p \implies q)$	$p \wedge \neg q$
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

9. (i) One formulation is " $\forall x, ((x \neq 0) \implies (\exists y \text{ s.t. } xy = 1))$ ".
(ii) The negation of the above formulation is " $\exists x \text{ s.t. } ((x \neq 0) \wedge (\forall y, xy \neq 1))$ ".

A.2. Sets

- (a) True, as both 1 and 2 appear in the set $\{1, 2, 3, 4\}$.
(b) False, since 3 does not appear in $\{1, 2, 4\}$.
(c) True. Any set is a subset of itself, including the empty set.
(d) False, the set S does not contain any element that is not in S . That is, $S \subseteq S$ but not $S \subset S$.
(e) True. S is indeed an element of $\{S, \emptyset\}$.
(f) True. The set containing S is not an element of $\{S, \emptyset\}$.
(g) False, the set S is not a subset of the set $\{S, \emptyset\}$.
(h) True. The set containing S is a subset of the set containing S and the empty set.
- (a) This is the set of rational numbers, $\mathbb{Q}$.
(b) This is the set of even integers.
(c) This is $\{2x - 1 \mid x \in \mathbb{Z}\}$.
(d) $\{\emptyset, \{\emptyset\}\}$.
- (a) True. The set of elements that are in either S or R is indeed $\{1, 2, 3, 4, 5\}$.
(b) False, $S \cup U = \{1, 2, 3, 4, (2, 2), (3, 3), (5, 5)\}$.
(c) True. The set of elements that are in both S and T is indeed $\{2, 3\}$.
(d) True. T and U share no elements in common, so their intersection is empty.

- (e) True. The elements that are in S but not in T are indeed 1 and 4.
 (f) False, $S \setminus \{1, 4\} = \{2, 3\}$, not $T = \{2, 3, 5\}$.
 (g) True.
 (h) True. $(S \cup T)^2 = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5)\}$, so

$$U = \{(2, 2), (3, 3), (5, 5)\} \subset (S \cup T)^2.$$

4. (a) Note that $S = \{-9, -7, -5, -3, -1, 1, 3, 5, 7, 9\}$, so $|S| = 10$.
 (b) Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 3, 4, 5, 6, 7\}$. Thus $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$ and so $|A \cup B| = 7$. Hence one sees that $|A \cap B| = |\{2, 3, 4, 5\}| = 4$.
 (c) $|A^p| = n^p$ since we are enumerating through the entirety of A each of the p times. Also $|\mathcal{P}(A)| = 2^n$. The reason is an element of A can either be in the subset or not, and that there are n such elements.

A.3. Proofs

1. *Proof.* Suppose $0 < x < 1$. Then $-1 < -x < 0$, meaning $0 < 1 - x < 1$. Therefore $x > 0$ and $1 - x > 0$, so their product $x(1 - x) > 0$. ■
2. *Proof.* Suppose that m and n have the same parity. We split into two cases.
- If both m and n are even, then we may write $m = 2a$ and $n = 2b$ where a and b are integers. Hence

$$\begin{aligned} m + n &= (2a) + (2b) \\ &= 2(a + b) \end{aligned}$$

which clearly means that $m + n$ is even.

- If both m and n are odd, then we may write $m = 2a + 1$ and $n = 2b + 1$ where a and b are integers. Hence

$$\begin{aligned} m + n &= (2a + 1) + (2b + 1) \\ &= 2a + 2b + 2 \\ &= 2(a + b + 1) \end{aligned}$$

which clearly means that $m + n$ is even.

Hence, in both cases, $m + n$ is even. ■

3. We consider a proof by contrapositive; the statement that we want to prove is "if **not** (a is even or b is odd) then $a(b^2 + 5)$ is **not** even". That is, "if a is **not** even **and** b is **not** odd then $a(b^2 + 5)$ is **not** even", meaning "if a is odd and b is even then $a(b^2 + 5)$ is odd".

Proof. Suppose that a is odd and b is even. Then we may write $a = 2m + 1$ and $b = 2n$ where m and n are integers. Hence

$$\begin{aligned} a(b^2 + 5) &= (2m + 1) \left((2n)^2 + 5 \right) \\ &= (2m + 1)(4n^2 + 5) \\ &= 8mn^2 + 10m + 4n^2 + 5 \\ &= 8mn^2 + 10m + 4n^2 + 4 + 1 \\ &= 2(4mn^2 + 5m + 2n^2 + 2) + 1 \end{aligned}$$

which clearly means that $a(b^2 + 5)$ is odd. ■

4. *Proof.* By way of contradiction assume there exist integers a and b such that $2a + 4b = 1$. Then dividing both sides by 2 leads to $a + 2b = \frac{1}{2}$. Note the left hand side is clearly an integer, while the right hand side is not an integer, a contradiction. ■

5. *Proof.* By way of contradiction assume that a and b are positive real numbers, and $\frac{a+b}{2} < \sqrt{ab}$. This means $a + b < 2\sqrt{ab}$. Squaring both sides yields $(a + b)^2 < 4ab$. Note

$$(a + b)^2 = a^2 + 2ab + b^2 < 4ab$$

which implies $a^2 + b^2 < 2ab$, leading to $a^2 - 2ab + b^2 < 0$. However $a^2 - 2ab + b^2 = (a - b)^2 \geq 0$ for all positive real numbers a and b . Hence we have $(a - b)^2 < 0$ and $(a - b)^2 \geq 0$, a contradiction. ■

6. We note that a positive odd number is of the form $2n - 1$ where n is a positive integer.

Proof. Set $a = 2n - 1$; we induct on n .

When $n = 1$, we have $a^2 - 1 = (2(1) - 1)^2 - 1 = 1 - 1 = 0$ which is clearly a multiple of 8.

Assume that the statement holds for some positive integer k , i.e. $(2k - 1)^2 - 1 = 8m$ for some integer m . We show that the statement holds for $k + 1$.

We note that $(2k - 1)^2 - 1 = 4k^2 - 4k$. Observe

$$\begin{aligned}
 (2(k + 1) - 1)^2 - 1 &= (2k + 1)^2 - 1 \\
 &= 4k^2 + 4k + 1 - 1 \\
 &= (4k^2 - 4k) + 8k \\
 &= 8m + 8k && \text{(by hypothesis)} \\
 &= 8(m + k)
 \end{aligned}$$

which means that $(2(k + 1) - 1)^2 - 1$ is a multiple of 8, proving that the statement holds for $k + 1$. By mathematical induction, $a^2 - 1$ is a multiple of 8 for all positive odd integers a . ■

7. *Proof.* We prove the forward direction using direct proof. Assume n is one more than a multiple of 5. Then we may write $n = 5a + 1$ where a is an integer. Note $5a + 1 = 5a + (5 - 4) = (5a + 5) - 4 = 5(a + 1) - 4$. Setting $k = a + 1$ yields required result.

We now prove the reverse direction using direct proof as well. Assume $n = 5k - 4$. Observe $5k - 4 = 5k - 5 + 1 = 5(k - 1) + 1$, meaning n is one more than a multiple of 5. ■

8. *Proof.* The number 7 satisfies this as $7 = 2^3 - 1$ and $7 = 3^2 - 2$. ■
9. (i) We use a proof by contradiction to prove this claim.

Proof. Seeking a contradiction, assume y is rational. Write $y = \frac{p}{q}$ where p and q are integers. Note $2^y = 2^{\frac{p}{q}}$ and $2^y = 2^{2 \log_2 3} = 9$. Hence $2^{\frac{p}{q}} = 9$ meaning $2^p = 9^q$. However 2^p is always even and 9^q is always odd, a contradiction. ■

- (ii) *Proof.* Note that

$$x^y = (\sqrt{2})^{2 \log_2 3} = \left(\sqrt{2}^2\right)^{\log_2 3} = 2^{\log_2 3} = 3$$

which is rational. ■

A.4. The Language of Predicate Logic

1. (a) The alphabet of L must contain the symbols a, b, c, d , and r .
- (b) Łukasiewicz notation: " $+ 4 \times - 5 6 7$ ". Reversed Łukasiewicz notation: " $4 5 6 - 7 \times +$ ".
- (c) The formal grammar of L contains only one formation rule.

If $s \in \Sigma$ is n -ary and $\phi_1, \phi_2, \phi_3, \dots, \phi_n$ are words in L , then

$$\phi_1\phi_2\phi_3 \dots \phi_ns$$

is also a word in L .

This is similar to Łukasiewicz notation, except that the s is placed at the end instead of at the front.

2. To argue which of the strings are words and which are not, observe that we may only have at most one a that may come after a b , regardless of the number of b s that are in between.

(a) Word. $ba \xrightarrow{1} aba \xrightarrow{2} abab$.

(b) Word. $\varepsilon \xrightarrow{1} a \xrightarrow{1} aa \xrightarrow{1} aaa \xrightarrow{2} aaab \xrightarrow{2} aaabb \xrightarrow{2} aaabbb$.

(c) Not a word by above observation. From the 1st b there are 2 a s that come after it.

(d) Word. $ba \xrightarrow{1} aba \xrightarrow{2} abab \xrightarrow{2} ababb \xrightarrow{2} ababbb$.

3. All of them are strings, but most are not words.

4. (a) This is a term. From **T0** we know that x is a term, and **T1** tells us that 7, being a nullary function symbol, is also a term. Thus **T1** tells us that $f(x, 7)$ is a term.

(b) This is a WFF, but not an atomic formula since quantifiers are being used in the string.

(c) Neither a term nor a WFF. Observe that both $f(x)$ and 3 are terms by **T1**, but there is no rule (for terms and for WFFs) that allows concatenating the two terms together into the form $3(f(x))$.

(d) This is a WFF, but not an atomic formula.

(e) This is a WFF, but not an atomic formula.

(f) Neither a term nor a WFF. The string $x = y$ is a WFF by **F1**, so it is not a term. Hence having " $x = y$ " being used as an argument for the predicate symbol P is invalid.

5. (a) Bound.

(b) Both bound and free. The left side (i.e., " $x = f(y)$ ") has x being a free variable, while the right side (i.e., " $\forall x \exists y (y = x)$ ") has x being a bound variable.

(c) Free.

(d) Neither. The variable x does not appear in the WFF.

A.5. Axiomatic Set Theory

1. We induct on n .

The base case of $n = 1$ is readily proven by [Proposition 5.4.3](#).

Suppose the theorem holds true for any positive integer k , i.e. the set $\{a_1, \dots, a_k\}$ exists for any $a_1, \dots, a_k$. We show that $\{a_1, \dots, a_k, a_{k+1}\}$ exists for any $a_1, \dots, a_k, a_{k+1}$.

We know that the singleton $\{a_{k+1}\}$ exists by [Proposition 5.4.3](#). Thus [Axiom 5.4.1](#) tells us that $\{\{a_1, \dots, a_k\}, \{a_{k+1}\}\}$ exists. Hence applying [Axiom 5.5.1](#) to this set tells us that $\{a_1, \dots, a_k, a_{k+1}\}$ exists, proving the theorem for $k + 1$.

Hence, by mathematical induction, the theorem is proven.

A.6. Defining Natural Numbers

1. We note

$$\begin{aligned} \mathcal{S}(\mathcal{S}(\{x, y\})) &= \mathcal{S}(\{x, y\} \cup \{\{x, y\}\}) \\ &= \mathcal{S}(\{x, y, \{x, y\}\}) \\ &= \{x, y, \{x, y\}\} \cup \{\{x, y, \{x, y\}\}\} \\ &= \{x, y, \{x, y\}, \{x, y, \{x, y\}\}\}. \end{aligned}$$

2. (i) $5 = \{0, 1, 2, 3, 4\}$.

- (ii) We see

$$\begin{aligned} 5 = \{ & \\ & \{\}, \\ & \underbrace{\{\{\}\}}_0, \\ & \underbrace{\{\{\}\}}_0, \underbrace{\{\{\{\}\}\}}_1, \\ & \underbrace{\{\{\}\}}_0, \underbrace{\{\{\{\}\}\}}_1, \underbrace{\{\{\{\{\}\}\}\}}_2, \\ & \underbrace{\{\{\}\}}_0, \underbrace{\{\{\{\}\}\}}_1, \underbrace{\{\{\{\{\}\}\}\}}_2, \underbrace{\{\{\{\{\{\}\}\}\}\}}_3, \\ & \}. \end{aligned}$$

- (iii) We claim that the counter set of n requires 2^n open curly braces. We prove this using induction.

The counter set of 0 is $\{\}$ which requires $1 = 2^0$ open curly braces.

Suppose the claim holds true for some natural number k , i.e. the counter set of k uses 2^k open curly braces. We prove that the counter set of $k + 1$ uses 2^{k+1} open curly braces.

Recall that $k + 1 = \mathcal{S}(k) = k \cup \{k\}$. Our induction hypothesis says that the counter set of k uses 2^k open curly braces. Internally, we know that

$$k = \{0, 1, 2, \dots, k - 1\}$$

which means that the part “ $0, 1, 2, \dots, k - 1$ ” uses $2^k - 1$ open curly braces (since the beginning of the set starts with an open curly brace). So “ $0, 1, 2, \dots, k - 1, k$ ” will use $(2^k - 1) + 2^k = 2^{k+1} - 1$ open curly braces. When this is placed in a set, this increases the number of open curly braces by 1, thus making the counter set of $k + 1$ have 2^{k+1} open curly braces.

This proves the claim via induction. Hence the number of open curly braces for the counter set of 65536 is

$$2^{65536} = 2^{2^{16}} = 2^{2^{2^4}} = 2^{2^{2^{2^2}}}.$$

A.7. Relations I

1. (a) Since $\emptyset \subseteq A \times B$ and $\emptyset \subseteq A^2$, thus R_1 is a relation from A to B as well as a relation on A .
- (b) Note $(3, 3) \notin A \times B$, so R_2 is not a relation from A to B . However $R_2 \subseteq A^2$ so R_2 is a relation on A .
- (c) Observe $R_3 \subseteq A \times B$ so R_3 is a relation from A to B . But $(4, 8) \notin A^2$ so R_3 is not a relation on A .
- (d) Note $2 \notin A \times B$ and $2 \notin A^2$, so R_4 is neither a relation from A to B nor a relation on A .
- (e) One sees $(6, 8) \notin A \times B$ and $(6, 8) \notin A^2$ and thus R_5 is neither a relation from A to B nor a relation on A .
- (f) Note that $(5, 5) \notin A \times B$ so R_6 is not a relation from A to B . However clearly $A^2 \subseteq A^2$, which means R_6 is a relation on A .

(g) Since $(6, 6) \notin A \times B$ and $(6, 6) \notin A^2$, thus R_7 is neither a relation from A to B nor a relation on A .

2. (i) We will do this step by step.

- For $x = 2$, notice that $2 D 4$ (using $m = 2$), $2 D 6$ (using $m = 3$), and $2 D 8$ (using $m = 4$).
- For $x = 3$, notice that only $3 D 6$ (using $m = 2$).
- For $x = 4$, notice that only $4 D 8$ (using $m = 2$).
- For $x = 5$, $x = 6$, $x = 7$, and $x = 8$, there are no valid y that is related to them.

Thus $D = \{(2, 4), (2, 6), (2, 8), (3, 6), (4, 8)\}$. Hence we see

$$\begin{aligned} D^{-1} &= \{(4, 2), (6, 2), (8, 2), (6, 3), (8, 4)\} && \text{(Swap } x \text{ and } y) \\ &= \{(4, 2), (6, 2), (6, 3), (8, 2), (8, 4)\}. && \text{(Rearranging)} \end{aligned}$$

(ii) (a) Domain of D is $\{2, 3, 4\}$.

(b) Codomain of D is $S = \{2, 3, 4, 5, 6, 7, 8\}$.

(c) Range of D is $\{4, 6, 8\}$.

3. Note that $(R^{-1})^{-1} \subseteq A \times B$. One sees

$$\begin{aligned} (a, b) &\in (R^{-1})^{-1} \\ \iff (b, a) &\in R^{-1} && \text{(Definition of inverse relation)} \\ \iff (a, b) &\in R && \text{(Definition of inverse relation)} \end{aligned}$$

Thus any element in $(R^{-1})^{-1}$ is also in R and vice versa. Hence $(R^{-1})^{-1} = R$.

4. (a) True. Let $A = \{1\}$, $B = \{2, 3\}$, $C = \{4\}$, $R = \{(1, 2)\}$, and $S = \{(3, 4)\}$. Thus $S \circ R = \emptyset$ and so the domain of $S \circ R$ is empty.

(b) False, see (a).

(c) True. Let $A = \{1\}$, $B = \{2\}$, $C = \{3\}$, $R = \{(1, 2)\}$, and $S = \{(2, 3)\}$. Thus $S \circ R = \{(1, 3)\}$ and so the domain of $S \circ R$ is $\{1\} = A$.

(d) False. The codomain of $S \circ R$ is C which is given to be non-empty.

(e) True. This is the definition of the codomain of a relation.

- (f) False. If the codomain is indeed B , that means $C = B$ (since C is the actual codomain). But B and C are distinct, so this cannot happen.
- (g) True, see (a).
- (h) False, see (a).
- (i) True, see (c).
5. (a) False. Let $A = \{1\}$ and $B = \{2\}$. Let $R = \{(1,2)\}$ and $S = \{(2,1)\}$. We see $S \circ R = \{(1,1)\} \neq \{(2,2)\} = R \circ S$. Thus it is not always the case that $S \circ R = R \circ S$.
- (b) False. See (c).
- (c) True. Let $A = B = \{1,2\}$. Let $R = S = \{(1,2), (2,1)\}$. Thus $S \circ R = \{(1,1), (2,2)\} = R \circ S$. So it is possible for $S \circ R$ and $R \circ S$ to be the same.
6. We work step by step.

$$\begin{aligned}
 & (d, a) \in (S \circ R)^{-1} \\
 \iff & (a, d) \in S \circ R && \text{(Definition of inverse relation)} \\
 \iff & \exists x \in B \cap C \text{ s.t. } (a R x \wedge x S d) && \text{(Definition of composition)} \\
 \iff & \exists x \in B \cap C \text{ s.t. } (x R^{-1} a \wedge d S^{-1} x) && \text{(Definition of inverse relation)} \\
 \iff & \exists x \in B \cap C \text{ s.t. } (d S^{-1} x \wedge x R^{-1} a) && \text{(Commutativity of } \wedge \text{)} \\
 \iff & (d, a) \in R^{-1} \circ S^{-1} && \text{(Definition of composition)}
 \end{aligned}$$

Thus any element in $(S \circ R)^{-1}$ is also in $R^{-1} \circ S^{-1}$ and vice versa, thereby meaning that $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$.

7. The statement is false. Consider $A = B = \{1,2,3\}$ and the relation $R = \{(1,3), (2,3)\}$. We see that $R^{-1} = \{(3,1), (3,2)\}$. However, note that $(1,2) \in R^{-1} \circ R$ since $1 R 3$ and $3 R^{-1} 2$. Thus $R^{-1} \circ R \not\subseteq \{(x,y) \in A^2 \mid x = y\} = \{(1,1), (2,2), (3,3)\}$.

A.8. Functions

1. Yes. Look at conditions **F1** and **F2**. Both use the universal quantifier $\forall$. Since there is nothing in $\emptyset$, thus the two conditions **F1** and **F2** are vacuously true, meaning that $f : \emptyset \rightarrow A$ is indeed a function.
2. (a) $f(1) = x$, $f(2) = u$, and $f(4) = y$.

- (b) Domain is $A = \{1, 2, 3, 4, 5\}$, codomain is $B = \{u, v, w, x, y, z\}$, and range is $\{u, x, y\}$.
- (c) $f = \{(1, x), (2, u), (3, x), (4, y), (5, x)\}$.
3. (a) Ill-defined. 2 is not related to 1 or 8, so condition **F1** in **Definition 8.1.1** is not satisfied.
- (b) Well-defined. We address each of the conditions separately.
- **F1:** Obviously true. Suppose $x \in \mathbb{N}^2$. Thus we can write x as (m, n) where $m, n \in \mathbb{N}$. So their sum, $m + n$, must be a natural number and we can choose it to be y .
 - **F2:** $x \in \mathbb{N}^2$. Thus we can write x as (m, n) where $m, n \in \mathbb{N}$. Suppose we have $y_1, y_2 \in \mathbb{N}$ such that $x \ g \ y_1$ and $x \ g \ y_2$. So $y_1 = m + n$ and $y_2 = m + n$ by definition of g , which immediately means $y_1 = y_2$.
- Since both conditions are true, the function is well-defined.
- (c) Ill-defined. Note that $1 \ h \ (0, 1)$ and $1 \ h \ (1, 0)$, so condition **F2** in **Definition 8.1.1** is not satisfied.
4. (a) Is injective. Every element in A has a unique image in B .
- (b) Not injective. We see $f(0) = f(2) = 3$.
- (c) Is injective. Every element in A has a unique image in B .
- (d) Not injective. We see $f(1) = f(3) = 2$.
5. (i) The function H has to be injective.
- (ii) (a) No collisions. We see that $H(12) = 12$, $H(34) = 34$, and $H(56) = 56$.
- (b) Has collision. We see that $H(12) = H(1112) = 12$.
- (c) No collision. We see that $H(18323) = 23$, $H(389892) = 92$, $H(92349589) = 89$, and $H(8942894281) = 81$.
- (d) No collision. We see that $H(128) = 28$, $H(7327) = 27$, $H(83279) = 79$, $H(8934289) = 89$, $H(842892) = 92$, and $H(824379490) = 90$.
6. (a) Not surjective. 1 has no preimage.
- (b) Is surjective. All elements in B has a preimage in A .
- (c) Is surjective. All elements in B has a preimage in A .
- (d) Not surjective. 2 has no preimage.

7. There are many solutions, one of which is

$$f = \{(1, 2), (2, 3), (3, 4), (4, 1)\}.$$

One sees that f is a bijection with

$$f^{-1} = \{(2, 1), (3, 2), (4, 3), (1, 4)\},$$

which is not $\text{id}_A = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$.

8. (a) True. Suppose $h \circ f = h \circ g$. Thus $h(f(x)) = h(g(x))$ for all $x \in X$. As h is injective, this means $f(x) = g(x)$ for all $x \in X$. Thus $f = g$.

(b) False. Let $X = \mathbb{N}$, and consider the functions

$$f = \text{id}_X,$$

$$g = \{(n, n) \mid n \text{ is even}\} \cup \{(n, 0) \mid n \text{ is odd}\}, \text{ and}$$

$$h = \{(n, 2n) \mid n \in \mathbb{N}\}.$$

We first show that h is injective. Suppose $n_1, n_2 \in \mathbb{N}$ such that $h(n_1) = h(n_2)$. So $2n_1 = 2n_2$ which clearly means $n_1 = n_2$. Thus h is injective.

Now we show $f \circ h = g \circ h$. We see that

$$\begin{aligned} (f \circ h)(n) &= f(h(n)) && \text{(Definition of composition)} \\ &= f(2n) && \text{(Definition of } h) \\ &= 2n && \text{(Definition of } f) \\ &= g(2n) && \text{(Definition of } g, \text{ since } 2n \text{ is even)} \\ &= g(h(n)) && \text{(Definition of } h) \\ &= (g \circ h)(n) && \text{(Definition of composition)} \end{aligned}$$

for any $n \in \mathbb{N}$, which thus means $f \circ h = g \circ h$.

But one sees clearly that $(1, 1) \in f$ but $(1, 1) \notin g$, so $f \neq g$.

9. *Proof.* Suppose $f : A \rightarrow B$ and $g : C \rightarrow D$ are surjective functions. Let $d \in D$. Since g is surjective, there exists a $c \in C$ such that $g(c) = d$. In particular, since $g \circ f$ exists, we know that this c is in the range of f , which means that $c \in B$. Since f is surjective, we know that there exists an $a \in A$ such that $f(a) = c$ (remember, this c is also in B). We note that this element $a \in A$ satisfies

$$g(f(a)) = g(c) = d$$

so a preimage of d under $g \circ f$ is a . It follows that $g \circ f$ is surjective. ■

10. *Proof.* For the forward direction we assume [Axiom 8.6.1](#) holds. Consider a family of sets $\mathcal{S} = \{S_i\}_{i \in I}$ indexed by I . Since $\mathcal{S}$ is non-empty, there exists a choice function $f : \mathcal{S} \rightarrow \bigcup_{S \in \mathcal{S}} S$ such that $f(S) \in S$ for all $S \in \mathcal{S}$. For each $i \in I$, let the set $s_i = f(S_i)$. We clearly see $s_i \in S_i$. Thus we have found a corresponding family of sets $\mathfrak{S} = \{s_i\}_{i \in I}$ such that $s_i \in S_i$ for all $i \in I$. ■

A.9. Counting

1. *Proof.* Suppose there is an injection from B to A , say g . Define a function $f : A \rightarrow B$ such that

$$f(a) = \begin{cases} b \text{ where } g(b) = a & \text{if } a \text{ is in the range of } g \\ b_0 \text{ where } b_0 \text{ is a fixed element in } B & \text{otherwise} \end{cases}$$

We show that f is a surjection. Suppose $b \in B$. Note that $g(b)$ is the preimage of b under f since the first case in the definition of f clearly reduces to $f(g(b)) = b$. Thus f is a surjection, and therefore $|A| \geq |B|$ by [Definition 9.2.7](#). ■

2. (a) 7125.
 (b) 373267.
 (c) Any singleton would work. For example, $\{0\}, \{a\}, \{1235\}$ etc. would all work.
 (d) Note $13 = \{0, 1, 2, \dots, 12\}$ and $14 = \{0, 1, 2, \dots, 12, 13\}$, so $13 \cup 14 = \{0, 1, 2, \dots, 12, 13\} = 14$. Thus the cardinality of $13 \cup 14$ is 14.

3. First we note that

$$\mathcal{P}(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}.$$

Thus $\mathcal{P}(\mathcal{P}(\{a, b\}))$ is

$$\begin{aligned} &\{ \\ &\quad \emptyset, \\ &\quad \{\emptyset\}, \{\{a\}\}, \{\{b\}\}, \{\{a, b\}\}, \\ &\quad \{\emptyset, \{a\}\}, \{\emptyset, \{b\}\}, \{\emptyset, \{a, b\}\}, \\ &\quad \{\{a\}, \{b\}\}, \{\{a\}, \{a, b\}\}, \{\{b\}, \{a, b\}\}, \\ &\quad \{\emptyset, \{a\}, \{b\}\}, \{\emptyset, \{a\}, \{a, b\}\}, \{\emptyset, \{b\}, \{a, b\}\}, \{\{a\}, \{b\}, \{a, b\}\}, \\ &\quad \{\emptyset, \{a\}, \{b\}, \{a, b\}\} \\ &\} \end{aligned}$$

which has a cardinality of 16.

4. For any set A , we define the function $f : A \rightarrow \mathcal{P}(A)$ by $f(x) = \{x\}$ for all $x \in A$. One sees that for any $x_1, x_2 \in A$, if $f(x_1) = f(x_2)$ we have $\{x_1\} = \{x_2\}$ which means $x_1 = x_2$ by equality of sets. Thus f is injective as required.

A.10. Operations Involving the Natural Numbers

1. The set difference operation on a set A can be defined as the function $f : \mathcal{P}(A)^2 \rightarrow \mathcal{P}(A)$ where

$$f(S, T) = \{x \in A \mid x \in S \wedge x \notin T\}$$

for all $S, T \in \mathcal{P}(A)$.

2. *Proof.* Note $\mathcal{S}(n) = \mathcal{S}(n+0)$ by **A1**, which is the same as $n + \mathcal{S}(0)$ by **A2**. But since $\mathcal{S}(0) = 1$, we see that $\mathcal{S}(n) = n + 1$ as needed. ■

3. *Proof.* We induct on n .

When $n = 0$, the left-hand side is $\mathcal{S}(d) + 0 = \mathcal{S}(d)$ by **A1**. The right-hand side similarly simplifies to $\mathcal{S}(d+0) = \mathcal{S}(d)$ by **A1** too. Thus the base case is proven.

Suppose $\mathcal{S}(d) + k = \mathcal{S}(d+k)$ for some $k \in \mathbb{N}$. We are to prove that $\mathcal{S}(d) + \mathcal{S}(k) = \mathcal{S}(d + \mathcal{S}(k))$.

One sees that

$$\begin{aligned} \mathcal{S}(d) + \mathcal{S}(k) &= \mathcal{S}(\mathcal{S}(d) + k) && \text{(by A2)} \\ &= \mathcal{S}(\mathcal{S}(d+k)) && \text{(induction hypothesis)} \\ &= \mathcal{S}(d + \mathcal{S}(k)) && \text{(by A2)} \end{aligned}$$

which proves the case for $\mathcal{S}(k)$. ■

4. *Proof.* We see that

$$\begin{aligned} (a+b) + c &= a + (b+c) && \text{(by Theorem 10.3.6)} \\ &= a + (c+b) && \text{(by Theorem 10.3.5)} \\ &= (a+c) + b && \text{(by Theorem 10.3.6)} \end{aligned}$$

which proves the corollary. ■

5. *Proof.* We only prove the case that if $a + b = b$ then $a = 0$, since the other is easily implied by commutativity of addition ([Theorem 10.3.5](#)).

We can write $a + b = b$ as $a + b = b + 0$ by **A1**. Using commutativity of addition on the right-hand side we have

$$a + b = 0 + b.$$

Thus, using [Proposition 10.3.8](#) we get $a = 0$ as required. ■

6. (i) *Proof.* We consider induction on n .

The base case of $n = 0$ is easily proven since the left-hand side, being 0×0 , simplifies to 0 using **M1**.

Suppose that $0 \times k = 0$ for some $k \in \mathbb{N}$. We show that $0 \times \mathcal{S}(k) = 0$.

Note that

$$\begin{aligned} 0 \times \mathcal{S}(k) &= (0 \times k) + 0 && \text{(by M2)} \\ &= 0 \times k && \text{(by Lemma 10.3.3)} \\ &= 0 && \text{(induction hypothesis)} \end{aligned}$$

which proves the case for $\mathcal{S}(k)$. ■

- (ii) *Proof.* We induct on n .

When $n = 0$, the left-hand side is $m \times 0$ which is 0 by **M1**, and the right-hand side is $0 \times m$ which is 0 by [Lemma 10.4.3](#). So the base case is proven.

Suppose $m \times k = k \times m$ for some $k \in \mathbb{N}$. We show $m \times \mathcal{S}(k) = \mathcal{S}(k) \times m$.

Note

$$\begin{aligned} m \times \mathcal{S}(k) &= (m \times k) + m && \text{(by M2)} \\ &= (k \times m) + m && \text{(induction hypothesis)} \\ &= \mathcal{S}(k) \times m && \text{(by Lemma 10.4.4)} \end{aligned}$$

which proves the case for $\mathcal{S}(k)$. ■

7. *Proof.* We only prove the case that if $b \neq 0$ and $a \times b = b$ then $a = 1$, since the other is easily implied by commutativity of multiplication ([Theorem 10.4.5](#)).

Note $a \neq 0$ since, otherwise, the left-hand side will be 0 (by [Lemma 10.4.3](#)) and the right-hand side is $b \neq 0$, a contradiction. So we can

write $a \times b = b$ as $a \times b = 1 \times b$ by [Corollary 10.4.6](#). Thus, using [Proposition 10.4.10](#) we get $a = 1$ as required. ■

8. (a) False. One sees

$$\begin{aligned} 0 \wedge 1 &= 0^1 \\ &= (0^0) \times 0 && \text{(by E2)} \\ &= 0 && \text{(by M1)} \end{aligned}$$

but $1 \wedge 0 = 1^0 = 1$ by E1.

(b) False. One sees $(2 \wedge 3) \wedge 2 = (2^3)^2 = 64$ but $2 \wedge (3 \wedge 2) = 2^{(3^2)} = 512$.

9. *Proof.* We see that

$$\begin{aligned} n^1 &= n^{S(0)} \\ &= n^0 \times n && \text{(by E2)} \\ &= 1 \times n && \text{(by E1)} \\ &= n, && \text{(by Corollary 10.4.6)} \end{aligned}$$

proving this proposition. ■

10. *Proof.* We induct on n .

When $n = 0$ we have

$$\begin{aligned} a^0 \times b^0 &= 1 \times 1 && \text{(by E1)} \\ &= 1 && \text{(by Corollary 10.4.6)} \\ &= (a \times b)^0 && \text{(by E1)} \end{aligned}$$

so the base case is proven.

Suppose $a^k \times b^k = (a \times b)^k$ for some $k \in \mathbb{N}$. We show that $a^{S(k)} \times b^{S(k)} = (a \times b)^{S(k)}$.

Note that

$$\begin{aligned} a^{S(k)} \times b^{S(k)} &= (a^k \times a) \times (b^k \times b) && \text{(by E2)} \\ &= (a^k \times b^k) \times (a \times b) && \text{(properties of multiplication)} \\ &= (a \times b)^k \times (a \times b) && \text{(induction hypothesis)} \\ &= (a \times b)^{S(k)} && \text{(by E2)} \end{aligned}$$

which proves the case for $S(k)$. ■

A.11. Relations II

1. (a) Not reflexive since $0 \not\sim 0$.
 (b) Not symmetric since $0 \sim 1$ but $1 \not\sim 0$.
 (c) Is transitive. The condition for transitivity is vacuously satisfied.
2. (a) Not an equivalence relation since A is not reflexive (e.g., $1 \not\sim 1$).
 (b) Is an equivalence relation.

- **Reflexive:** We have $1 \sim 1$, $2 \sim 2$, and $3 \sim 3$. Thus B is reflexive.
- **Symmetric:** Suppose $x, y \in S$ where $x \sim y$. We split into two cases.
 - If $x = y$, reflexivity immediately implies $y \sim x$.
 - Otherwise symmetry is vacuously satisfied.

Thus B is symmetric.

- **Transitive:** Suppose $x, y, z \in S$ where $x \sim y$ and $y \sim z$. We split into two cases.
 - If $x = y = z$, we clearly have $x \sim z$ by reflexivity.
 - Otherwise transitivity is vacuously satisfied.

Thus B is transitive.

Since B is reflexive, symmetric, and transitive, it is an equivalence relation.

- (c) Not an equivalence relation since it is not symmetric. In particular we have $1 \sim 2$ but $2 \not\sim 1$.
 - (d) Not an equivalence relation since it is not transitive. For example we have $1 \sim 2$ and $2 \sim 3$ but $1 \not\sim 3$.
3. (a) *Proof.* We need to show that $\sim$ is reflexive, symmetric, and transitive.
 - **Reflexive:** Let $A \in \mathcal{P}(X)$. Clearly $|A| = |A|$ by using the identity function on A (i.e., id_A), so $A \sim A$, proving that A is reflexive.
 - **Symmetric:** Let $A, B \in \mathcal{P}(X)$ such that $A \sim B$. This means $|A| = |B|$, which tells us that there exists a bijection $f : A \rightarrow B$. But $f^{-1} : B \rightarrow A$ is also a bijection, so $|B| = |A|$, i.e. $B \sim A$, proving that $\sim$ is symmetric.

- **Transitive:** Suppose $A, B, C \in \mathcal{P}(X)$ such that $A \sim B$ and $B \sim C$. This means that $|A| = |B|$ and $|B| = |C|$, i.e. there exist bijections $f : A \rightarrow B$ and $g : B \rightarrow C$. But $g \circ f : A \rightarrow C$ is also a bijection by **Proposition 8.5.10**, so $|A| = |C|$, i.e. $A \sim C$, proving that $\sim$ is transitive.

Thus we see that $\sim$ is an equivalence relation. ■

- (b) Note that two sets have are in the same equivalence class if and only if they have the same cardinality. So the distinct equivalence classes of $\sim$ are

- $\{\emptyset\}$ (each set has cardinality 0);
- $\{\{1\}, \{2\}, \{3\}\}$ (each set has cardinality 1);
- $\{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$ (each set has cardinality 2); and
- $\{\{1, 2, 3\}\}$ (each set has cardinality 3).

4. *Proof.* Suppose $[x] \cap [y] \neq \emptyset$. This means that there is an $a \in [x] \cap [y]$, i.e. $a \in [x]$ and $a \in [y]$. By definition of equivalence classes, this tells us that $a \sim x$ and $a \sim y$. We hence know that $x \sim a$ (by symmetry) and thus $x \sim y$ (by transitivity). ■

5. Yes. Consider $A = \{0\}$ and $R = \{(0, 0)\}$. Note that R is symmetric since, for all x (which must only be 0) and y (which must also be 0), we see $x R y$ means $y R x$ (which is obvious as $0 R 0$). We also see that R is antisymmetric – for all x and y (which are both 0), if $x R y$ and $x R y$ then $x = y$.

6. (a) *Proof.* Suppose R is asymmetric. Seeking a contradiction, suppose there is an $x \in A$ such that $x R x$. But asymmetry tells us that $x \not R x$, which is a clear contradiction. Thus $x \not R x$ for all $x \in A$, i.e. R is irreflexive. ■

- (b) *Proof.* Suppose R is irreflexive. Seeking a contradiction, suppose there exists $x, y \in A$ such that $x R y$ and $y R x$. However, transitivity tells us that $x R x$, contradicting the assumption that R is irreflexive. Therefore, for all $x, y \in A$, if $x R y$ then $y \not R x$, i.e. R is antisymmetric. ■

7. *Proof.* We need to show $|$ is reflexive, antisymmetric, and transitive.

- **Reflexive:** Suppose $n \in \mathbb{N}$. Choose $q = 1$ and we see that $n = 1n$, i.e. $n \mid n$. This shows that $|$ is reflexive.
- **Antisymmetric:** Suppose $m, n \in \mathbb{N}$ such that $m \mid n$ and $n \mid m$. This means that there exists $q_1, q_2 \in \mathbb{N} \setminus \{0\}$ such that $n = q_1 m$

and $m = q_2n$. This thus means that

$$\begin{aligned} n &= q_1m \\ &= q_1(q_2n) \\ &= q_1q_2n \end{aligned}$$

which means $q_1q_2 = 1$ which hence means $q_1 = 1$ and $q_2 = 1$ by [Proposition 10.4.12](#), i.e. $m = n$.

- **Transitive:** Suppose $l, m, n \in \mathbb{N}$ such that $l \mid m$ and $m \mid n$. This means that there exists $q_1, q_2 \in \mathbb{N} \setminus \{0\}$ such that $m = q_1l$ and $n = q_2m$. One clearly sees that

$$\begin{aligned} n &= q_2m \\ &= q_2(q_1l) \\ &= (q_2q_1)l \end{aligned}$$

and, since $q_1 \neq 0$ and $q_2 \neq 0$, thus $q_2q_1 \in \mathbb{N} \setminus \{0\}$. Thus $l \mid n$, proving that $\mid$ is transitive.

Therefore $\mid$ is a partial order relation. ■

8. (a) *Proof.* We are to prove reflexivity, antisymmetry, and transitivity.

- **Reflexive:** Let $a \in A$. Since $a \preceq a$, thus $(a, a) \in \preceq$ (by definition of relation). So $(a, a) \in \preceq^{-1} = \succ$ (definition of inverse relation) which means $a \succ a$.
- **Antisymmetric:** Suppose $x, y \in A$ such that $x \succ y$ and $y \succ x$. So $y \preceq x$ and $x \preceq y$ (by definition of $\succ$), which hence means $x = y$ (by antisymmetry of $\preceq$).
- **Transitive:** Suppose $x, y, z \in A$ such that $x \succ y$ and $y \succ z$. This means $y \preceq x$ and $z \preceq y$, which therefore means $z \preceq x$ (transitivity of $\preceq$). Hence $x \succ z$.

Thus $\succ$ is also a partial order on A . ■

- (b) *Proof.* We prove irreflexivity and transitivity since asymmetry will be implied by [Proposition 11.2.8](#).

- **Irreflexive:** Suppose $a \in A$. Clearly $a \not\prec a$ (since $\prec$ is irreflexive) which means $(a, a) \notin \prec$ (definition of relation). Thus $(a, a) \notin \prec^{-1} = \succ$ (definition of inverse relation) which hence tells us $a \not\succ a$.
- **Transitive:** Similar argument as (a), just replacing $\succ$ with $\prec$.

Thus $\prec$ is also a strict partial order on A . ■

9. One sees that

$$\preceq = \{(0,0), (0,2), (0,3), (0,5), (1,1), (1,4), (1,6), \\ (2,2), (2,5), (3,3), (3,5), (4,4), (4,6), (5,5), (6,6)\}$$

10. We will prove all the statements.

- (a) Suppose $m_1, m_2 \in A$ are greatest elements. That means that, for all $x \in A$, we have $x \preceq m_1$ and $x \preceq m_2$. For the first relation, we can choose $x = m_2$ and see $m_2 \preceq m_1$. For the second relation, we can choose $x = m_1$ and get $m_1 \preceq m_2$. Thus, by antisymmetry, we have $m_1 = m_2$. Therefore the greatest element is unique.
- (b) Similar argument as (a).
- (c) Suppose $m_1, m_2 \in A$ are distinct maximal elements. Seeking a contradiction, suppose $M \in A$ is the greatest element (which we know is unique by (a)). By definition of greatest element we must have $m_1 \preceq M$ and $m_2 \preceq M$. But, by definition of maximal elements, this must mean that $M = m_1$ and $M = m_2$, a clear contradiction since $m_1 \neq m_2$. Therefore, if there is more than one maximal element, there is no greatest element.
- (d) Similar argument as (c).

A.12. Defining the Integers

1. Let $\sim$ be the relation on $\mathbb{N}^2$ where

$$(a, b) \sim (c, d) \iff a + d = b + c$$

for all $(a, b), (c, d) \in \mathbb{N}^2$. We show that $\sim$ is indeed an equivalence relation.

- **Reflexive:** Clearly $a + b = b + a$ for any $a, b \in \mathbb{N}$ (commutativity of addition, [Theorem 10.3.5](#)), so $(a, b) \sim (a, b)$ by definition of $\sim$, proving reflexivity.
- **Symmetric:** Suppose $(a, b), (c, d) \in \mathbb{N}^2$ are such that $(a, b) \sim (c, d)$. This means $a + d = b + c$ by definition of $\sim$. Hence $c + b = d + a$ (commutativity of addition), so $(c, d) \sim (a, b)$, proving symmetry.
- **Transitive:** Suppose $(a, b), (c, d), (e, f) \in \mathbb{N}^2$ such that $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$. This means

$$a + d = b + c \text{ and } c + f = d + e.$$

Adding f to both sides of the first equation we get $a + d + f = b + c + f$. Note $b + c + f = b + d + e$ by the second equation, so we have

$$a + d + f = b + d + e.$$

Therefore $a + f = b + e$ by the cancellation property of addition ([Proposition 10.3.8](#)), which means $(a, b) \sim (e, f)$, thus proving transitivity.

Thus $\sim$ is an equivalence relation.

2. (a) *Proof.* We are to prove that $\leq$ on $\mathbb{Z}$ is reflexive, antisymmetric, and transitive.

- **Reflexive:** Let $[(a, b)] \in \mathbb{Z}$. Clearly $a + b \leq a + b$ (since $\leq$ on $\mathbb{N}$ is a partial order and is thus reflexive), so $[(a, b)] \leq [(a, b)]$ by definition of $\leq$ on $\mathbb{Z}$.
- **Antisymmetric:** Suppose we have $[(a, b)], [(c, d)] \in \mathbb{Z}$ such that $[(a, b)] \leq [(c, d)]$ and $[(c, d)] \leq [(a, b)]$. This means

$$a + d \leq b + c \text{ and } c + b \leq d + a$$

by definition of $\leq$ on $\mathbb{Z}$. But clearly the second relation is the same as $b + c \leq a + d$ (commutativity of addition), which thus means $a + d = b + c$ (as $\leq$ on $\mathbb{N}$ is antisymmetric). Therefore $[(a, b)] = [(c, d)]$ by definition of $\sim$ on $\mathbb{Z}$.

- **Transitive:** Suppose $[(a, b)], [(c, d)], [(e, f)] \in \mathbb{Z}$ are such that $[(a, b)] \leq [(c, d)]$ and $[(c, d)] \leq [(e, f)]$. Hence we have

$$a + d \leq b + c \text{ and } c + f \leq d + e.$$

Adding f to both sides of the first relation we get $a + d + f \leq b + c + f$. Note $b + c + f \leq b + d + e$ by the second equation, which thus means

$$a + d + f \leq b + d + e$$

by transitivity of $\leq$ on $\mathbb{N}$. Therefore $a + f \leq b + e$, which means $[(a, b)] \leq [(e, f)]$.

Hence $\leq$ is indeed a partial order on $\mathbb{Z}$. ■

- (b) *Proof.* We will prove that $<$ is irreflexive and transitive, since asymmetry is implied by [Proposition 11.2.8](#).

- **Irreflexive:** Let $[(a, b)] \in \mathbb{Z}$. Clearly $a + b \not< a + b$ (since $<$ on $\mathbb{N}$ is a strict partial order and is thus irreflexive), so $[(a, b)] \not< [(a, b)]$ by definition of $<$ on $\mathbb{Z}$.

- **Transitive:** Similar argument as in **(a)**, replacing $\leq$ with $<$.
Thus $<$ is a strict partial order on $\mathbb{Z}$. ■

APPENDIX B

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